

Unified Theory Based on Time-Wave Dynamics (The P-model)

Part II: Astrophysical and Cosmological Tests

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This paper is the application-validation volume of the P-model (cosmic/quantum program), and its symbol conventions, minimum assumptions, and mappings follow Part I (the core theory).

Abstract

This paper (Part II) is the astrophysical validation volume of the P-model. Assuming the mapping frozen in Part I ($P \rightarrow \phi \rightarrow \text{gravity/clocks/light}$) and the shared parameter β , it fixes reproducible comparison metrics from public primary data. The validation logic is condensed into the validation summary, and each topic is written in the order input (primary source) $\rightarrow$ process (fixed conditions) $\rightarrow$ metric $\rightarrow$ figure $\rightarrow$ notes (systematics), so that a third party can reach the same conclusion from the same inputs.

This paper does not claim a fit in which β is adjusted to match the observations. Instead, under the frozen mapping, it states explicit differential predictions (for example, the coefficient difference in EHT and the cosmological distance mapping), and fixes as output the falsification conditions that specify which observation rejects the model once which threshold is crossed. The velocity-saturation parameter δ_0 is not part of the Part I core. Here it is organized only as an extension hypothesis together with its upper bounds and rejection conditions. Within the early thermal history, BBN is fixed in Part III (the quantum volume) as **raw watch** and, after propagated derivation uncertainty, pass for the minimal He-4 chain ($Y_p \approx 0.25$). It is therefore referenced here as a boundary condition rather than re-derived. The acceptance or rejection of the full multi-isotope consistency (D/H, He-3/He-4, Li-7) is governed by the Watch verdict in Part III.

1 Introduction

The role of this paper (Part II) is to follow the Part I reference document (symbol conventions, minimum assumptions, mappings, and frozen values) and fix the validation procedures so that public primary data in astrophysics reproduce "the same input $\rightarrow$ the same output." The targets span weak fields (the Solar System), dynamical phenomena (binary radiation and gravitational-wave chirps), strong fields (EHT), and cosmological scales. This paper, however, does not immediately conclude that the reference framework has been replaced by a final theory. In cosmology in particular, the target is the **late-universe audit** of SNe Ia/BAO/AP/CMB peaks. Derivations on the early-thermal-history side (the minimal He-4 chain in BBN and the blackbody origin of the CMB) are separated into Part III (the quantum volume), and are referenced here only as boundary conditions. Likewise, the growth-rate mapping for structure formation (RSD-derived $f\sigma_8$) is fixed here only in its initial minimal form including delayed response. The current verdict is limited to the three low-redshift BOSS DR12 consensus points ($z = 0.38, 0.51, 0.61$), while the final decision across the full growth history, including the high-redshift side, is treated as a separate audit task.

What this paper fixes are the definitions of observables, processing conditions (inputs, preprocessing, windows, and outlier rules), and metrics (including the decomposition of uncertainty). The conclusions are then reduced to a validation summary and falsification-condition packs that can be tracked mechanically. This policy foregrounds falsifiability, namely where the decision is made, rather than the strength of the claim. Future improvements should clarify, under the same metrics, whether statistics (covariance) or systematics (procedure dependence) dominate, and then connect that result to required precision estimates and observational proposals.

This trilogy is meant to be read in the order **Part I** $\rightarrow$ **Part II** $\rightarrow$ **Part III**. Part I establishes the mappings, freezing rules, and falsification form; Part II accumulates application tests from classical to cosmological scales under the same framework; and Part III treats the quantum domain with the same interface, so that the P-model can be assessed not as a "claim" but as a falsifiable validation program.

The approach of this study is not to modify or extend the parameters of the standard cosmological model (Λ CDM). Instead, it deductively applies the P-model-specific assumptions established in Part I, including the minimal model of the static background P, and compares the resulting consequences with observational data. The standard theory is used only as a comparison target in the interpretation of distance indicators and black-hole shadows, not as the theoretical foundation.

Style rule: the main text closes using only the P-model's own definitional mapping procedure. Comparisons with reference frameworks (existing theories) are isolated in reference notes at the ends of sections and are not allowed to replace definitions or derivations in the main text.

Reference note: when this paper refers to GR or the standard cosmology as comparison rulers, the grammatical subject remains the P-model throughout. The reference frameworks are restricted to the comparison of final expressions of results.

Advance declarations:

- **GW polarization network:** at present scalar_exclusion is not yet satisfied, and the decision is deferred to incremental high-S/N data from three or more detectors (4.11).
- **EHT strong-field constraint:** the κ systematics are the rate-limiting term, and Watch is maintained until the differences in ring/shadow definition and calibration systematics are further reduced (see Sections 4.8 and 4.17).
- **Cosmological DDR:** the verdict is deferred until the interface audit of distance indicators (the

assumption dependence of the defining equations) is completed. It is treated not as immediate rejection but as a reconnection condition (5.3).

2 Theoretical Framework

The symbol conventions, minimum assumptions, and mappings used here follow Part I. The theoretical equations are repeated only minimally, and only the content required for Part II to be independently readable is summarized here.

2.1 P Field and Potential ϕ

Let $P(x)$ denote the stationary limit of time-wave density and P_∞ the far-field reference value. The adopted convention is $P > P_\infty$ **near mass**. For systems with rotation or flow, the extension $P_\mu = (P_t, P_1, P_2, P_3)$ introduced in Part I 2.8 is used. In the stationary limit, the exact reduction condition $J^i = 0 \Rightarrow P_i = 0$, $P_t \equiv P$ is imposed so that the covariant form reduces exactly to the scalar mappings of Part I 2.1-2.4. This paper strictly distinguishes P_∞ (the far-field reference) from P_t (the time component of the four-vector field). Only the ratio P/P_∞ , which is dimensionless, enters observables. The absolute value or dimensional identification of P lies outside the scope of this paper, although P and P_∞ must share the same dimensions for $\ln(P/P_\infty)$ to be defined. The gravitational potential is defined by

$$\phi \equiv -c^2 \ln \left(\frac{P}{P_\infty} \right)$$

so that it is zero at infinity and negative near mass.

2.2 Gravity (Equation of Motion)

The gravitational acceleration is unified as

$$\mathbf{a} = -\nabla\phi = c^2 \nabla \ln \left(\frac{P}{P_\infty} \right)$$

thereby fixing in equation form the statement that motion "slides down the P gradient."

2.3 Clocks (Gravitational Term + Velocity Term)

The rate of a clock is decomposed as

$$\frac{d\tau}{dt} = \frac{P_\infty}{P} \left(\frac{d\tau}{dt} \right)_v$$

and the velocity term is taken, as the standard recovery term, to be

$$\left(\frac{d\tau}{dt} \right)_v = \sqrt{1 - \frac{v^2}{c^2}} \quad (1)$$

The velocity-saturation parameter δ_0 is not included in the Part I core and is treated in Section 5.2 only as an extension hypothesis.

Reference note: the velocity term in this section is isomorphic to the special-relativistic time-dilation factor in the reference framework.

2.4 Light (Refraction Toward Higher P)

The refractive index for light is defined by

$$n(P) = \left(\frac{P}{P_\infty} \right)^{2\beta}$$

and light is taken to refract **toward the higher-P side** according to Fermat's principle. β follows the three-layer definition in Part I 3.4. The operational frozen value is $\beta = 1.0000105$ (Layer 2), and

Parts II and III do not alter this value locally. If an update becomes necessary, it is consolidated into a revision of Part I. The terminal verdict is governed by `policy_D` in Part IV (the LLR + MESSENGER β_{lt} chain), while VLBI is retained only as an auxiliary audit as long as `comparatoreligibility=false` (details in Section 4.4).

Reference note: when comparison is made in PPN language, the coefficient on the light side can be compared through the correspondence $1 + \gamma = 2\beta$; the factor of 2 is carried by β .

3 Data and Methods

This paper starts from primary sources (the observations themselves or official public datasets), and fixes both the storage of the acquired data and the processing conditions (windows, outlier rules, and so on) together with the output filenames. The primary sources (URLs and access dates) are collected in the appendix "Data Sources (Primary Sources)." This structure allows one to track "which output came from which input" even when data-provider updates or environment differences exist.

Some public primary sources do contain preprocessing based on the standard theory (geodetic systems, time systems, derived products, and so on). Therefore, at the present stage, this paper does not claim full raw-data reanalysis. Priority is given to fixing the inputs (primary sources) and visualizing corrections and systematics through fixed outputs.

The metrics are condensed into the validation summary, and each section states under which inputs, processing conditions, and uncertainties (statistical/systematic) the result was obtained. Statistical uncertainty is evaluated by fixed procedures such as bootstrap, while systematics are treated separately as sensitivities to procedural knobs such as window choice, offsets, and model selection. The key point is not to move β afterward in order to force agreement, but to keep it frozen as defined in Part I and, under that frozen value, output differential predictions and rejection thresholds.

Parameter operation classes (allowlist of movable parameters):

To avoid after-the-fact tuning (overfitting), movable parameters are fixed operationally in the following three classes.

Class	Representative parameters	Role	Re-tuning allowed?	Tracking target
Core (frozen)	β , metric_choice=effective	Core constants of the mapping frozen in Part I	No (not re-estimated)	Part I frozen registry / Part IV hash lock
Nuisance (systematic knobs)	κ_{fit} , window/offset, scatter budget	Procedural parameters used only to separate observational-interface systematics	Yes, but only as systematic-sensitivity audits	The sys sensitivity figures and score_raw in each section
Extension (extension hypotheses)	δ_0 , λ_H , τ_{eff}	Branches used only for differential prediction and future discrimination	Conditionally yes, but only as separated extension branches	Section 5 differential predictions and the falsification pack

Standard result outputs (stat /sys /sys-stat):

Each results section from 4.2 through 4.17 must present the following three items, in the same order, for its principal metric.

- **stat**: statistical uncertainty from a fixed procedure (bootstrap, fit covariance, jackknife, and so on)
- **sys**: sensitivity width to procedural knobs (window, offset, definition differences, calibration differences)
- **sys/stat**: $|\text{sys}|/\sigma_{\text{stat}}$, namely the degree to which systematics dominate over the statistical error

If a quantity cannot be defined, it is recorded as N/A , together with the reason (missing data, out-of-scope estimate, gate-only item, and so on).

3.1 Shared Rejection Procedure

Each validation block in this paper fixes its falsifiability through the following common form. From this point onward, every section follows this order, so that the content functions not as loose discussion but as a reproducible and rejectable protocol.

Purpose: what is validated or discriminated in the section. **Input:** the primary source (or a derived public product), its origin, time span, version, and the local fixed storage path. **Process:** preprocessing (selection, outlier rules, windows, offsets) and frozen parameters. **Equation:** the equations used in the section (definitions of variables only; no derivations). **Metric:** the statistics adopted (for example, RMS, χ^2 , z) and, if needed, the rejection threshold (for example, beyond 3σ). **Result:** the principal numbers, with **stat** and **sys** separated. Representative figures are inserted when needed, with 2-4 lines stating what the reader should look at. **Discussion:** interpretation of the result, including alternative explanations. **Notes:** the reference framework (GR, and so on), dominant systematics, limitations, and the update policy.

3.2 Primary Data and Preprocessing Dependence (Explicit Statement of Dependencies)

This paper claims a primary-source-first workflow, but public astrophysical data often contain practical preprocessing. To avoid overstatement, the dependence points are stated first, together with the inputs that would be needed to remove those dependencies.

- **GPS (IGS products)**: the default inputs are IGS precise orbit/clock products (SP3/CLK and related files), which already arise from an estimation procedure based on raw observation data (RINEX observations from each station). To remove preprocessing dependence, one would need to fix the observation RINEX files together with the network model and the estimation procedure (weights, outliers, and corrections). At the present stage, priority is given instead to visualizing the dependency points themselves.
- **Cassini (DSN tracking; PDS TDF)**: TDF is close to a primary tracking observable, but can still include operational processing such as frequency generation, time-scale handling, and ground-station corrections. Removing the dependency would require raw receiver-side logs and a fully fixed correction pipeline (time scales, media corrections, antenna motion, and so on).
- **LLR (CRD Normal Point)**: the Normal Point is a compressed product of the raw photon sequence. Removing preprocessing dependence would require the full-rate photon time series and the fixed Normal Point construction rules (binning and outlier treatment).
- **EHT (black-hole shadow)**: the default inputs are measured quantities published in papers and public tables, which already include preprocessing from calibration, imaging, and model fitting. Removing that dependency would require public visibility data together with fixed calibration procedures and imaging knobs.

4 Results

This chapter gives a minimal summary for each validation corresponding to the rows of the validation summary, using the primary source, processing conditions, metrics, and the principal numbers ($stat/sys/sys - stat$). The purpose is not to make an impressionistic claim of "success" or "failure," but to fix comparison points that can be recomputed under the same interface, so that the grounds for adoption or rejection remain traceable.

4.1 Validation Summary Table (Scoreboard)

Purpose: to provide an overview of the total set of frozen validation items in this paper, before entering the detailed sections 4.2-4.17.

The verdict status (and the figure legend colors) is unified here into **four aggregate categories plus one operational state**.

- **Pass:** the theoretical prediction agrees with the primary data within the adopted threshold.
- **Watch:** systematics are the rate-limiting term, and the present data do not yet support a clean decision of acceptance or rejection.
- **Reject:** the item violates the falsification condition (threshold) and is therefore explicitly rejected.
- **Reference:** the item serves only as a comparison value or upper-limit benchmark, rather than as a direct verdict channel.
- **Closed (ineligible):** the comparator has been judged ineligible as an operational state. It is tracked in the relevant section but excluded from the total counts of the four aggregate categories.

At the ledger layer in Part IV (Verification Materials), the reserved term **Inconclusive** is retained for states that are neither rejected nor decidable because the required information is still missing. In the present chapter, it is omitted because there are no such rows in the current aggregate.

Result (overall score): The total verdict counts are as follows.

Verdict	Count	Summary
Pass	14	The main frozen gates in weak-field and strong-field tests are satisfied.
Watch	5	These include the low-degree-of-freedom boundary in the Bullet Cluster, the lack of ΔAIC support for the strong-field higher-order term λ_H , and the EHT κ systematics, all of which require further observational-analysis precision.

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Verdict	Count	Summary
Reject	3	Flat-background metric caseA, the pure-scalar limit, and the direct Pantheon+ fit. These are fixed as explicit mismatches against the falsification conditions.
Reference	5	Velocity saturation δ_0 , GPS clock residuals, and comparisons with published estimated values (light deflection γ^* , redshift ϵ^* , and frame-dragging summary values).

Population definition (27 items): The aggregate counts consist of 19 section-mapped items from 4.2-4.17 (20 rows minus one closed row), plus 7 differential-prediction/cosmology items from Chapter 5, plus 1 imported boundary item (the BBN He-4 boundary condition from Part III). The closed (**ineligible**) state is managed separately as a comparator-eligibility state and is not counted in the four aggregate categories above.

The number of items for which a σ -style score is defined (meaning `score_kind=abs_z` in the machine-readable validation ledger, excluding gate-only items) is $N = 7$. (The DDR interface-audit row and the pure-scalar rejection row are not included in this denominator.) Within 1σ : 5/7 items, within 2σ : 7/7 items, within 3σ : 7/7 items. (Breakdown: 5 items from Chapter 4, 1 item from Chapter 5 (RSD), and 1 imported item (the BBN He-4 boundary condition).)

Items with a defined σ score (N=7)	$ z $	Section
Solar light deflection (VLBI)	0.735	4.4
Gravitational redshift (GP-A/Galileo)	0.077	4.9
Double pulsar (orbital decay)	1.067	4.10
Rotation (frame dragging)	0.278	4.12
Structure formation (RSD $f\sigma_8$)	0.308	5.3.2
BBN (early thermal history)	0.625	5.5.1
EHT (black-hole shadow)	1.174	4.8

4.1.1 Item Map (Section Map + Decision Variables)

This table shows only the section mapping in the main text for 4.2-4.17 (20 rows including one closed row). Nineteen of these contribute to the four-category aggregate. The differential-prediction/cosmology items in Chapter 5 and the imported boundary item are governed by the verdict tables in their own sections and by the Part IV ledger.

Item	Verdict	score_kind (decision variable)	Threshold (Pass/Watch/Reject)	Section
LLR (lunar laser ranging)	Pass	weighted_rms_ns, abs_z	weighted RMS ≤ 5 ns and the $ z $ distribution within the operational threshold / boundary gives Watch / excess gives Reject	4.2
Cassini (solar conjunction)	Pass	shape_rmse, abs_z_beta	shape RMSE within threshold and the β consistency not significantly displaced / boundary gives Watch / displacement gives Reject	4.3
Solar light deflection (VLBI)	Reference	abs_z_gamma	comparison only against published γ^* (outside the primary verdict path)	4.4
Direct VLBI β estimate	Closed (ineligible)	comparator_eligible	retain as an auxiliary audit while comparator eligibility remains false (excluded from total counts)	4.4.2
LLR β (κ_{LLR})	Pass	abs_z_kappa	$abs(z(\kappa_{\text{LLR}} - 1))$ within the operational threshold	4.4 (see Part IV)
MESSENGER β (β_{lt})	Pass	abs_z_beta_lt	$abs(z(\beta_{\text{lt}} - 1))$ within the operational threshold	4.4.3
β terminal audit (policy_D)	Pass	policy_D_governance	policy_D governance conditions satisfied and active policy fixed to D	4.4 (see Part IV)
Viking (Shapiro delay)	Pass	delta_peak_us	representative delay difference within the allowed range gives Pass / boundary gives Watch / systematic excess gives Reject	4.5
Mercury (perihelion precession)	Pass	delta_arcsec_per_century	$\Delta\omega$ difference within threshold gives Pass / boundary gives Watch / excess gives Reject	4.6
GPS (satellite clocks)	Reference	rms_ns_reference	comparison only (no direct acceptance/rejection verdict)	4.7
EHT (black-hole shadow)	Watch	abs_z_kappa, sys_stat_ratio	κ within threshold and acceptable sys/stat gives Pass / systematic-limited gives Watch / clear displacement gives Reject	4.8
Gravitational redshift (GP-A/Galileo)	Reference	abs_z_epsilon	comparison only against published ϵ^* (outside the primary verdict path)	4.9

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Item	Verdict	score_kind (decision variable)	Threshold (Pass/Watch/Reject)	Section
Double pulsar (orbital decay)	Pass	abs_z_ratio, dipole_gate	the quadrupole-leading ratio satisfies $ z \leq 3$ and the dipole gate remains respected	4.10
Gravitational waves (chirp phase + polarization audit)	Watch	r2_sanity, scalar_overlap_proxy	simultaneous satisfaction of the chirp sanity and polarization gates gives Pass / failure gives Watch / clear breakdown gives Reject	4.11
Rotation (frame dragging)	Reference	abs_z_r_drag	comparison only against published GP-B/LAGEOS summary values	4.12
Pure-scalar limit (no rotational effect)	Reject	scalar_limit_abs_z	$ z > 5$ gives Reject (direct rejection evidence)	4.12.1-4.12.2
XRISM (high-resolution X-ray spectroscopy)	Pass	sys_stat_ratio, detected_obsids	simultaneous satisfaction of the sys/stat and detected-obsid gates gives Pass / partial failure gives Watch	4.13
Galaxy rotation curves (SPARC)	Pass	delta_chi2_count, median_chi2_nu	the improved-galaxy fraction and median indicator remain within the operational threshold	4.14
Cluster collision offsets (Bullet-like systems)	Watch	chi2_dof, p_value	low-degree-of-freedom boundary gives Watch (currently $p \approx 0.076$) / strong mismatch gives Reject	4.15
Joint strong-field higher-order constraint (EHT + GW + Pulsar + Fe-K α)	Watch	delta_aic_joint, abs_z_joint	reaching $\Delta AIC(= AIC_{\text{baseline}} - AIC_{\text{P-model}}) > 2$ gives Pass / failure to reach it gives Watch	4.17

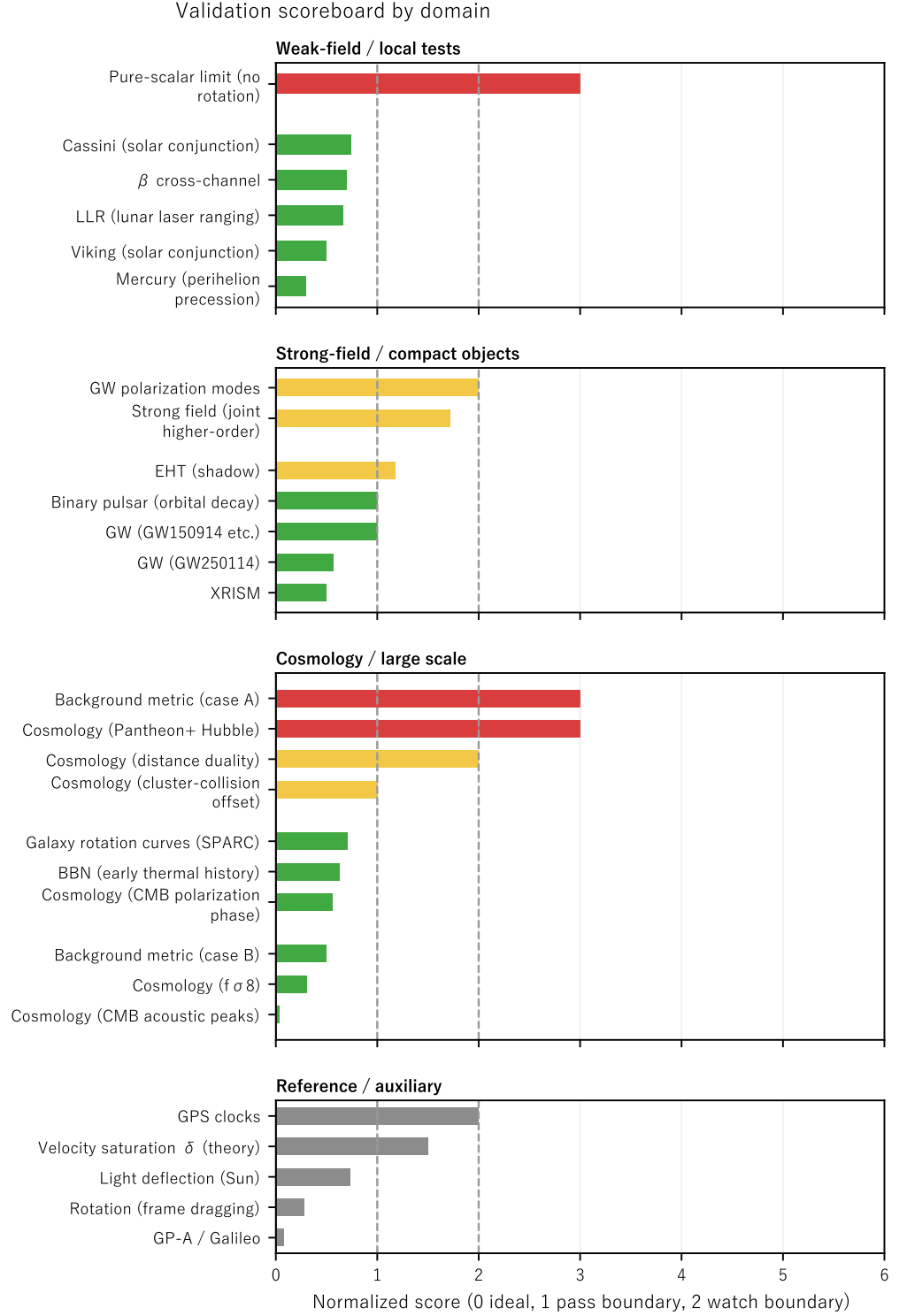


Figure 1: Validation scoreboard. It gives a color-coded overview of the status of all items (Pass / Watch / Reject / Reference).

Caution: the colors in the figure are not the conclusion itself. Their purpose is to visualize which systematics and precision gaps should be worked on next. The rigorous decision criteria are the metrics and rejection conditions stated in each section. The bar lengths are normalized to the verdict bands (Pass/Watch/Reject/Reference), while the raw measured values are tracked in the indicator values for each row (validation summary / score_raw). score_raw is fixed in the Part IV verification

ledger (public summary JSON), and is tracked through *rows*[], *score_{raw}* (raw value) together with *rows*[], *score_kind* (evaluation type).

4.2 LLR (Lunar Laser Ranging)

Purpose: to confirm, in the weak-field Solar-System regime, that the P-model (geometry + solar Shapiro term) is not inconsistent with the primary data (CRD Normal Points), using operational metrics such as weighted RMS, normalized residuals, and chi-square-like indicators.

Input: publicly released CRD Normal Points from the EDC (many station×reflector combinations; Normal Point products). [20][19]

Process: atmospheric delay, tides (solid Earth + lunar body), and ocean loading are added step by step on top of geometry + solar Shapiro. Outliers (spikes) are excluded from the statistics while retaining the original data rows; no automatic reassignment is performed.

Equation: define the observed-minus-model residual r_i in ns. $r_i = t_{i,\text{obs}} - t_{i,\text{model}}$ Operationally, the NP bin RMS (σ_{NP}) is used as the weight, and weighted RMS is used together with the normalized residual $z_i = |r_i|/\sigma_{\text{NP},i}$. The median RMS over station×reflector is retained as an auxiliary metric.

Metric: the primary metrics are weighted RMS (with NP bin RMS weights), the $|z|$ distribution, and a chi-square-like indicator. Auxiliary metrics include the median RMS over station×reflector and the outlier fraction.

Result:

Metric target	Value	Notes
weighted RMS (all points)	2.2185 ns	NP bin RMS weighting
weighted RMS (excluding nglr1)	2.2183 ns	no artificial improvement from nglr1 mixing
modern APOL weighted RMS	4.9994 ns	nglr1 excluded after 2023; within the estimated model floor (5.0624 ns)
median RMS (unnormalized)	4.3542 ns	N=13132
modern APOL median $ z $	22.07	$p_{95} = 69.59$ (roughly equivalent to 3.97 ns)
full-sample median $ z $	17.30	$p_{95} = 83.54$ (roughly equivalent to 3.11 ns)

Decomposing the causes of the 4 ns exceedance into the three layers station/target/group shows that the excess in the group-weighted RMS (4.4699 ns) is concentrated mainly in APOL and GRSM (station contribution: APOL 54.78%, GRSM 40.07%, MDOL 5.15%). The main residual source is therefore treated not as a single global offset, but as a systematics pattern that depends on station-reflector combinations, namely the overlap of troposphere-driven tails with APOL-specific systematics.

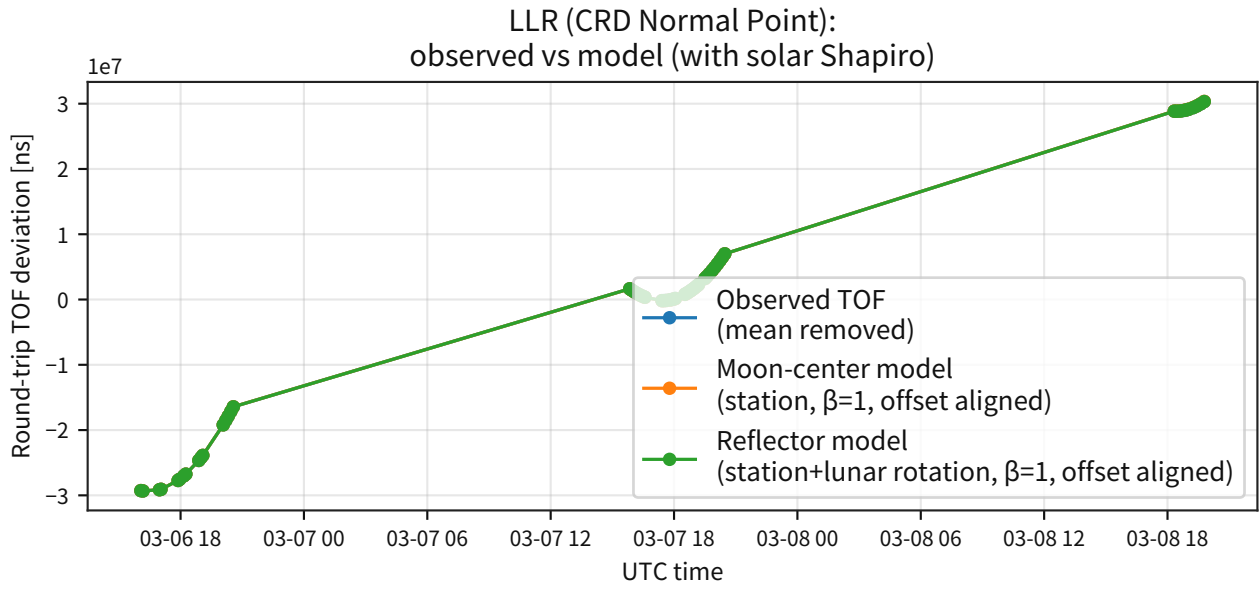


Figure 2: Overlay of the observed ToF and model ToF, used to confirm agreement on the overall scale.

Supplement: because the difference is hard to see in this plot, the next residual figure should be read as the primary figure.

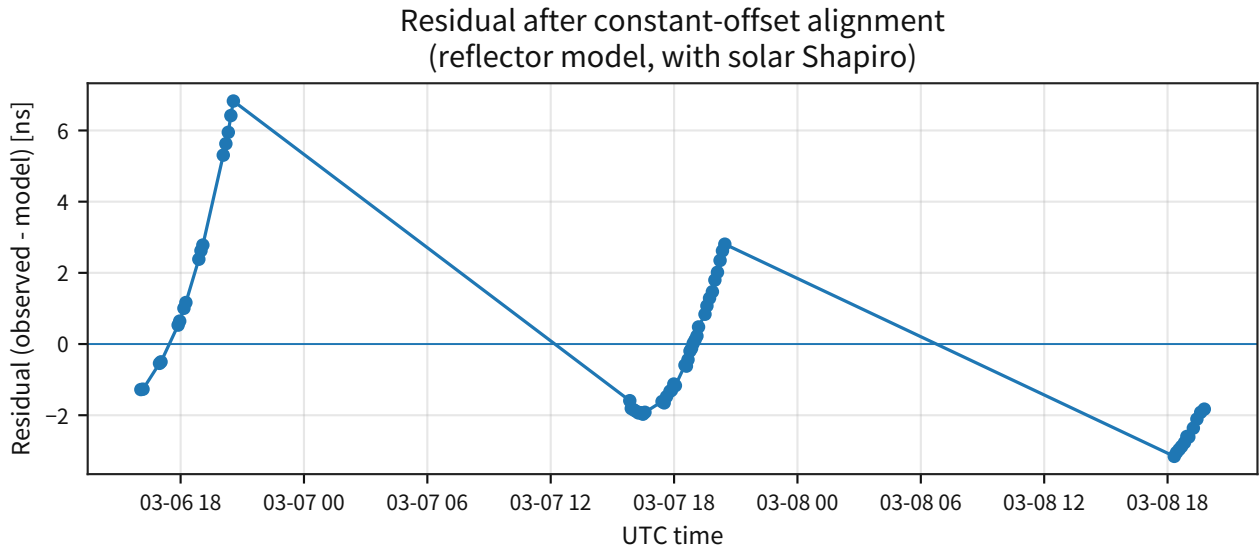


Figure 3: Representative observed-minus-model residual figure, directly visualizing the difference.

Supplement: the direction of improvement from the stepwise additions (troposphere / tides / ocean loading) can be confirmed in the same figure.

Lunar Laser Ranging (LLR):
observed-P-model residuals (inliers only, $n=13133$)

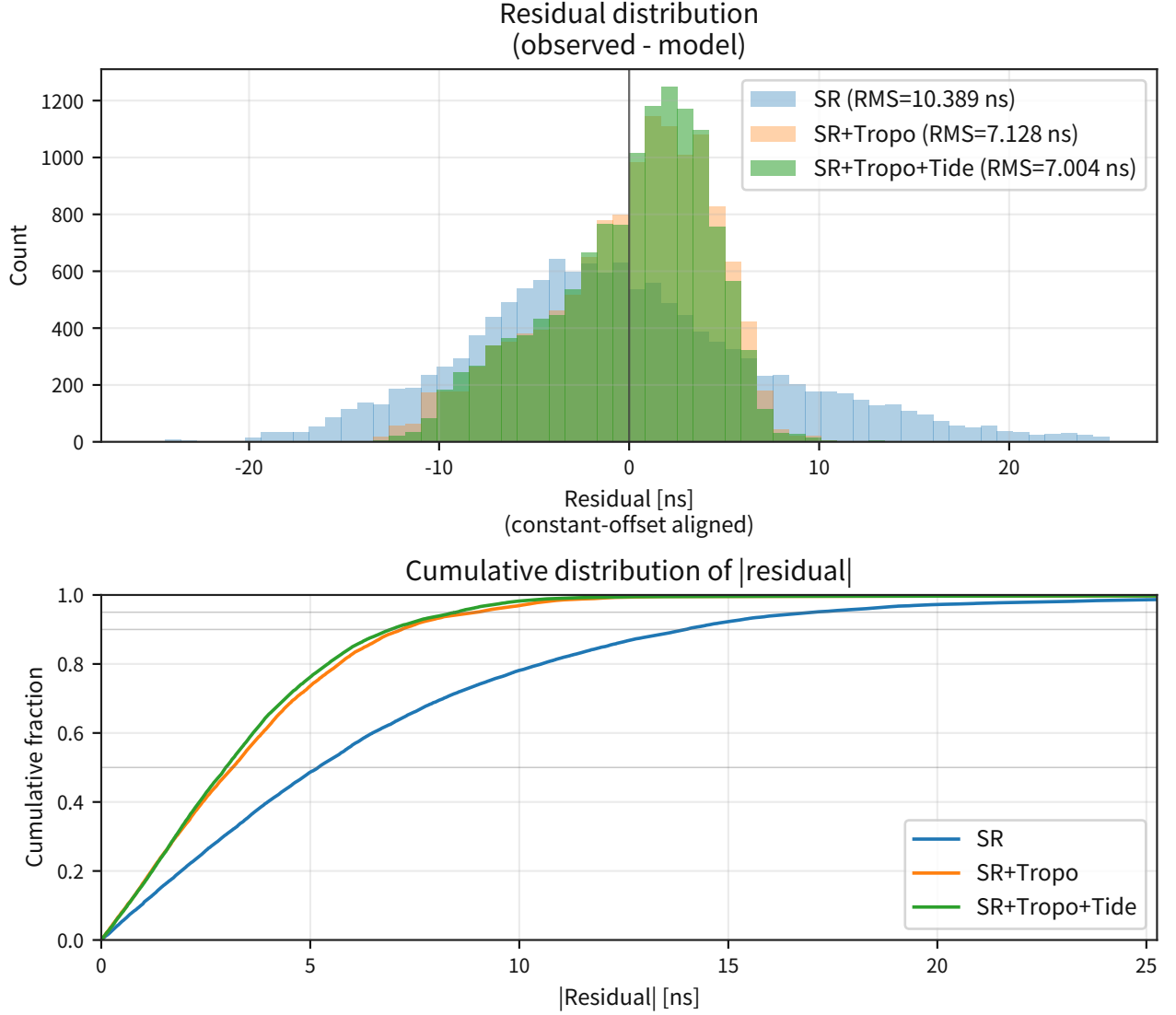


Figure 4: Residual distribution used to compare RMS and outlier behavior under the stepwise additions, and to confirm convergence and tail evolution.

Supplement: in this paper, not only “RMS reduction” but also “explainability of the outlier tail” is treated as part of the acceptance logic.

Discussion: ns-level residuals are determined by multiple overlapping systematic factors, so it is important that the contributions of the added terms can be separated step by step.

Notes: geometry (EOP, station coordinates, lunar ephemerides and rotation models, and so on) depends on the standard model. This section therefore does not aim at an independent ephemeris estimate, but rather at fixing the framework in which residuals and systematics are decomposed under fixed geometry. [21, 24, 22, 23] Additional diagnostic figures (elevation dependence, tide ablation, station-coordinate source differences, and so on) are also used to separate the systematics.

4.3 Cassini (Solar Conjunction / Doppler)

Purpose: to reproduce the Doppler residual series (TDF y) during solar conjunction from the primary data, and confirm whether the shape and scale remain consistent under the P-model with frozen β .

Input: Cassini TDF (y) from PDS SCE1. [8]

Process: acknowledging that y is a pseudo-residual of “observation minus prediction,” the comparison fixes the assumptions for restoring the reference Shapiro term, handling missing intervals, and the sign convention.

Equation: compute the RMSE and the correlation **corr** of y in the indicator window (for example, ± 10 days). RMSE: $\text{RMSE} \equiv \sqrt{\frac{1}{N} \sum_i (y_{i,\text{obs}} - y_{i,\text{P}})^2}$ Correlation (Pearson):

$$\text{corr} \equiv \frac{\sum_i (y_{i,\text{obs}} - \bar{y}_{\text{obs}})(y_{i,\text{P}} - \bar{y}_{\text{P}})}{\sqrt{\sum_i (y_{i,\text{obs}} - \bar{y}_{\text{obs}})^2} \sqrt{\sum_i (y_{i,\text{P}} - \bar{y}_{\text{P}})^2}}$$

Metric: RMSE of y and the correlation **corr**.

Result:

Window	RMSE	corr	N	Notes
± 10 days	7.069e-11	0.96308	190	β frozen

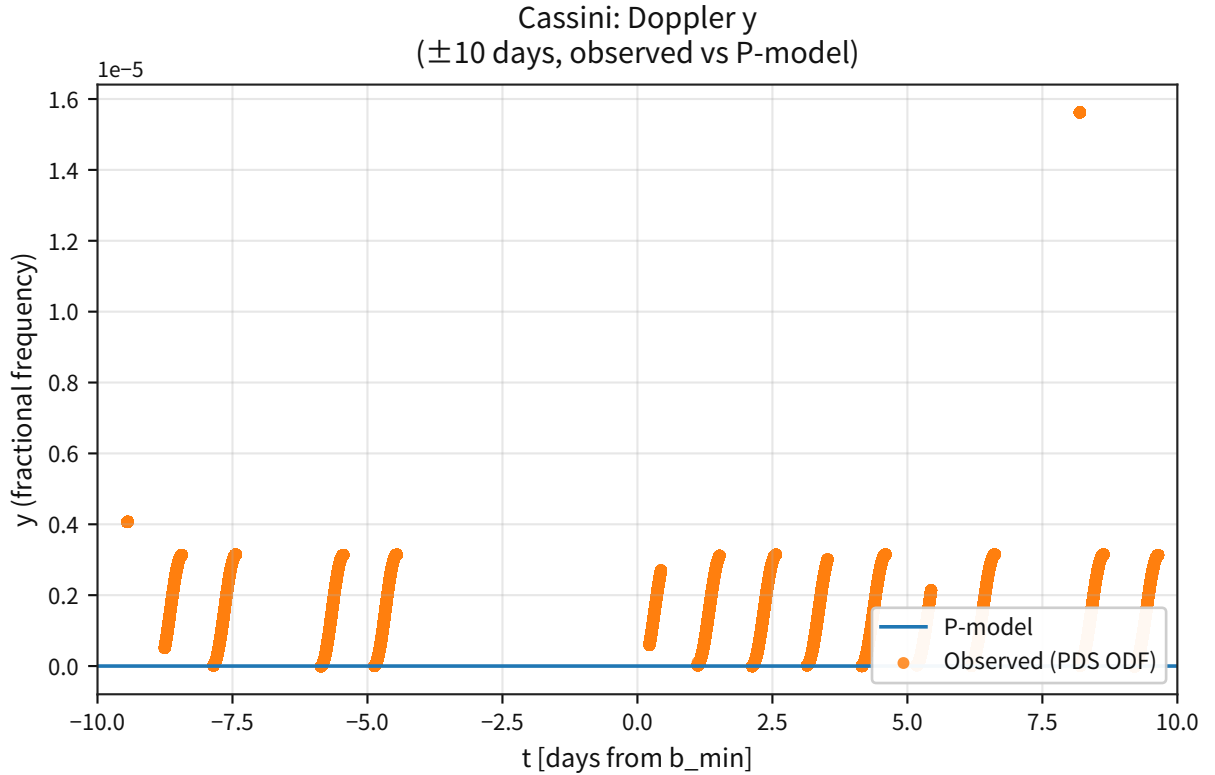


Figure 5: Overlay of the observed y and the P-model y near solar conjunction (± 10 days).

Supplement: the blue curve is the P-model and the orange points are the observed series. Starting from PDS TDF, the observational series prioritizes the Ka+X plasma correction, fills missing data

with a single-band pseudo-residual and restored reference Shapiro term, and then applies smoothing and detrending. This figure is meant primarily to confirm shape agreement; absolute-level differences should be read from the residual plot.

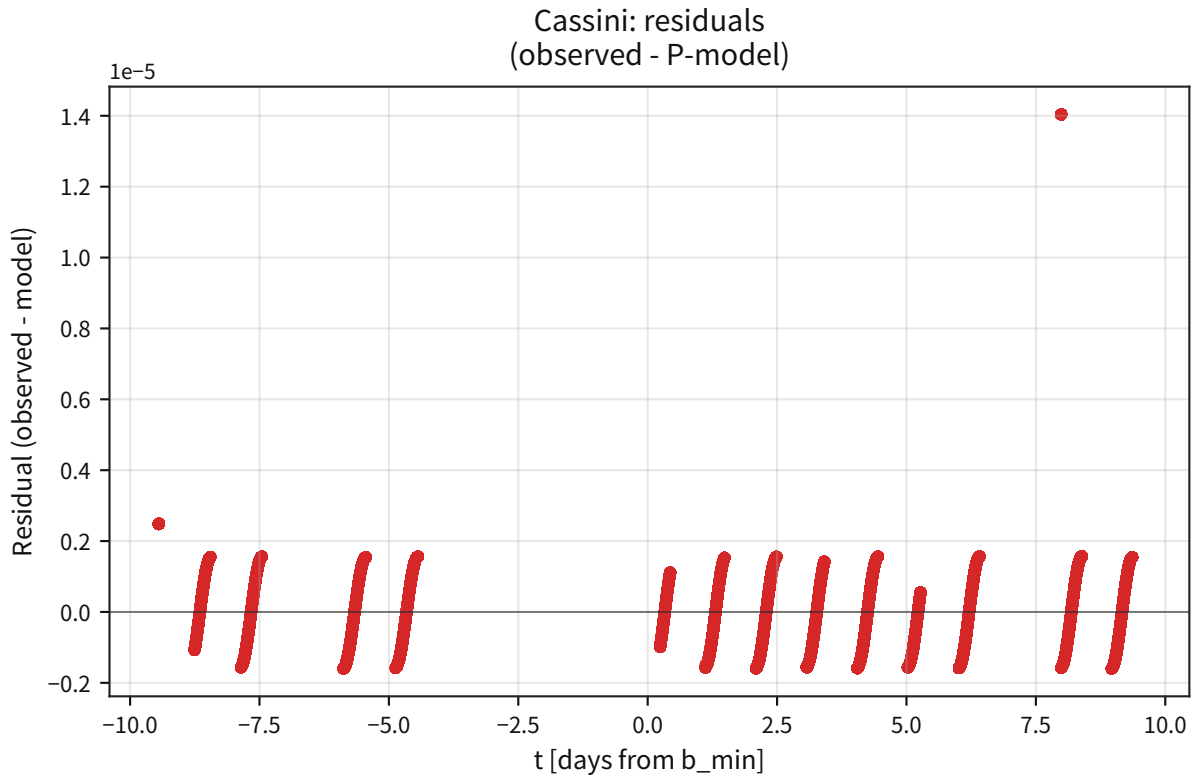


Figure 6: Residuals (observation minus P-model) as a function of time, used to diagnose bias and systematic drift.

Supplement: the effect of missing intervals and endpoint behavior on the indicators (RMSE/corr) can also be checked here.

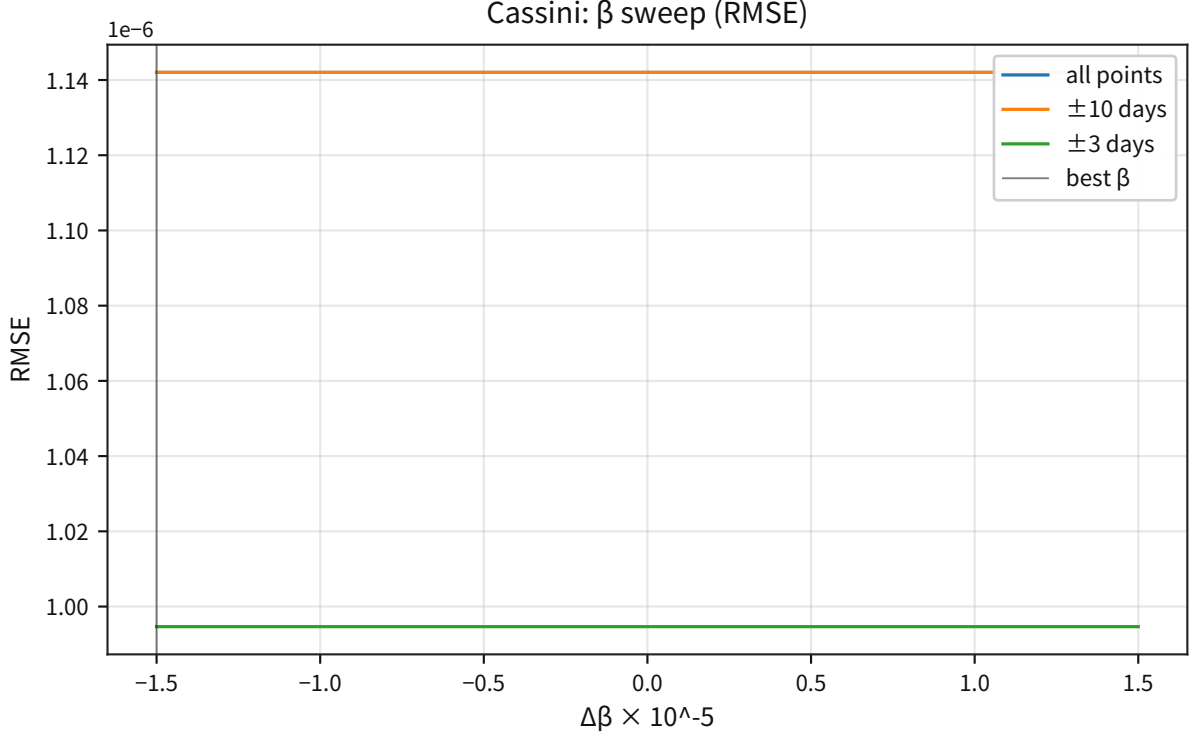


Figure 7: RMSE sensitivity when β is scanned, used to show how strongly Cassini alone constrains β .

Supplement: under the present window and processing conditions, the RMSE is fixed to be nearly flat in β .

Discussion: the reach of this section is to confirm “consistency with the primary data.” It does not claim a precision estimate of β .

Notes: y depends on the TDF definition and is not a pure raw observation, since it can include operational processing such as frequency generation, time-scale handling, and other corrections. The separation of those dependency points is treated as a separate task. [2][7] In the background-metric comparison, caseA (flat $\eta_{\mu\nu}$) is fixed as Reject and caseB (effective metric $g_{\mu\nu}(P)$) as Pass; the β -frozen evaluation in this section is therefore operated only on the effective-metric branch (caseB). The integrity of the PDS primary-data processing (agreement with the digitized paper figure) is fixed in Part IV.

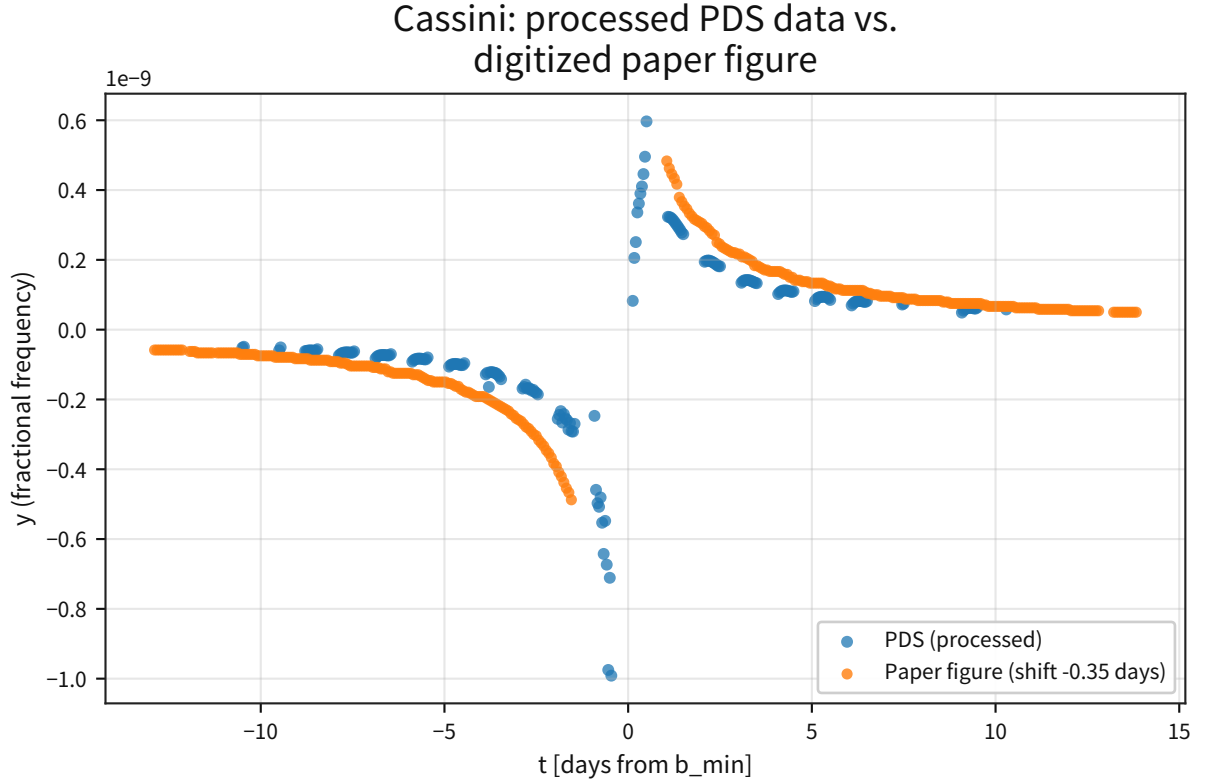


Figure 8: Consistency check against the digitized paper figure (reproducibility of the PDS primary data).

4.3.1 Direct β Fit to Cassini ODF raw (Primary-data-direct / PPN-independent audit)

Purpose: to fit a single template directly to PDS ODF raw $y(t)$ and audit the behavior of the resulting $\Delta\beta$ estimate without passing through a published PPN γ value, as a bias audit.

Input: PDS SCE1 ODF raw Doppler (`--sourcepds_odf_raw`) and the unit template $y_{\text{unit}}(t; \beta = 1)$ built from the same geometry. [8]

Process: with the window fixed to ± 10 days around conjunction, fit the observed series by

$$y_{\text{obs}}(t) = a_0 + \Delta\beta y_{\text{unit}}(t; \beta = 1)$$

using least squares, and define $\beta_{\text{est}} = \beta_{\text{ref}} + \Delta\beta$ with $\beta_{\text{ref}} = 1$.

Metric: N , $\Delta\beta$, β_{est} , RMSE, and corr.

Result:

Input series	N	$\Delta\beta$	β_{est}	RMSE	corr
ODF raw (± 10 days)	421841	-615.109	-614.109	1.139×10^{-6}	0.0693

Supplement: the direct-fit figure for ODF raw (observed y , linear fit, residuals) and its indicator JSON are governed by the public snapshot ledger in Part IV.

Discussion: direct fitting to ODF raw is useful as a bias-avoidance route because it does not use a published PPN value. At present, however, $|\beta_{\text{est}}| \gg 1$ and the correlation is low, so the physical interpretation of an absolute β is unstable. This route is therefore kept as a **primary-data-direct audit entry (Watch)** and is not used in the main verdict path.

Notes: because ODF raw y can include practical differences in receiver system, band choice, and correction definitions, an additional audit is required in which the ODF-side preprocessing definitions (media correction, time scale, frequency reference) are fixed under a common interface before an absolute β can be frozen.

4.3.2 Preprocessing Dependence of the Cassini Direct Fit (ODF/TDF cross-comparison)

Purpose: to quantify how much $\Delta\beta$ changes when the input series (TDF plasma / TDF raw / ODF raw) is swapped while keeping the direct-fit equation and window fixed.

Input: the three Cassini direct-fit branches (tdf_plasma, tdf_raw, odf_raw).

Process: for all three series, fit

$$y_{\text{obs}}(t) = a_0 + \Delta\beta y_{\text{unit}}(t; \beta = 1)$$

in the same window (± 10 days), and generate the cross_source comparison table.

Metric: N , $\Delta\beta$, corr, RMSE, and the inter-series difference $\Delta\beta(\text{tdf_plasma}) - \Delta\beta(\text{tdf_raw})$.

Result:

Series	N	$\Delta\beta$	corr	RMSE
TDF processed plasma	190	+0.10565	0.4880	6.42×10^{-11}
TDF raw	844707	+0.02860	0.2769	1.25×10^{-11}
ODF raw	421841	-615.109	0.0693	1.14×10^{-6}

In the same window, the differences become $\Delta\beta(\text{tdf_plasma}) - \Delta\beta(\text{tdf_raw}) = +0.0770$ and $\Delta\beta(\text{tdf_plasma}) - \Delta\beta(\text{odf_raw}) = +615.2146$, confirming strong dependence on the preprocessing definition of the input series.

Discussion: between the two TDF branches (plasma/raw), the sign and order of magnitude agree, but ODF raw deviates strongly in absolute value. Therefore, before an absolute β can be used as a primary verdict channel, the ODF-side interface (band definition, media correction, time scale, and sign convention) must be fixed by the same procedure.

Notes: this comparison is an auxiliary audit for bias control. The main verdict gates remain the TDF shape-consistency branch and the scalar-limit gate. The comparison figures and machine-readable JSON are governed by the public ledger in Part IV.

4.3.3 Normalized I/F for ODF raw (Promotion condition for absolute β)

Purpose: to formalize the common interface for band, sign, media correction, and time scale as the prerequisite for promoting ODF raw to an absolute- β decision path.

Input: PDS SCE1 ODF raw, the same geometry template, and the available metadata for ground-side correction definitions.

Process: absolute- β evaluation is activated only when the four keys defined in the I/F item table immediately below are all fixed.

Metric: the fixed-state status of the I/F (Pass/Watch) and the stability of $\Delta\beta$ after re-fitting once the I/F is fixed, namely whether the inter-series difference converges within the operational threshold.

Result:

I/F item	Fixing rule	Current state
Band (band_id)	Match X/Ka and related band identifiers between the observed series and the template	Pass
Sign (doppler_sign_convention)	Fix the DSN Doppler sign convention (receive minus transmit) to a single convention	Watch
Media correction (media_correction_state)	Fix record-level media state by joining TRK/SCE1 and ancillary payloads	Pass (external_csp_join_ready, coverage_any=1.00)
Time scale (time_scale_id)	Fix UTC/TDB and related conversion rules to a single time interface	Pass

Under this fixed I/F, the same-window, same-equation refit was rerun and the following convergence gates were audited.

Convergence gate	Threshold	Measured	State
internal TDF difference $ \Delta\beta_{\text{plasma}} - \Delta\beta_{\text{raw}} $	≤ 0.10	0.0770	Pass
ODF-TDF convergence $z_{\min}$	≤ 3	45.10	Watch
ODF shape correlation corr_{odf}	≥ 0.25	0.0693	Watch

The sign-I/F candidates (same sign / inverted sign / constant-offset removal) were refit in the same window, and even the best candidate failed to satisfy the convergence threshold.

Sign candidate (same window)	$z_{\min}$	corr	Verdict
same_sign	45.10	0.0693	Watch
invert_sign	45.09	0.0693	Watch
same_sign + offset removal	33.03	0.0693	Watch
invert_sign + offset removal (best)	33.01	0.0693	Watch

The sign candidates were then re-audited at record level (station / band / arc); details are governed by the public audit ledger in Part IV.

Record-level audit indicator	Measured	Verdict
Number of audit groups (station/band/arc)	14	-
Number of groups reaching closure ($z_{\min} \leq 3$ and $\text{corr} \geq 0.25$)	2	Watch
Closure-achievement ratio	0.1429	Watch

At the same time, the conjunction window and quality condition (outlier trim) were refixed under a common rule, and *window_days* in 10, 8, 6, 4 together with *trim_quantile* in 0, 0.005, 0.01 were exhaustively reevaluated.

Window/quality reevaluation (best scenario)	Measured	Verdict
Window width	4.0 days	-
Trim fraction (upper-tail removal in $ residual $)	0.01	-
Number of used points	117484	-
Best candidate	same_sign	-
$z_{\min}$	0.0141	Pass
corr	5.07×10^{-5}	Watch

In the TRK/SCE1 cross-match audit, the media correction on the *TRK* – 2 – 23 side was confirmed to be defined as the following external CSP correction: `ADJUST / DELETE, MODEL (CHPART / WET NUPART / DRY NUPART / DRVID)`. The primary payloads (*.ion/.tro*) were then obtained from *sce1_ancillary/ion* and *sce1_ancillary/tro* in PDS SCE1, and the CSP command windows were joined onto the ODF rows using *time_utc*, *station_rx*, *station_tx*, *downlink_band_id*, *uplink_band_id*, *data_type_id*, and *source_file*. The join audit (governed by the public audit ledger in Part IV) confirmed *record_level_media_state_extractable*=true, *coverage_any_ratio*=1.0, *coverage_on_mapped_ratio*=1.0, and *status*=pass, so *media_correction_state* was promoted to Pass. This removed the hard watch that had originated from the missing external CSP join.

As a supplement, ODF labels still contain a *TRK* – 2 – 18 reference note (for FORMAT ID=1), but the target ODF has FORMAT ID=2, and the current specification match is therefore fixed to *TRK* – 2 – 23.

Discussion: the large inter-series differences in 4.3.2, especially for ODF raw, indicate that the dominant source of instability in absolute β lies not in the physical equation but in the I/F mismatch. The media-correction key has now been promoted to Pass, so three of the four keys (band, media correction, and time scale) are fixed. On the other hand, record-level decomposition reaches closure in only 2 of the 14 groups, and even the best window/quality scenario leaves *corr* far below the threshold. The *doppler_sign_convention* therefore remains Watch, and promotion of ODF raw to an absolute- β decision path is deferred. The β terminal decision itself is already fixed to Pass by *policy_D*, so this Watch only marks the remaining work on the ODF raw route and does not affect the β verdict.

Notes: the main verdict gates continue to be *cassini* (TDF shape consistency) and *scalar_limit_reject* (hard gate). The machine-readable ledgers for the manifest and convergence gates are governed by the public snapshot ledger in Part IV.

4.4 Solar Light Deflection (Observed γ vs P-model)

Purpose: to evaluate the strength of light deflection (published γ^*) as a reference comparison against the frozen β , outside the primary verdict path.

Input: published light-deflection strengths γ^* estimated from VLBI and related analyses. [4, 5, 6]

Process: freeze β and compare the result against the observed $\gamma \pm \sigma$. The weak-field deflection curve $\alpha(b)$ is also shown.

Equation: weak-field deflection angle for impact parameter b :

$$\alpha(b) \approx \frac{4\beta GM}{c^2 b} \quad (2)$$

Correspondence between β and γ for comparison:

$$\gamma_{\text{pred}} = 2\beta - 1 \quad (3)$$

Metric: $z = (\gamma_{\text{obs}} - \gamma_{\text{pred}})/\sigma$, with the solar-limb deflection angle α_{limb} listed as an auxiliary quantity.

Result: using the published value with the smallest uncertainty, $\gamma^* = 0.99983 \pm 0.00026$, the reference comparison for the P-model with frozen β is as follows.

Metric model	γ_{pred}	z score	Verdict
caseA (flat $\eta_{\mu\nu}$)	0	3845.5	Reference (large mismatch)
caseB (effective metric $g_{\mu\nu}(P)$)	1.000021	-0.73	Reference (within 1σ)

The difference between the P-model prediction (caseB) and the published value stays within 1σ ($z \approx -0.7$). It is recorded here only as a reference comparison and is not used as a primary acceptance/rejection channel.

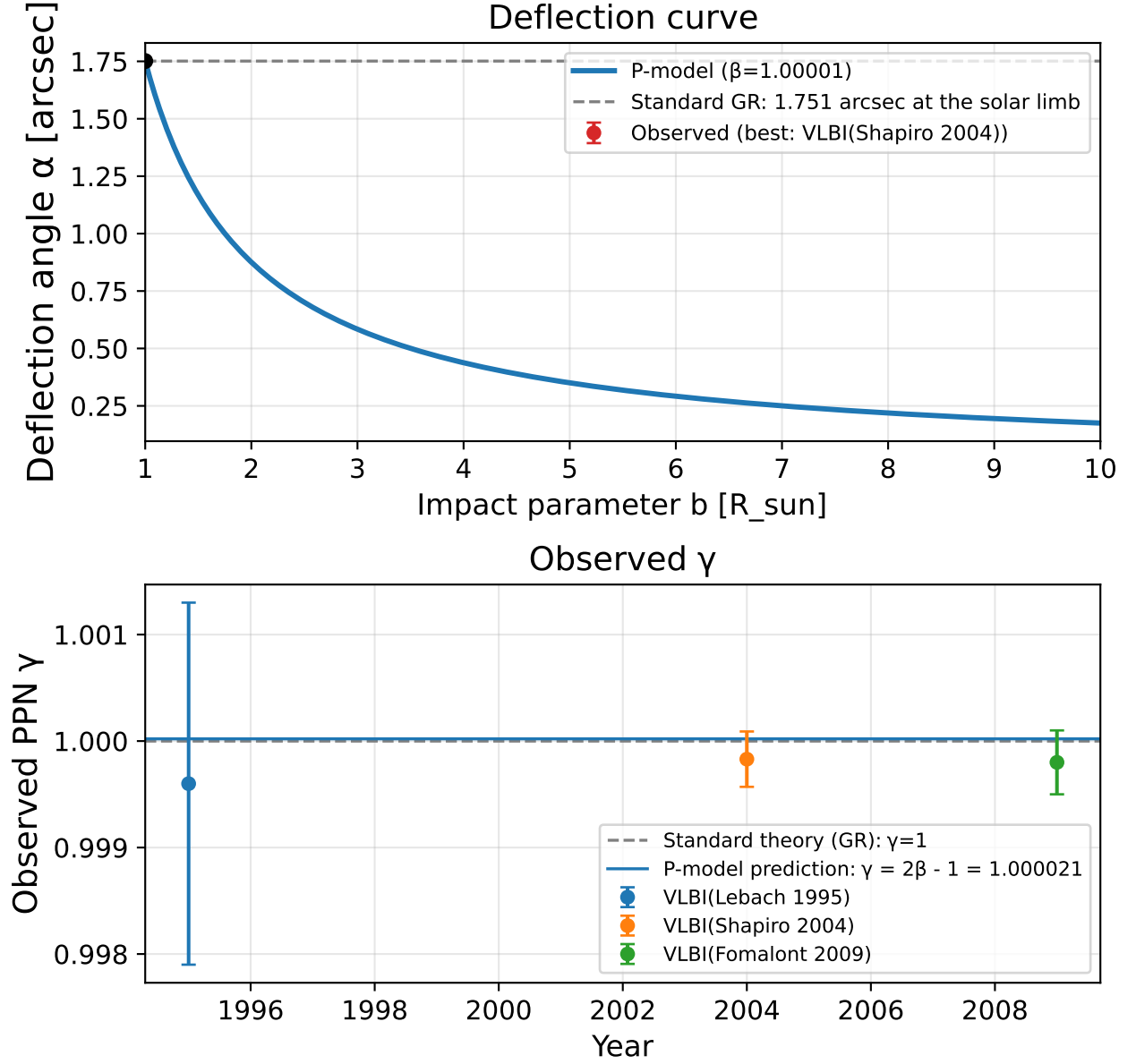


Figure 9: Upper panel: the $\alpha(b)$ curve under frozen β and the scale near the solar limb. Lower panel: comparison between observed $\gamma \pm \sigma$ and γ_{pred} , serving as the entry point to the z-score.

Discussion: the reach of this section is simply to confirm that the weak-field behavior is not inconsistent. It does not claim a precision estimate of β from this dataset alone.

Notes: the γ^* used here is a published estimated value, not a direct fit to a primary time series. This section is therefore treated as a Reference row outside the primary verdict path. [2] γ is the light-deflection parameter in the reference PPN framework, and $\gamma_{\text{pred}} = 2\beta - 1$ is used only as a comparison mapping. [2] The choice of background metric has already been fixed through the caseA/caseB comparison, so all later weak-field comparisons are operated on the $g_{\mu\nu}(P)$ branch.

4.4.1 Direct fit to IVS vgosDb (AUA020) group delay as a primary-data auxiliary audit

Purpose: independently of the published- γ^* comparison in 4.4, fix an entry route for directly estimating β from the primary data of session AUA020 and avoid dependence on pre-existing theory values.

Input: from the vgosDb inside 17MAY01XA.tgz (*Session* = AUA020), use *GroupDelayFull_bX/bS*, *DelayTheoretical*, *Cal – BendSun*, *FreqGroupIono*, *DelayFlag*, and *Source*.

Process:

1. Build the ionosphere-free group delay from *GroupDelayFull_bX/bS*.
2. Define Base=DelayTheoretical-Cal-BendSun and replace only the solar-gravity term by the P-model coefficient β .
3. Compare the nuisance-mode sensitivities none/baseline_intercept/baseline_intercept_linear under the same interface.
4. Compare the difference with and without the source filter (0229+131, 0235+164).

Equation:

$$\tau_{\text{obs,IF}} = \frac{f_X^2 \tau_X - f_S^2 \tau_S}{f_X^2 - f_S^2}, \quad \tau_{\text{obs,IF}} - \tau_{\text{base}} = c_0 + \beta \tau_{\text{BendSun}}^{\text{GR}} + \varepsilon$$

$$\tau_{\text{base}} = \tau_{\text{theory}} - \tau_{\text{BendSun}}^{\text{GR}}$$

Metric: session identification, $\beta \pm \sigma_\beta$, weighted RMSE, and the source-filter difference $\Delta\beta_{\text{all-selected}}$ together with $|z| = |\Delta\beta|/\sigma_{\text{comb}}$.

Result:

Audit item	Value	Verdict
Session identification (17MAY01XA – > AUA020)	206/206 nc files matched	Pass
selected (2 sources), baseline_intercept_linear	$\beta = 1.04023 \pm 0.01123$	
all sources, baseline_intercept_linear	$\beta = 1.18669 \pm 0.06961$	
source-filter difference (best mode)	$\Delta\beta_{\text{all-selected}} = +0.14646,$ $ z = 2.08$	Watch

The auxiliary outputs are governed by the public audit ledger in Part IV.

Discussion: without nuisance terms (**none**), the β estimate becomes divergently unstable, so baseline systematics must be estimated simultaneously. The agreement improves between baseline_intercept and baseline_intercept_linear, but a difference of about $|z| \approx 2.08$ remains with respect to the source filter. The present handling is therefore an auxiliary Watch audit.

Notes: this section does not replace the published- γ^* comparison in 4.4. It adds an independent audit route directly tied to the primary data. Promotion into the primary verdict path requires the same-interface reproduction of additional sessions, including confirmation that the source-filter dependence collapses.

4.4.2 Closure verdict for the VLBI β estimate (systematic limit of current data)

Purpose: to determine whether the VLBI-based, PPN-independent estimate of β is eligible as an adoptable comparator.

Input: the AUA020 primary-fit audit, the all-sky session-combination audit, the high-sensitivity source $\times$ session matrix audit, and the cross-channel terminal ledger.

Process:

1. Reevaluate the source-filter difference for AUA020 alone (4.4.1).
2. Judge inter-session consistency from the all-sky χ^2/dof .
3. Use the source $\times$ session matrix to compare near-Sun sources with distant reference sources.
4. Reflect the LLR cross-consistency and comparator eligibility into the terminal ledger.

Metric: $\Delta\beta$, $|z|$, χ^2/dof , and comparator eligibility.

Result:

Audit item	Value	Verdict
AUA020 alone (source-filter difference)	$\Delta\beta = +0.14646$, $ z = 2.08$	Watch
all-sky aggregate (53 sessions)	$\chi^2/\text{dof} = 271.86$	Reject
source $\times$ session (near-Sun source 0235+164)	$\chi^2/\text{dof} = 7.52$	Reject
source $\times$ session (near-Sun source 0229+131)	$\chi^2/\text{dof} = 22.01$	Reject
source $\times$ session (distant reference source 0955+476)	$\chi^2/\text{dof} = 0.51$	Pass
cross-channel comparator eligibility	false (reason=subset__consistency__reject)	Reject

Supplement: the Pass for the distant source indicates only internal consistency. It does not imply direct agreement with $\beta = 1$.

Discussion: in VLBI, the mismatch persists for near-Sun sources, and the all-sky aggregate also retains a large χ^2/dof . The main reason is that nondispersive solar-corona residuals and year-dependent source structure are superposed on the same observable, the group delay.

Notes: in the current snapshot, VLBI comparator eligibility is fixed to false, and VLBI is removed from the primary PPN-independent β -estimation channel. The detailed audit trace and reproducibility entry route are governed by the Part IV audit ledger.

4.4.3 MESSENGER theory-native direct β estimate (ODF main branch + TNF rescan)

Purpose: to fix the theory-native β estimate obtained from MESSENGER radiometric primary data as an independent channel in the main text after the VLBI route has been closed.

Input: the public audit outputs of Stages D/E/J using the ODF main branch and TNF replay, together with the β terminal ledger in Part IV.

Process:

1. Fix the principal estimate as β_{lt} (the light-time branch).
2. Separate β_{dyn} as a diagnostic value monitoring deficiencies in the nongravitational-acceleration model.
3. Reflect the LLR–MESSENGER pair gate and policy governance into the cross-channel terminal state.

Equation:

$$|z(\beta - 1)| = \left| \frac{\beta - 1}{\sigma_\beta} \right|$$

Metric: the stage_j final gate, $\beta_{lt} \pm \sigma$, $|z(\beta_{lt} - 1)|$, $\beta_{dyn} \pm \sigma$ (diagnostic), the policy_D pair gate, and the active policy.

Result:

Audit item	Value	Verdict
MESSENGER final gate	stage_j=pass, stage_e_replay=pass, bias_audit=pass	Pass
primary β (β_{lt})	$\beta_{lt} = 1.042153 \pm 0.983588$, $ z(\beta_{lt} - 1) = 0.04286$	Pass
diagnostic β (β_{dyn})	$\beta_{dyn} = -8.543545 \pm 0.679605$, $ z(\beta_{dyn} - 1) = 14.04$	Reject (diagnostic)
policy_D pair gate (LLR vs MESSENGER)	$ z(\beta_{llr} - \beta_{lt}^{(MESSENGER)}) = 0.04217$, $watch/pass = pass/pass$	Pass
beta terminal (active policy)	policy_D_exclude_plus_ messenger_fusion	Pass
active policy switch	already switched to policy_D after satisfying the LLR bias/promotion gate and the MESSENGER pair gate	Pass

Supplement: in final operation, the LLR bias/promotion gate and the MESSENGER pair gate have already been confirmed, and the active policy has been switched to policy_D. Here $\beta_{messenger}$ refers to β_{lt} , the light-time-side principal estimate. β_{dyn} is a diagnostic value for nongravitational-acceleration model insufficiency and is not used in the pair gate. The score detail has already been synchronized to the active policy policy_D_exclude_plus_messenger_fusion.

Discussion: MESSENGER achieves Pass as an independent channel through β_{lt} , which is consistent with $\beta = 1$. By contrast, β_{dyn} is highly sensitive to the nongravitational-acceleration model and is therefore restricted to diagnostic use rather than to primary coefficient estimation.

Notes: the official cross-channel terminal verdict is governed by the ledger in Part IV. At present the switch to policy D has already been reflected, and the β terminal status is fixed as Pass.

4.5 Viking (Solar Conjunction / Shapiro Delay)

Purpose: to confirm, through the peak value and peak time, whether the round-trip Shapiro delay during solar conjunction remains consistent in scale under the frozen β .

Input: Earth/Mars vectors (Sun-centered ICRF) obtained from JPL HORIZONS, from which the conjunction geometry is reconstructed. [11] The observation is taken from literature values for the representative round-trip Shapiro-delay peak (roughly 200-250 μ s). [9][10]

Process: compute the round-trip Shapiro delay from the geometry, with β frozen, and compare the resulting scale and peak time.

Equation: the round-trip Shapiro delay $\Delta t_{\text{Shapiro}}$ derived from the P-model refractive index $n(P) = (P/P_\infty)^{2\beta}$ is fixed, in weak-field approximation, as

$$\Delta t_{\text{Shapiro}} \approx \frac{4\beta GM_\odot}{c^3} \ln \left(\frac{r_\oplus + r_{\text{Mars}} + R_{\text{EM}}}{r_\oplus + r_{\text{Mars}} - R_{\text{EM}}} \right) \quad (4)$$

where $r_\oplus$ and r_{Mars} are heliocentric distances and R_{EM} is the Earth-Mars distance.

Metric: peak delay in μ s and the UTC time of the peak.

Result:

Item	Value	Notes
round-trip peak delay	247.96 μ s	
peak time	1976-11-25 UTC	

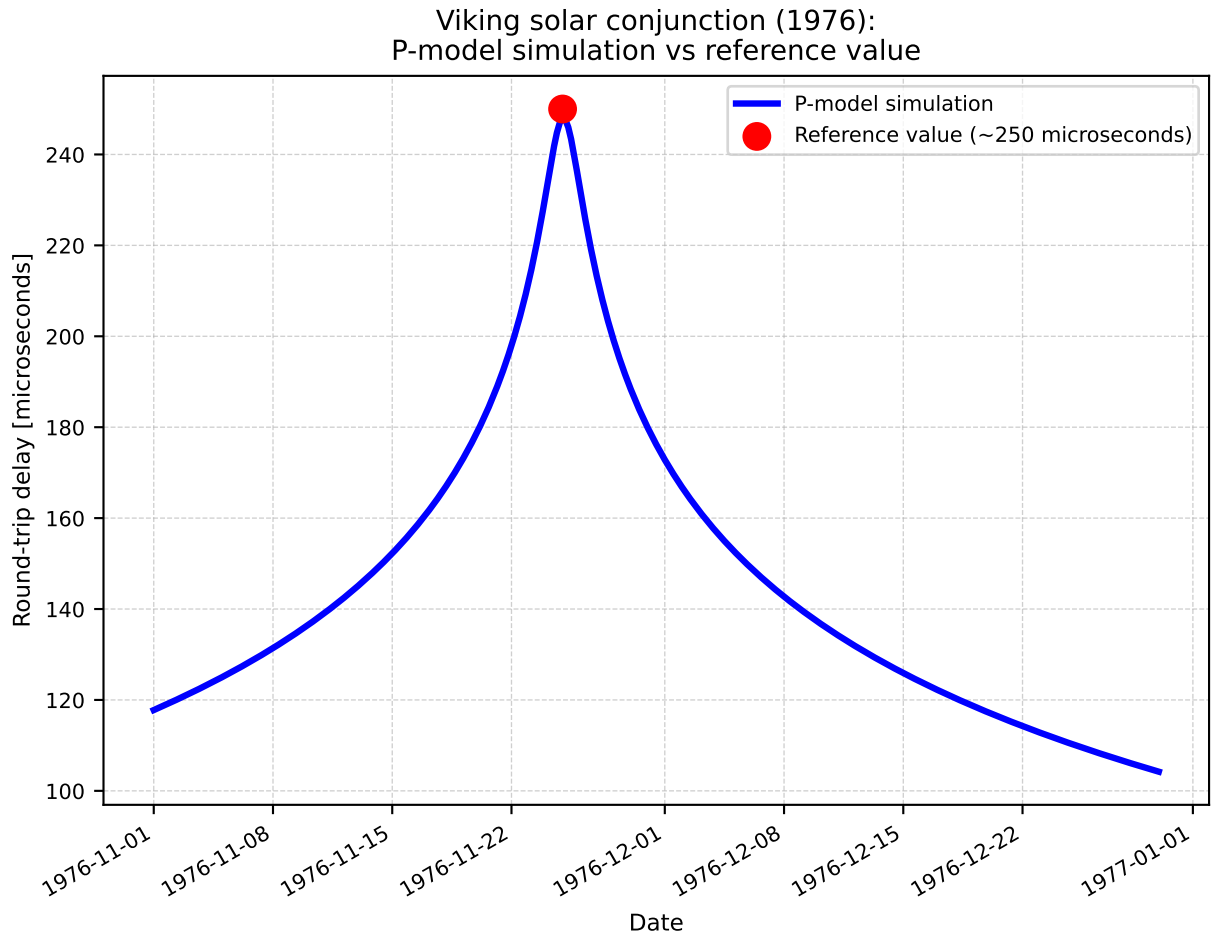


Figure 10: P-model round-trip delay curve compared in the same figure with the representative range quoted in the literature.

Supplement: the main point here is consistency of coefficient scale and peak location.

Notes: the “observation” used here is not a primary time series but a representative literature value. A direct fit to primary data remains a separate task. The geometry depends on HORIZONS (the standard ephemeris), so this section is positioned only as a scale check under fixed geometry.

4.6 Mercury (Perihelion Precession)

Purpose: as a representative weak-field orbital correction, to confirm whether Mercury’s perihelion advance can be reproduced within the P-model.

Input: the representative observed residual value for perihelion precession, **42.98 arcsec/century**. [12]

Process: numerically integrate Mercury’s orbit in the Sun-centered two-body approximation and estimate the perihelion advance per century from the perihelion direction angle of each orbit.

Equation: use the weak-field, slow-velocity acceleration derived from geodesics on the effective metric $g_{\mu\nu}(P)$ in the numerical integration.

$$\mathbf{a} = -\frac{GM_{\odot}}{r^3}\mathbf{r} + \frac{GM_{\odot}}{c^2r^3} \left[\left(\frac{4GM_{\odot}}{r} - v^2 \right) \mathbf{r} + 4(\mathbf{r} \cdot \mathbf{v})\mathbf{v} \right]$$

This form is fixed so that, in the 1PN limit of the weak-field expansion of the effective metric $g_{\mu\nu}(P)$ ($\phi/c^2 \ll 1$), it becomes isomorphic to the Schwarzschild 1PN acceleration. The analytically predicted perihelion shift per orbit is then

$$\Delta\omega \approx \frac{6\pi GM_{\odot}}{c^2a(1-e^2)} \quad (5)$$

Metric: perihelion advance (arcsec/century) and linearity with respect to orbit count (whether the cumulative advance remains nearly linear).

Result:

Metric	Value	Notes
Representative observed residual	42.98 arcsec/century	
P-model estimate	42.9929 arcsec/century	
Difference	+0.0129 arcsec/century	

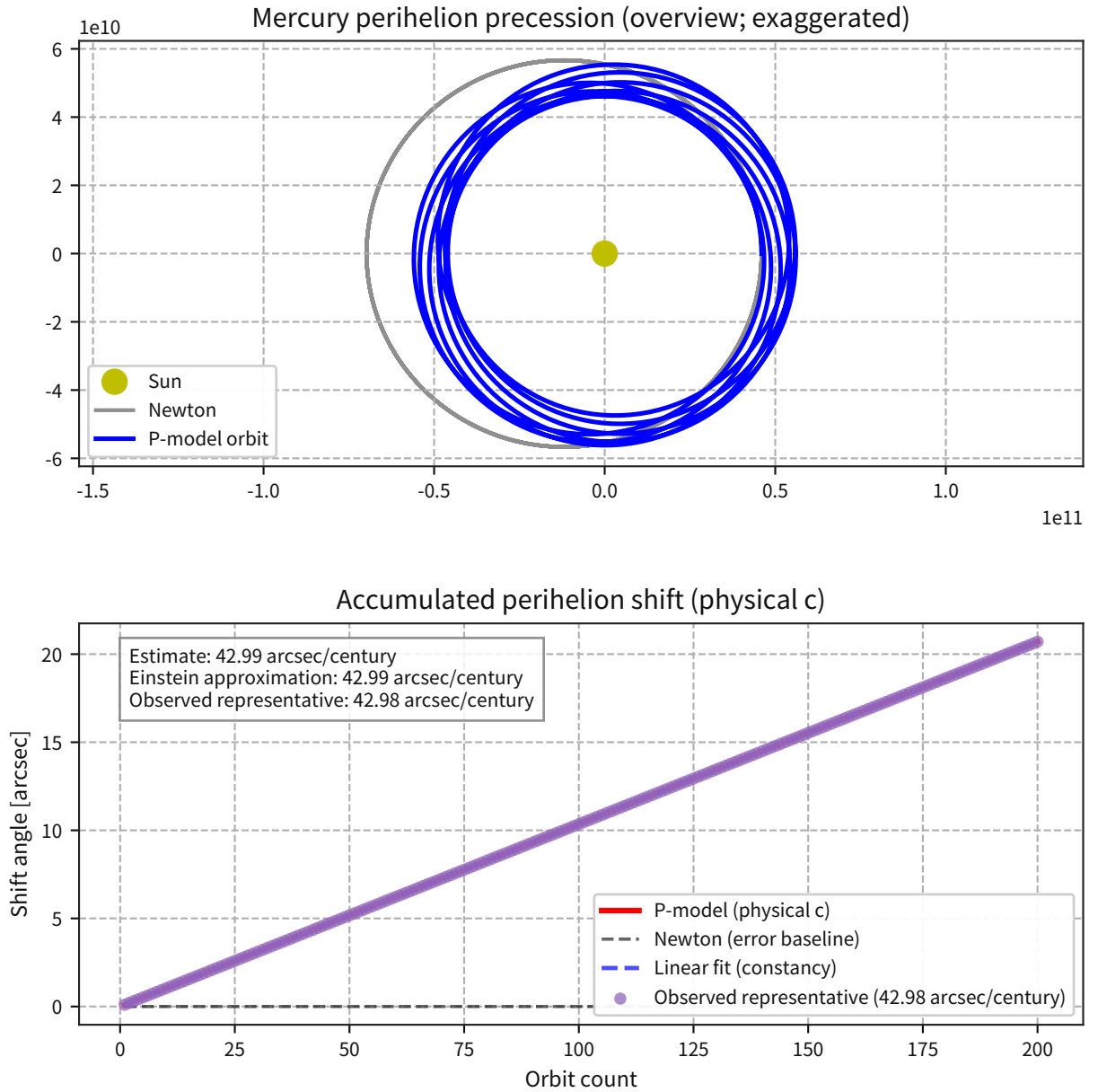


Figure 11: Mercury’s orbit (top view) and the cumulative change in the perihelion direction are shown in a single figure.

Supplement: confirm that the cumulative advance is nearly linear, and use that behavior as the scale comparison.

Caution: this is not a precision ephemeris model including planetary perturbations, the solar quadrupole, or minor-body perturbations. The agreement here is therefore limited to checking the scale of the weak-field correction.

4.7 GPS (Satellite Clocks)

Purpose: to fix the scale of relativistic corrections for satellite clocks as a residual comparison under the same input set (IGS Final), and to provide an entry point for systematics including procedural dependence.

Input: use IGS Final CLK/SP3 (quasi-observed) as the reference and compare it with the BRDC broadcast ephemeris for the same day. [13][14]

Process: for 31 satellites on 2025-10-01, compute clock residuals relative to IGS Final after removing bias and drift, and compare the RMS for BRDC and the P-model. [3]

Equation: the periodic relativistic correction term associated with eccentricity, derived from the P-model clock rate $d\tau/dt = (P_\infty/P)\sqrt{1 - v^2/c^2}$, is written as

$$\Delta t_{\text{rel}} = \frac{2\mathbf{r} \cdot \mathbf{v}}{c^2} = \frac{2\sqrt{GM_\oplus a}}{c^2} e \sin E \quad (6)$$

where $M_\oplus$ is the Earth mass and E is the eccentric anomaly. Residuals are defined by $\Delta t_i = t_{i,\text{obs}} - t_{i,\text{model}}$, and their RMS is used as the metric.

Metric: residual RMS for each satellite (ns) plus the median RMS over all satellites. Auxiliary metrics: correlation and RMSE of the periodic component.

Result:

Metric	BRDC	P-model	Notes
Median RMS (ns)	0.3278	0.4022	31 satellites, reference = IGS Final
Satellites for which “P-model is smaller”	—	6/31	versus BRDC
Periodic-component correlation	—	0.999818	standard formula vs P-model
Periodic-component RMSE (ns)	—	0.4776	same

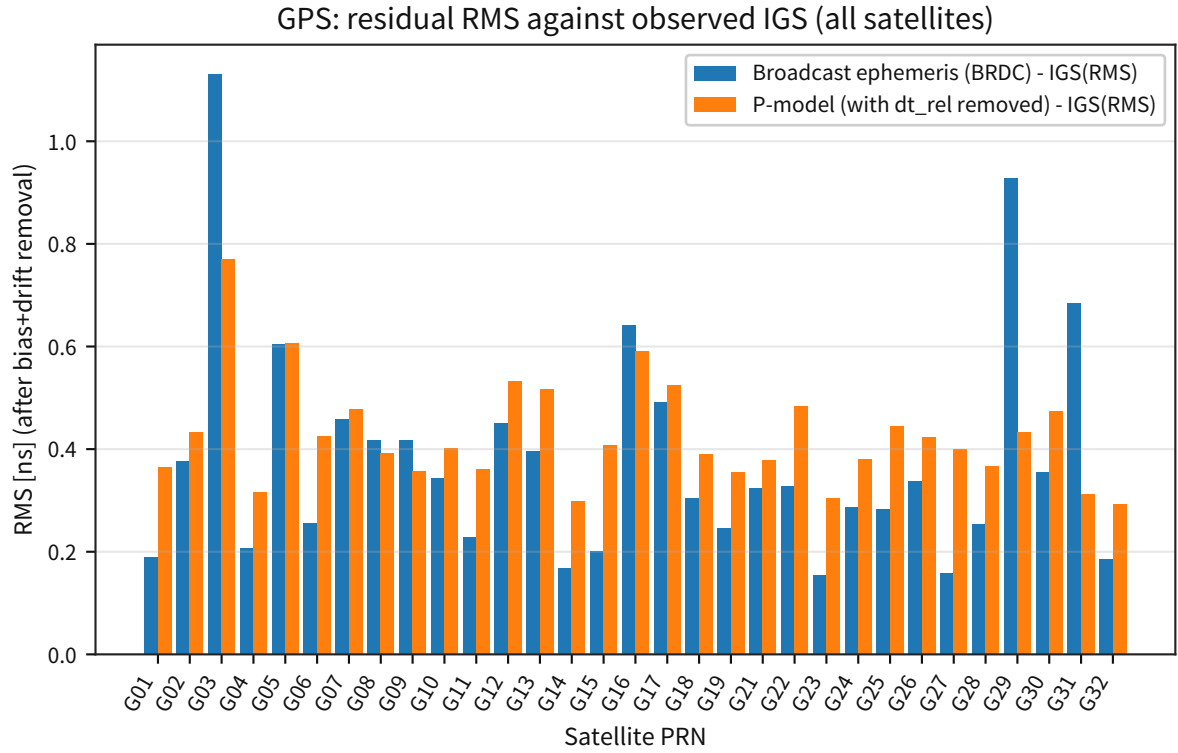


Figure 12: Residual RMS for each satellite (BRDC vs P-model), giving an overview of the distribution and the outlier satellites.

Supplement: it is important that the comparison uses the same reference (IGS Final).

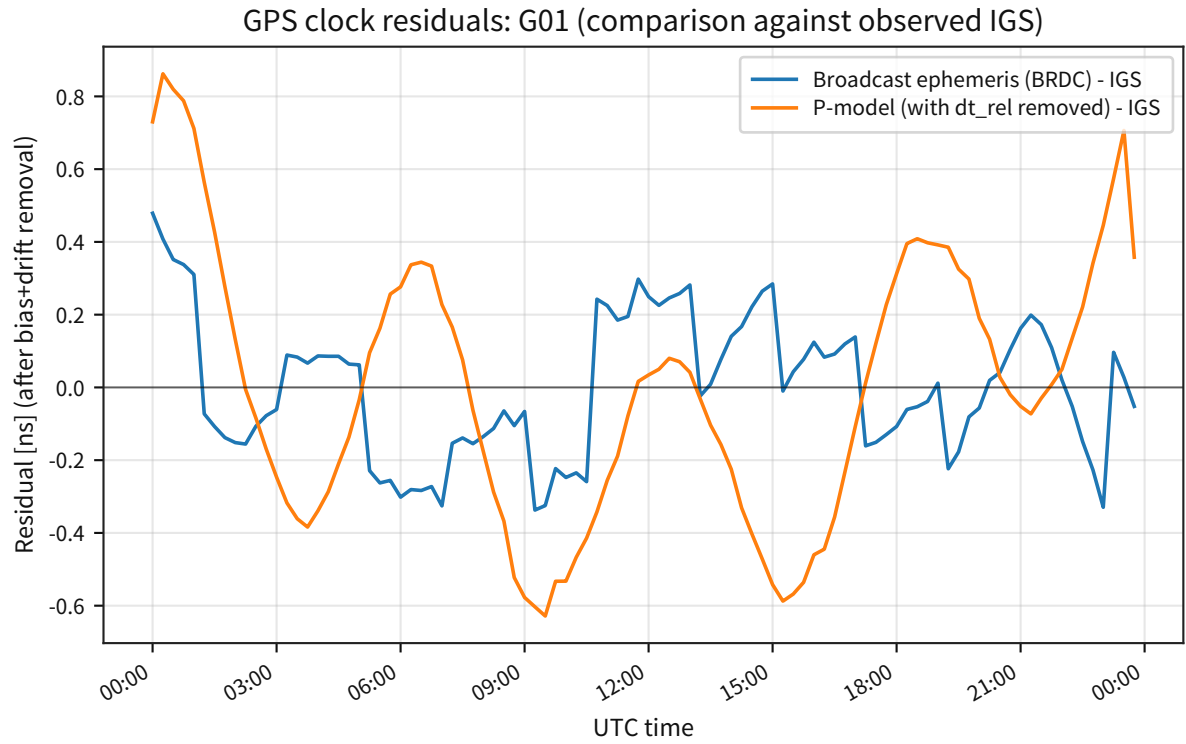


Figure 13: A representative single-satellite time-series residual example (trend and periodic shape).

Supplement: this provides an entry point for distinguishing whether the figure shape comes from a model difference or from a difference in the estimation product.

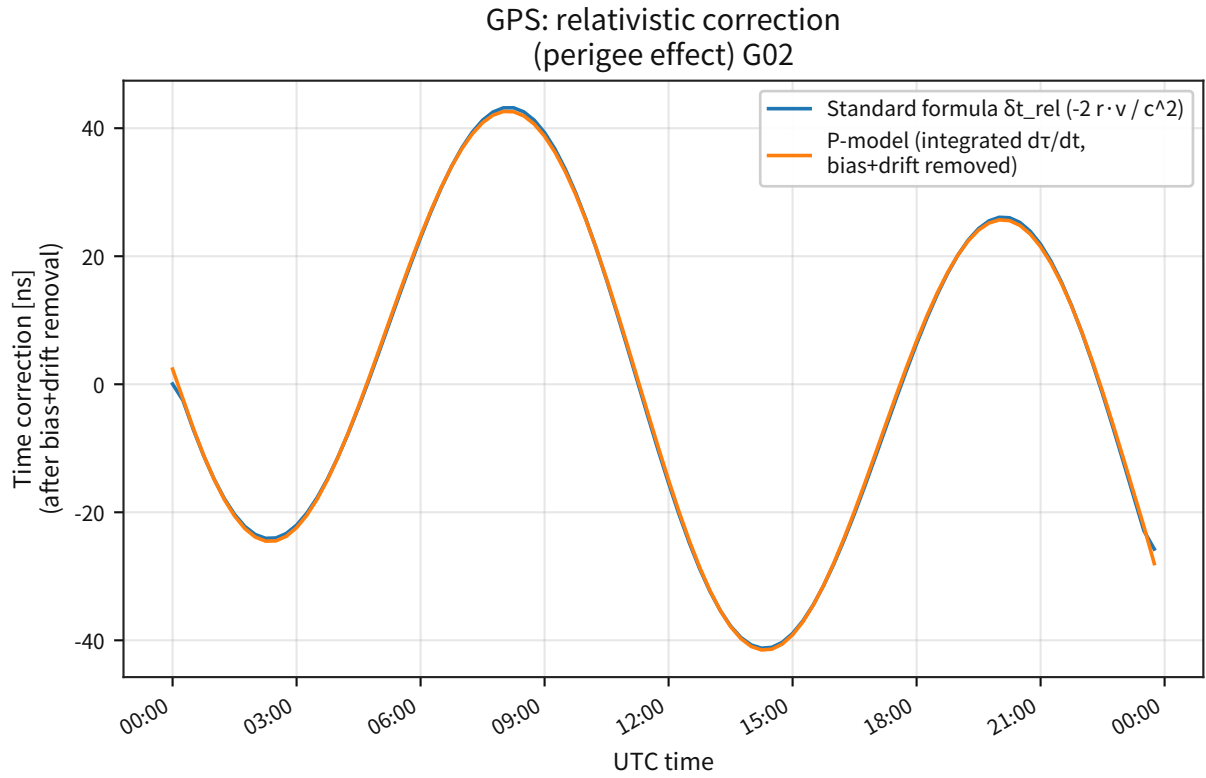


Figure 14: An example comparison of the periodic component in the standard formula and the P-model (checking correlation and phase offset).

Supplement: this shows the structure in which the periodic component agrees while the residual RMS difference remains.

Discussion: this section is not a replacement test. It fixes the comparison frame as a consistency check against an already estimated product (IGS Final).

Caution: GPS residuals contain multiple contributions, including clock models, orbit errors, ionosphere/troposphere effects, and antenna phase-center offsets. It is therefore not possible to conclude in a single metric that “the P-model is better than observation.” [3] The overview of BRDC–IGS residuals for all satellites is fixed in the verification materials (Part IV).

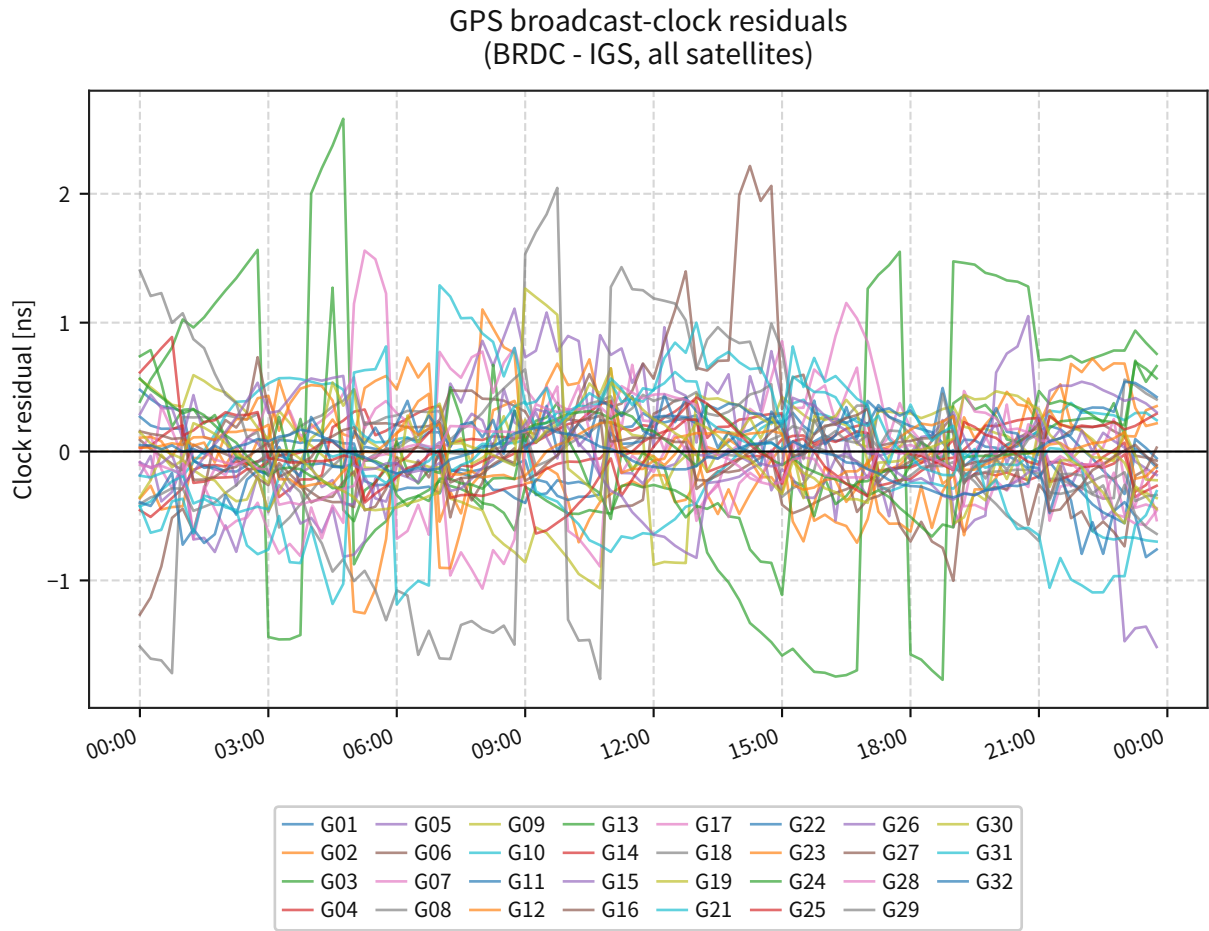


Figure 15: Overview of residuals for all satellites (BRDC–IGS, 31 spacecraft). Colored traces denote individual PRNs, and the English layout shows them with a compact external legend.

4.8 EHT (Black-Hole Shadow)

Purpose: to separate the mapping between the ring diameter (observation) and the shadow diameter (theory) into κ , and to fix the strong-field effective coefficient required by EHT and the dominant systematics under frozen β .

Input: use the EHT ring diameters (M87*: 2017/2018, Sgr A*: 2017). [25][26][27][28][29] For mass and distance, use the EHT estimate for M87* and the GRAVITY estimate for Sgr A*. [34]

Process: write $\theta_{\text{ring}} = \kappa \cdot \theta_{\text{sh}}$, compute θ_{sh} in the P-model with frozen β , and infer κ_{fit} from the observed θ_{ring} . In parallel, compute the envelope of κ_{fp} from the first-principles branch fixed in Part I 2.8.9 (light rays on $g_{\mu\nu}(P)$ plus radiative transfer with P_μ/J^μ), and judge its consistency with κ_{fit} under the same I/F.

Equation: ring/shadow mapping

$$\theta_{\text{ring}} = \kappa \theta_{\text{sh}} \quad (7)$$

Angular scale of the shadow diameter ($r_s \equiv 2GM/c^2$):

$$\theta_{\text{sh}} = C \frac{r_s}{D}$$

First-principles transport (invariant quantity):

$$\tilde{I}_\nu \equiv \frac{I_\nu}{\nu^3}, \quad k^\mu \nabla_\mu^{(g(P))} \tilde{I}_\nu = \tilde{J}_\nu - \tilde{A}_\nu \tilde{I}_\nu$$

$$\tilde{I}_{\nu,\text{obs}}(b) = \int \tilde{J}_\nu(\lambda, b) \exp[-\tau_\nu(\lambda, b)] d\lambda, \quad \tau_\nu = \int \tilde{A}_\nu d\lambda$$

Defining b_{peak} by $dI_{\nu,\text{obs}}/db = 0$ gives

$$\kappa \equiv \frac{b_{\text{peak}}}{b_{\text{sh}}(P)} \simeq 1 + \frac{1}{b_{\text{sh}}\Lambda_2} [\partial_b \ln(g_{\text{red}}^3 \tilde{J}_\nu) - \partial_b \tilde{\tau}_\nu]_{b=b_{\text{sh}}} \quad (8)$$

where g_{red} is the redshift factor and Λ_2 is the geometric curvature near the capture boundary.

Metric: κ_{fit} , κ_{fp} (first-principles envelope), $z = (\kappa_{\text{fit}} - \kappa_{\text{fp,center}})/\sigma_{\text{fp}}$, and the z-score under the assumption $\kappa = 1$.

Caution: the d_{sh} (shadow diameter) used in this section is a reference “shadow scale” inferred by EHT from the ring diameter, not an independent primary observable. It contains assumptions about radiative models, scattering, reconstruction, and definition differences. Accordingly, Fig. 17 separately shows the “observed ring” θ_{ring} and the “model shadow diameter” θ_{sh} with $\kappa = 1$, in order to distinguish whether the main source of the difference is κ (mapping/definition/model dependence) or θ_{unit} (M/D).

Result:

Target	$\theta_{\text{ring,obs}}$ (μas)	$\theta_{\text{sh,P}}$ (μas)	κ_{fit}	$\theta_{\text{sh,ref}}(d_{\text{sh}})$ (μas)	$\kappa_{\text{ref}}(d_{\text{sh}})$	Notes
M87*	42.0 $\pm$ 3.0	41.53 $\pm$ 4.89	1.011 $\pm$ 0.139	39.7 $\pm$ 4.7	1.058 $\pm$ 0.146	frozen β
Sgr A*	51.8 $\pm$ 2.3	54.52 $\pm$ 0.25	0.950 $\pm$ 0.042	48.7 $\pm$ 7.0	1.064 $\pm$ 0.160	frozen β

Consistency check for the first-principles transport branch (κ_{fp}):

Target	κ_{fp} center	σ_{fp} (1σ)	κ_{fp} 1σ interval	$z(\kappa_{\text{fit}} - \kappa_{\text{fp}})$	Verdict
M87*	1.000	0.076	[0.924, 1.076]	+0.150	Pass
Sgr A*	1.000	0.083	[0.917, 1.083]	−0.602	Pass

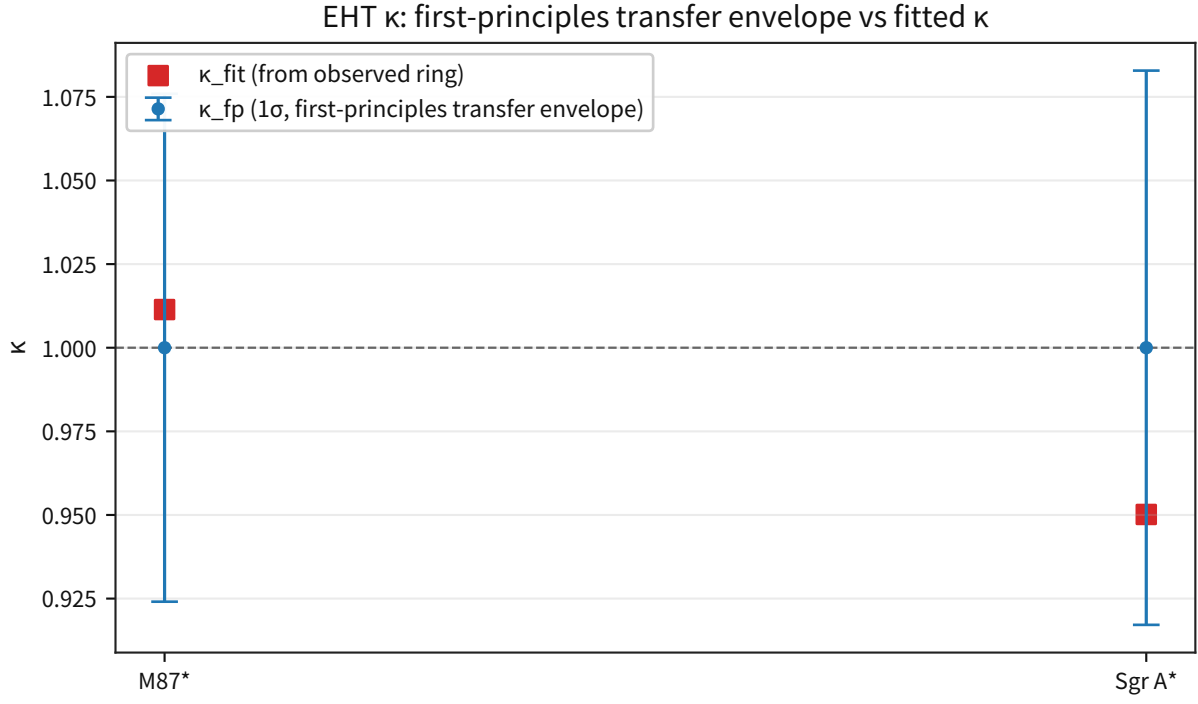


Figure 16: A comparison on the same axis between κ_{fp} (1σ envelope) obtained from the transport equation on $g(P)$ and κ_{fit} inferred from the observed ring.

Supplement: the central value of κ is fixed to the geometric limit $\kappa = 1$, and the contributions from radiative gradients, absorption, and scattering are collected into σ_{fp} .

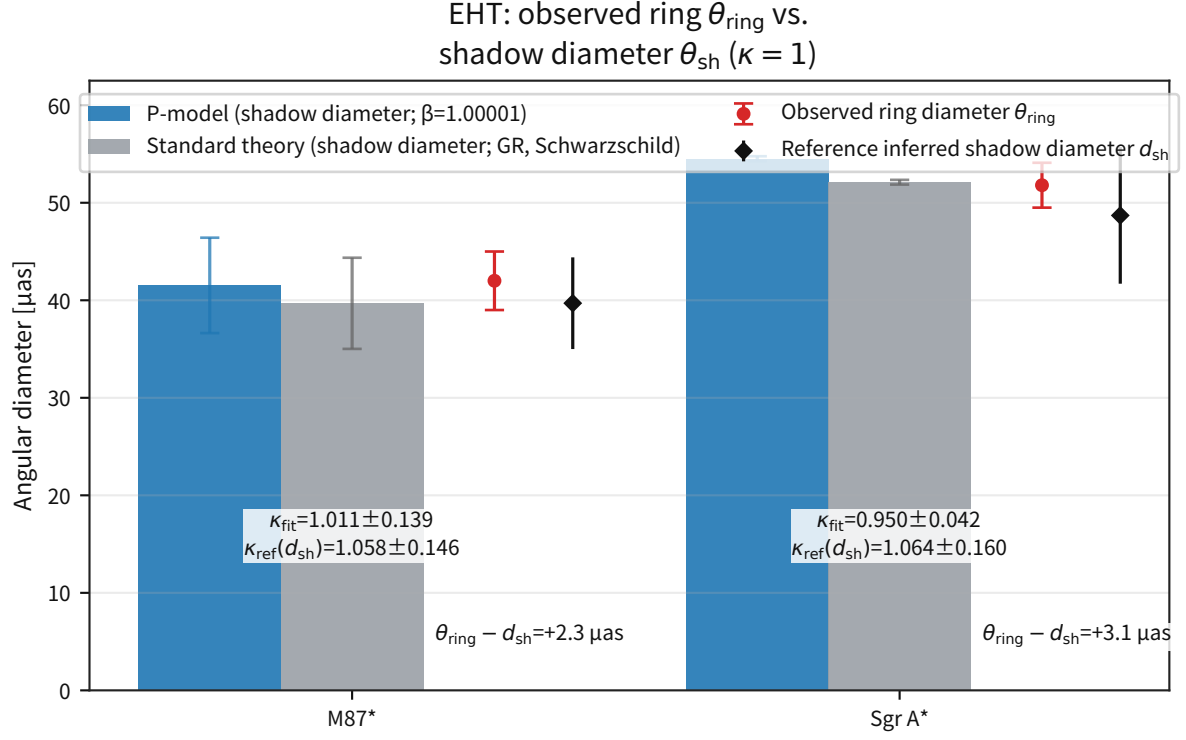


Figure 17: Direct comparison between the ring diameter (observation) and the shadow diameter (P-model, $\kappa = 1$).

Supplement: by “taking κ outside,” the main source of the difference can be separated into κ (radiation/scattering/definition) or θ_{unit} (M/D). The coefficient difference of +4.63% in this figure is the baseline value without strong-field higher-order corrections. For the final value including higher-order corrections (4.81%), see Sections 4.17 and 5.1.

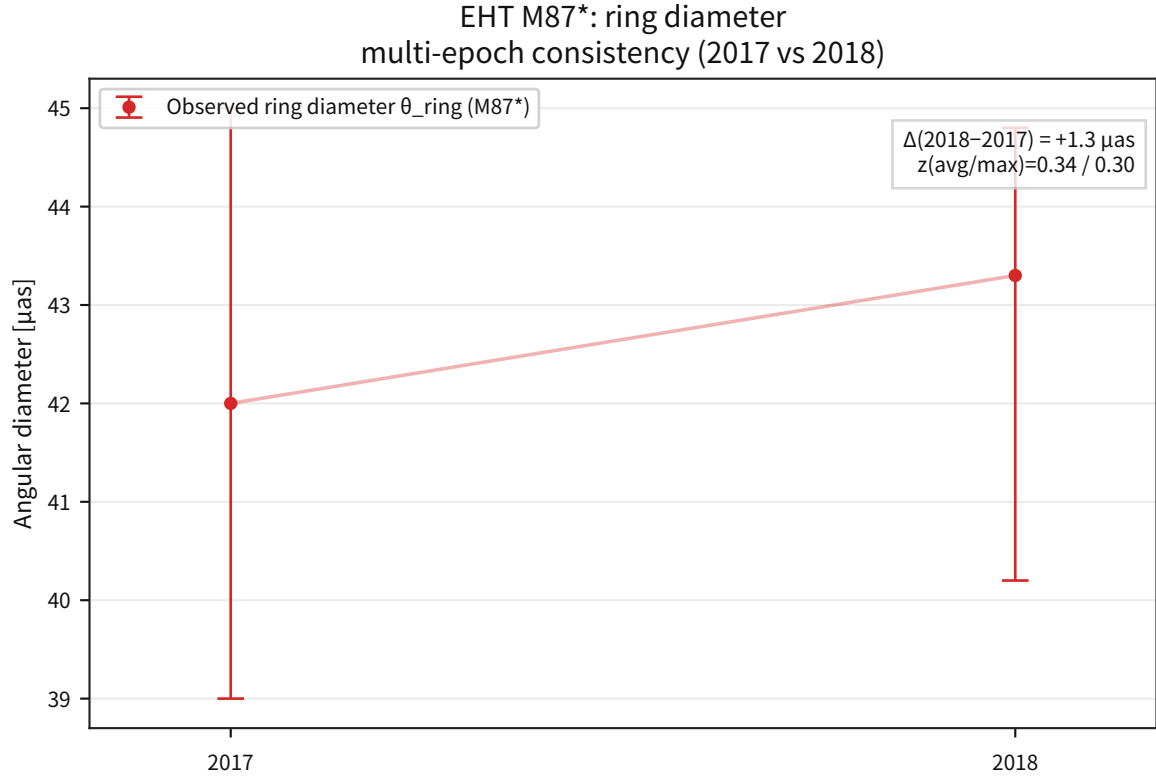


Figure 18: Independent 2017/2018 epochs for M87* shown together to confirm that the measured ring diameter is not fluctuating strongly.

Supplement: this is used to check the reproducibility of the observational input θ_{ring} before the discussion of κ (ring/shadow mapping) begins.

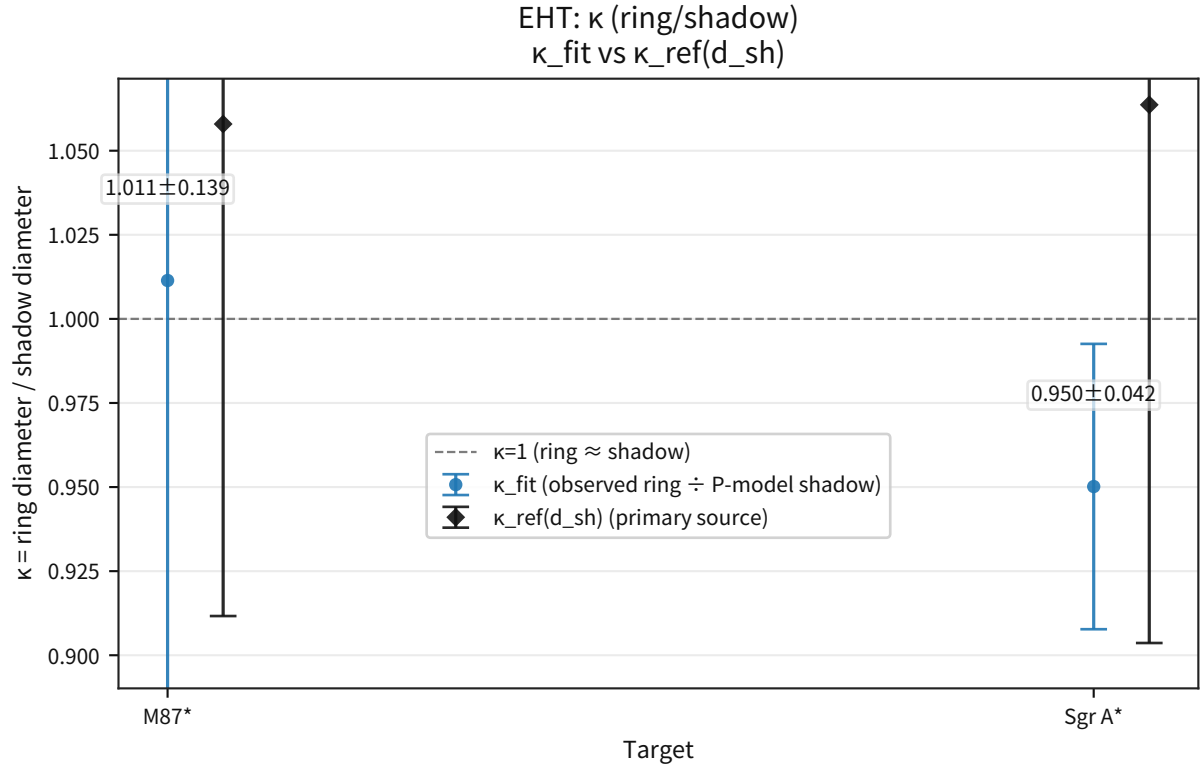


Figure 19: The inferred κ_{fit} from the observed θ_{ring} .

Supplement: the uncertainty in κ connects directly to the required precision for the differential prediction in Section 5.1.

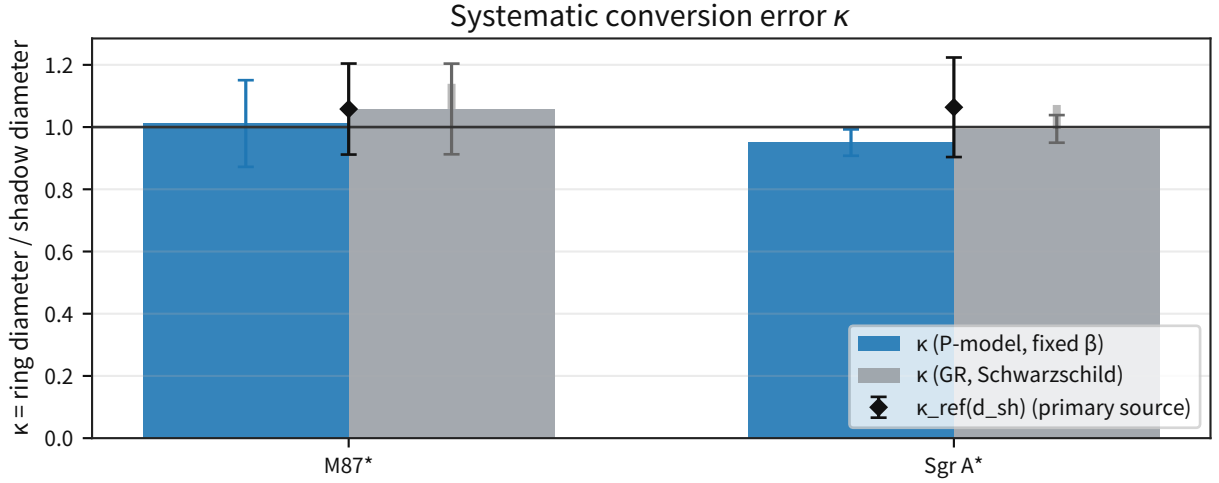
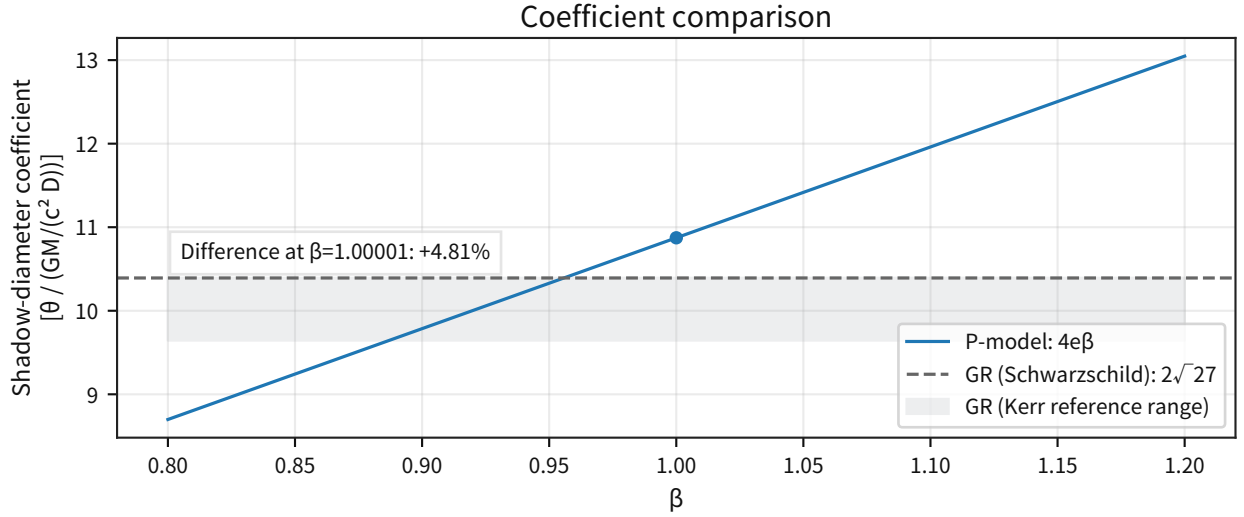


Figure 20: Shows κ and the coefficient range on the reference side (spin, inclination, definition differences, etc.) on the same axis, visualizing the dominant systematics.

Supplement: in this paper, κ is treated as the principal systematic knob, and the strong-field differential prediction depends on the precision with which κ is constrained.

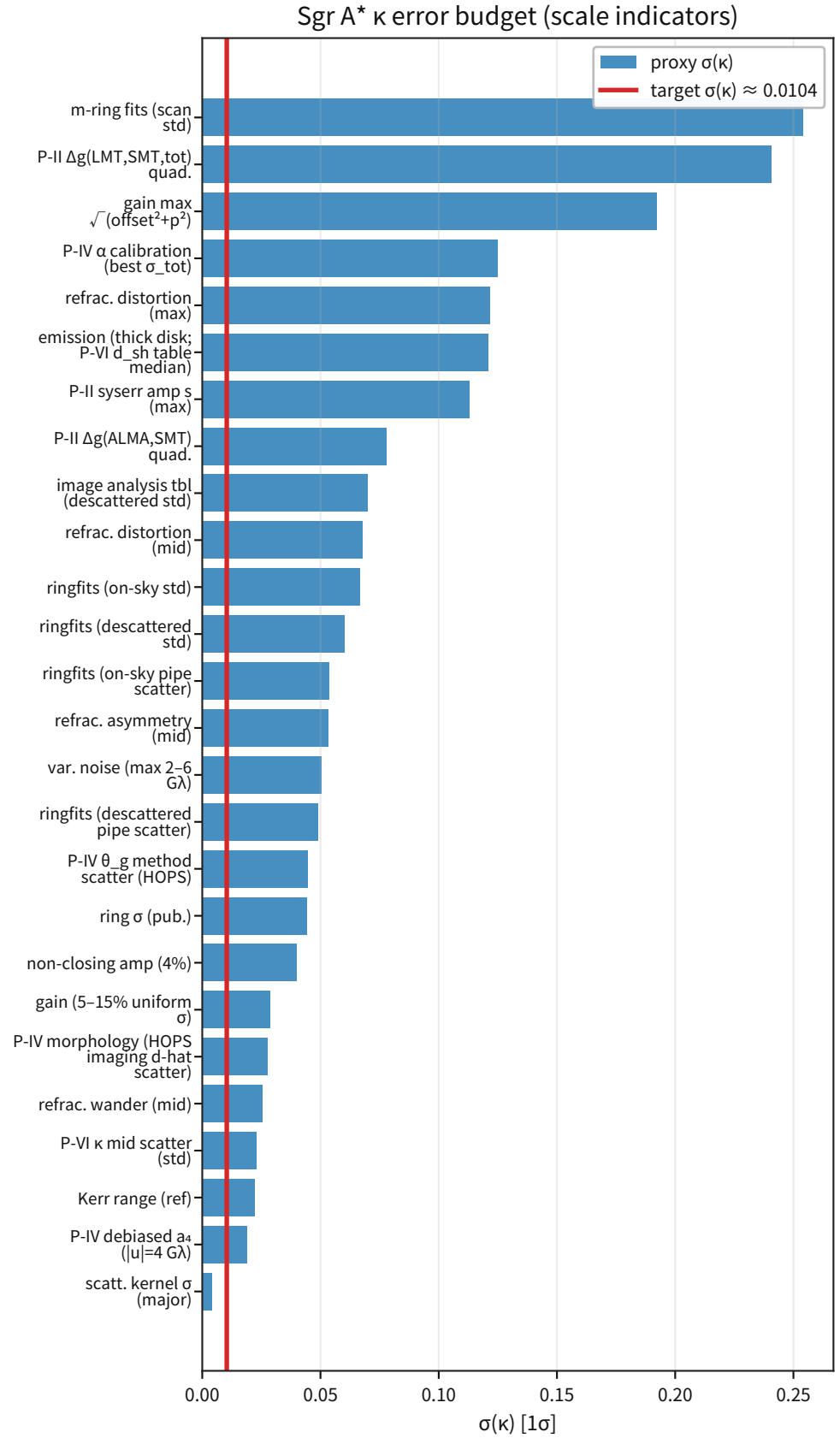


Figure 21: The error budget of κ shown on a common axis with multiple systematic proxies (calibration, scattering, imaging, model dependence, etc.).

Supplement: in particular, the possible fluctuation of the ring-to-shadow mapping due to radiative-model dependence (such as thick disks; [31]) is fixed as a conservative σ_κ by treating the spread of the inferred shadow-diameter table in Paper VI as a model-dependent floor.

Discussion: for M87*, the relative error in $\theta_{\text{unit}} (M/D)$ is the rate-limiting term, whereas for Sgr A* the dominant systematics are ring σ and κ (quantified in Section 5.1).

Caution: EHT has large systematics from radiative models, scattering, image reconstruction, and spin/inclination dependence. This section is therefore not a strong proof but a fixed entry point for the differential prediction, namely which precision is required. [40][39][41][31][32][33] Additional diagnostic figures (shape indicators, scattering budget, definition differences, z-scores, etc.) are used together to separate the dominant systematics. The observational reconnection conditions for Sgr A* (Paper V) are organized in the same way, clarifying how assumptions such as digitization and independence affect the p-value. [30][38][37] For the strong-field tests with S2/GRAVITY (f, f_{SP}), the difference from this model falls to 2PN (weak-field higher order), so differential detection is difficult at current precision. [35][36] The coefficients on the reference side (Schwarzschild/Kerr) follow the expression of the published observables, while the P-model side is compared with the mapping and frozen conditions kept fixed (the reference framework is isolated in “Caution”).

4.9 Gravitational Redshift (GP-A / Galileo)

Purpose: to compare weak-field, static-clock gravitational redshift with the published deviation parameter ϵ^* as a reference check outside the primary verdict.

Input: the published deviation parameter ϵ from the GP-A (1976) and Galileo 5/6 (GREAT) gravitational-redshift tests. [15][16]

Process: observations are written as $z \equiv \Delta f/f$, namely $z_{\text{obs}} = (1 + \epsilon)\Delta U/c^2$. In the P-model, for weak fields and static clocks, $d\tau/dt = P_\infty/P \approx 1 + \phi/c^2$, so the first-order prediction is $\epsilon = 0$.

Equation: observational definition of the deviation parameter

$$z_{\text{obs}} = (1 + \epsilon) \frac{\Delta U}{c^2}$$

Weak-field, static-clock P-model relation:

$$\frac{d\tau}{dt} = \frac{P_\infty}{P} \approx 1 + \frac{\phi}{c^2}$$

Metric: ϵ^* and σ for each experiment, together with the reference-comparison z-score $z = \epsilon^*/\sigma$.

Result:

Experiment	ϵ	z ($=\epsilon/\sigma$)	Notes
GP-A	$0 \pm 7.0 \times 10^{-5}$	0.00	for convenience, the paper-level expression for σ is treated as an approximate 1σ scale
Galileo 5/6	$(0.19 \pm 2.48) \times 10^{-5}$	+0.08	GREAT



Figure 22: Places the published ϵ values and their error bars side by side, together with the $\epsilon = 0$ reference line.

Supplement: the verdict in this section is only a reference comparison against the published ϵ^* and is not used as a primary accept/reject gate.

Discussion: in weak fields near the ground and near Earth, differential detection through ϵ is less central than differential predictions in stronger-field or separate probes such as cosmology and quantum phenomena.

Caution: the ϵ used here is the deviation parameter for gravitational-redshift tests and is distinct from the ϵ used in BAO/AP warping or the ϵ_0 used in DDR. Because GP-A is quoted in the literature as being at the “ 70×10^{-6} level,” this paper uses $\sigma = 70 \times 10^{-6}$ as an approximate 1σ scale for convenience (for strict statistical interpretation, see the primary source). [15]

4.10 Double Pulsars (Orbital Decay: Radiation Constraint)

Purpose: to verify from public primary values that far-zone weak-field radiation is quadrupole-dominated, or at least not dipole-dominated, by recalculating $\dot{P}_b$.

Input: for each system (NS-NS / NS-WD), extract orbital elements (P_b , e), masses (m_1 , m_2), and the intrinsic $\dot{P}_b^{(\text{int})}$ after external corrections (kinematic corrections, Galactic acceleration, etc.) from the primary sources, then derive R by recalculating $\dot{P}_b$ in the P-model weak-field quadrupole branch. [44][45][46][47]

Process: use the defense condition fixed in Part I Section 2.6, namely the universality of $g_{P,a}/m_a = \kappa$ for all matter. The dipole-forbidden condition is fixed by

$$D_P = \kappa M X_{\text{cm}}, \quad X_{\text{cm}} = 0 \Rightarrow \ddot{D}_P = 0 \quad (9)$$

so dipole radiation cancels exactly. Accordingly, in the far weak-field regime, the **quadrupole term is dominant**, so $\dot{P}_b$ is recalculated using the Peters–Mathews quadrupole formula, and the P-model (weak-field quadrupole) predicts $R = 1$. [42][43]

Equation:

$$\dot{P}_b^{\text{quad}} = -\frac{192\pi}{5} T_\odot^{5/3} \left(\frac{2\pi}{P_b} \right)^{5/3} \frac{m_1 m_2}{(m_1 + m_2)^{1/3}} \frac{1 + \frac{73}{24}e^2 + \frac{37}{96}e^4}{(1 - e^2)^{7/2}} \quad (10)$$

$T_\odot \equiv GM_\odot/c^3$, with masses expressed in $M_\odot$. Observed ratio:

$$R \equiv \frac{\dot{P}_{b,\text{int}}}{\dot{P}_b^{\text{quad}}} \quad (11)$$

The implementation is fixed so that the same R is reproducible from the same inputs.

Metric: $R = 1$ means consistency with quadrupole radiation in the weak-field, far-zone P-model. If **dipole radiation** or other additional energy loss were present, R would deviate systematically from 1 (and the observationally allowed deviation would provide an upper bound on such extra radiation).

Result:

System	Type	R (=Pdot_int / Pdot_quad)	1 σ (rough)	Upper bound on extra radiation (3 σ ; rough fraction)
PSR B1913+16	NS-NS	0.998292	0.0016	<0.65%
PSR J0737-3039A/B	NS-NS	0.999953	1.3×10^{-4} (95%)	<0.025%
PSR J1738+0333	NS-WD	0.943158	0.117	<41%
PSR J0348+0432	NS-WD	1.059881	0.174	<58%

Across the range above, all systems remain consistent with $R = 1$ (although the error widths differ greatly among systems).

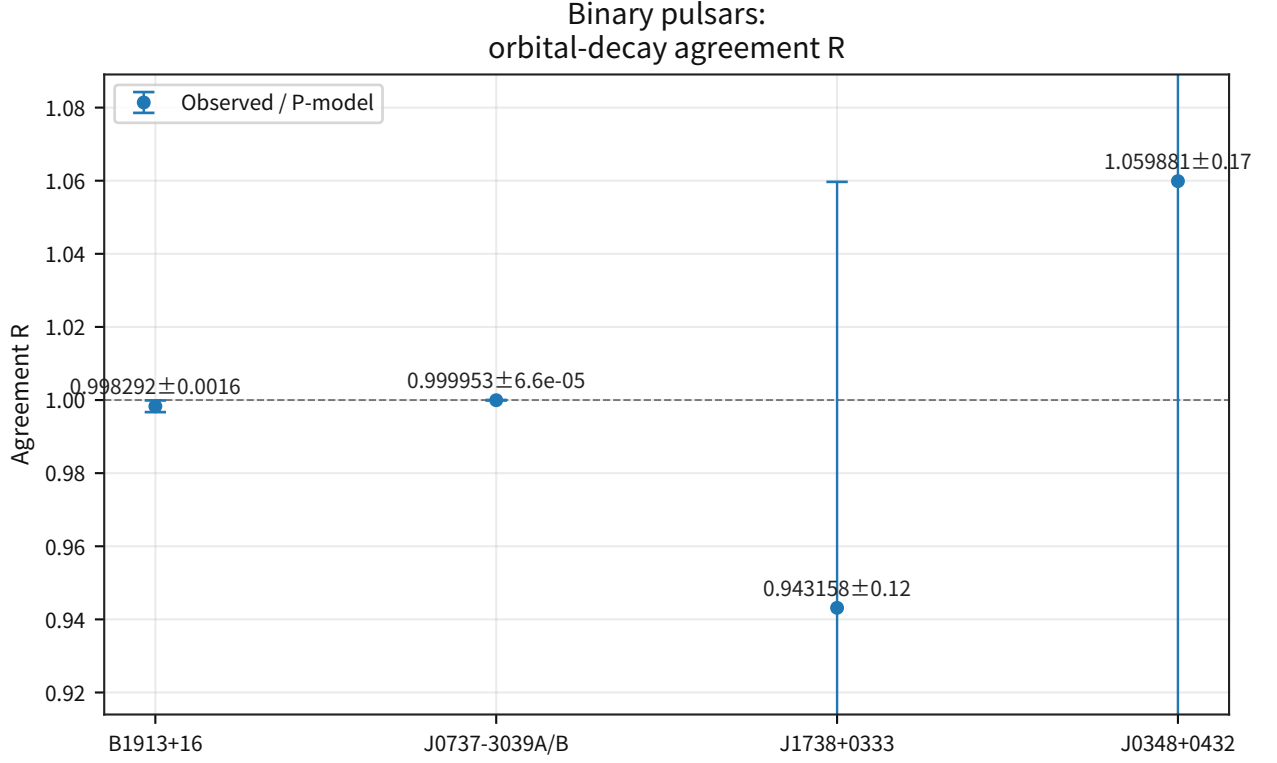


Figure 23: Overview of R (observation / quadrupole) for each binary, comparing deviations from $R = 1$ and their uncertainties.

Supplement: this section fixes “not inconsistent with $R = 1$ ” as the necessary condition, while listing the upper bound on extra radiation as a rough fraction.

Discussion: the NS-NS systems (B1913+16, J0737) strongly constrain R , so radiation beyond the quadrupole term is tightly limited, whereas NS-WD systems still have large uncertainties.

Caution: this section verifies only the necessary condition that the quadrupole term dominates in the weak-field far zone. It does not claim that the strong-field, near-merger regime or the full waveform has been independently rederived from the P-model (see the role defined in Part I Section 2.6). NS-WD binaries are sensitive to dipole radiation, but strict exclusion requires not only $|R - 1|$ but also mapping into scalar charge (sensitivity) and the theory parameter space.

4.11 Gravitational Waves (GW150914: Consistency of the Chirp Phase)

Purpose: using minimal statistics extractable from public strain data, to fix reproducible checks of (i) consistency with the quadrupole chirp law (phase), and (ii) ringdown/QNM, the area theorem, and IMR consistency without claiming to replace matched filtering.

Input:

- GW150914: GWOSC public strain (H1/L1, 32 s, 4 kHz). [49][48]
- GW250114: GWOSC Event API (O4_Discovery_Papers) plus the public data release referenced by GWOSC (Zenodo). [49]

Process:

- GW150914: bandpass $\rightarrow$ Hilbert phase $\rightarrow$ extract instantaneous frequency $f(t)$ and fit it to $t = t_c - A f^{-8/3}$ (Newtonian approximation).
- GW250114: use the scan outputs from the public data release (ringdown start time / inspiral truncation), make systematics such as t_{ref} and truncation explicit, and tabulate QNM, the area theorem, and IMR consistency.

Equation:

$$\frac{df}{dt} = \frac{96}{5} \pi^{8/3} \left(\frac{GM_c}{c^3} \right)^{5/3} f^{11/3} \quad (12)$$

Quadrupole chirp law (integrated form):

$$t_c - t = \frac{5}{256} \left(\frac{GM_c}{c^3} \right)^{-5/3} (\pi f)^{-8/3} \equiv A f^{-8/3}$$

Ringdown damping time (from the damping rate g): $\tau = 1/g$ Area-theorem significance (Gaussian proxy): $\sigma = \frac{\mu_r - \mu_i}{\sqrt{\sigma_r^2 + \sigma_i^2}}$

Metric:

- GW150914: fitted R^2 and the inferred M_c from A (simple estimate).
- GW250114: QNM (f_{220} , τ_{220}), area-theorem significance σ , and IMR consistency (proxies z_f , z_τ , p).

Result:

GW150914 (chirp phase; Hilbert, fixed window)

Detector	R^2	M_c ($M_\odot$)	Notes
H1	0.473	27.7	sanity check

Continued on next page

Detector	R^2	M_c ($M_\odot$)	Notes
L1	0.618	4.38	chirp track unstable due to phase wrapping and contamination from the low-S/N interval; not used as a physical constraint

The M_c value for L1 is not adopted as a physical estimate because the combination of post-bandpass phase wrapping and a short effective track length causes the fit $t = t_c - Af^{-8/3}$ to fall into a local solution.

GW250114 (ringdown/QNM, area theorem, IMR; scan treated as a systematic)

Metric	Value	Notes
f_{220} (Hz)	247 (p16–p84: 243–251)	ringdown start-time scan
τ_{220} (ms)	4.52 (p16–p84: 4.11–4.95)	same
Area-theorem σ (combined)	4.45	
Area-theorem σ (pyRing only)	4.70	
IMR z_f	−0.14	proxy
IMR z_τ	+0.54	proxy
IMR p	0.74	proxy

The area theorem reaches $\sigma \geq 5$ when inspiral truncation is moved earlier (first at $-10M$), and this choice is frozen as a systematic knob.

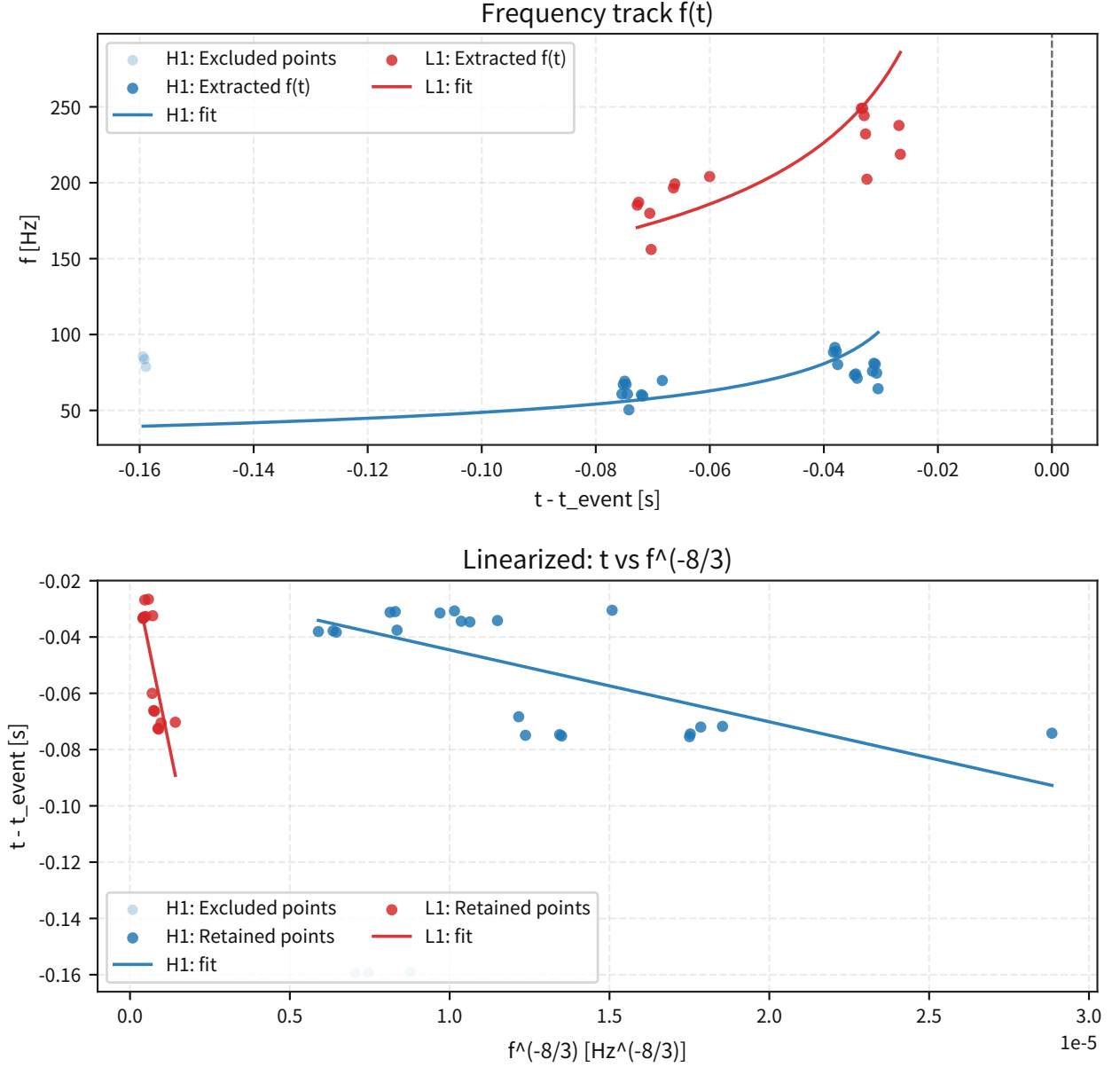


Figure 24: Visualization of the extracted $f(t)$ and its fit to the linear relation $t = t_c - Af^{-8/3}$.

Supplement: this figure is not intended as a replacement for matched filtering, but as a minimal consistency check from the public strain data. The lower in-figure annotations that previously overlapped the plot (detector-by-detector M_c , R^2 , number of adopted points, and σ_t) were moved into the main text to prioritize readability.

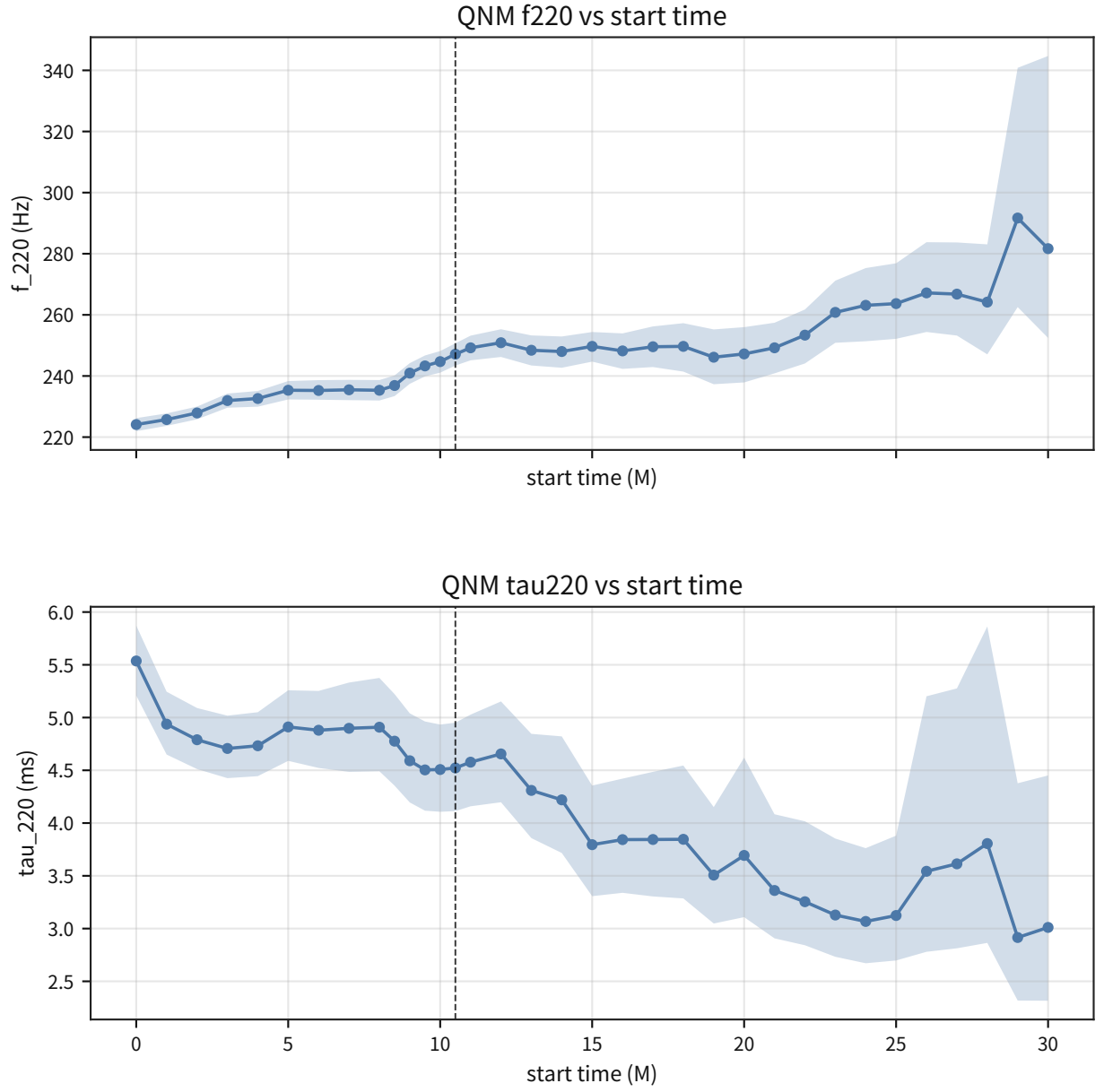


Figure 25: Estimated QNM(220) f and τ as functions of the ringdown start-time scan.

Supplement: this makes explicit that the choice of start time is a systematic and compares results at fixed t_{ref} .

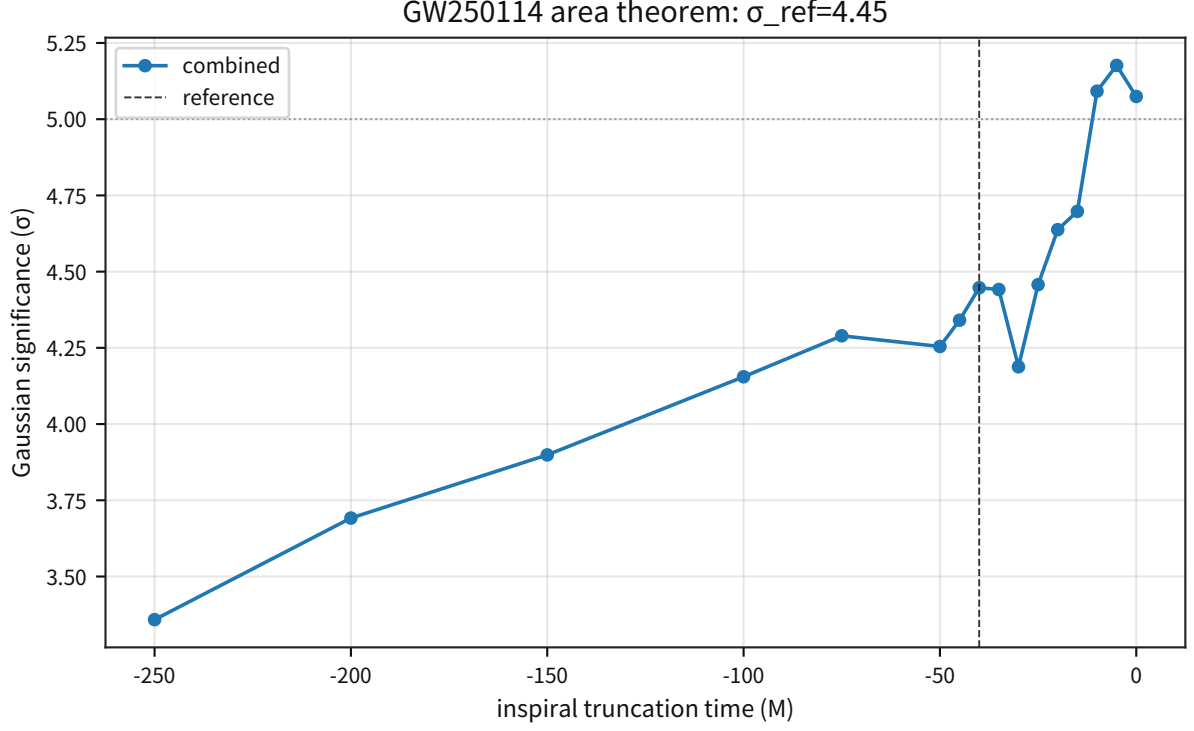


Figure 26: Evolution of the area-increase significance σ under the inspiral truncation scan.

Supplement: the truncation at which 5σ is reached is fixed as a systematic knob in the ledger.

Discussion: the low R^2 values (0.4–0.6) obtained for GW150914 reflect the noise amplification limit caused by extracting instantaneous phase directly from public strain via a Hilbert transform without matched filtering. Accordingly, the chirp metric in this section does not claim precise parameter estimation such as M_c , but is limited to a first-isomorphism sanity check confirming that the macroscopic form of the quadrupole chirp law is not grossly broken. For ringdown, the area theorem, and IMR, the choice of start time and truncation is the dominant systematic, so it is essential to freeze the result together with the systematics ledger.

Caution: this section does not replace the standard LIGO/Virgo matched-filtering workflow based on templates; it is a sanity check. A multi-event summary (R^2 and match) is also tabulated to provide an overview of consistency across several events (match may be missing depending on the window definition).

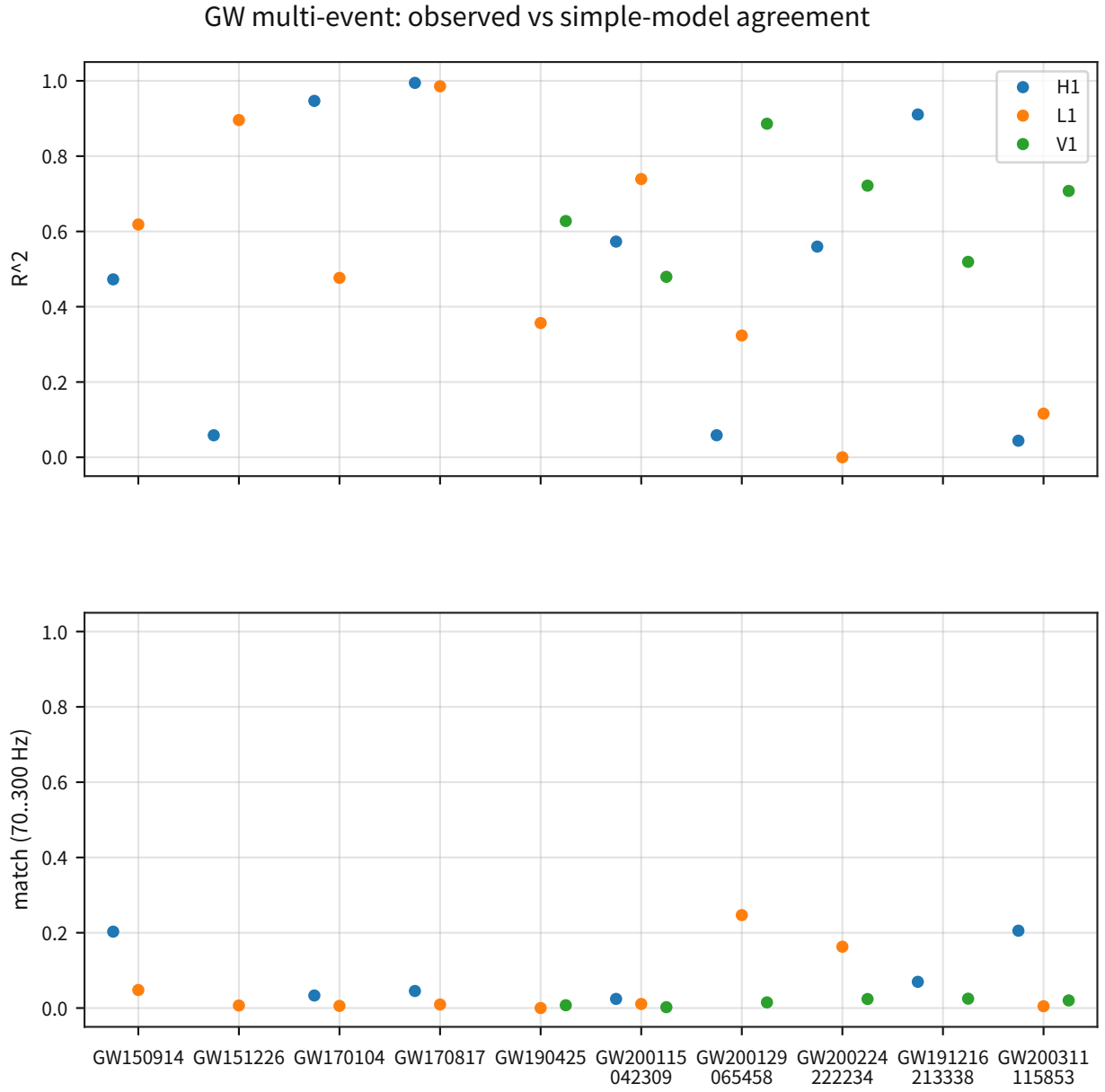


Figure 27: Summary figure over multiple events (R^2 and match).

Supplement: the possibility that match may be missing because of the analysis-window definition is treated as a procedure-dependent systematic.

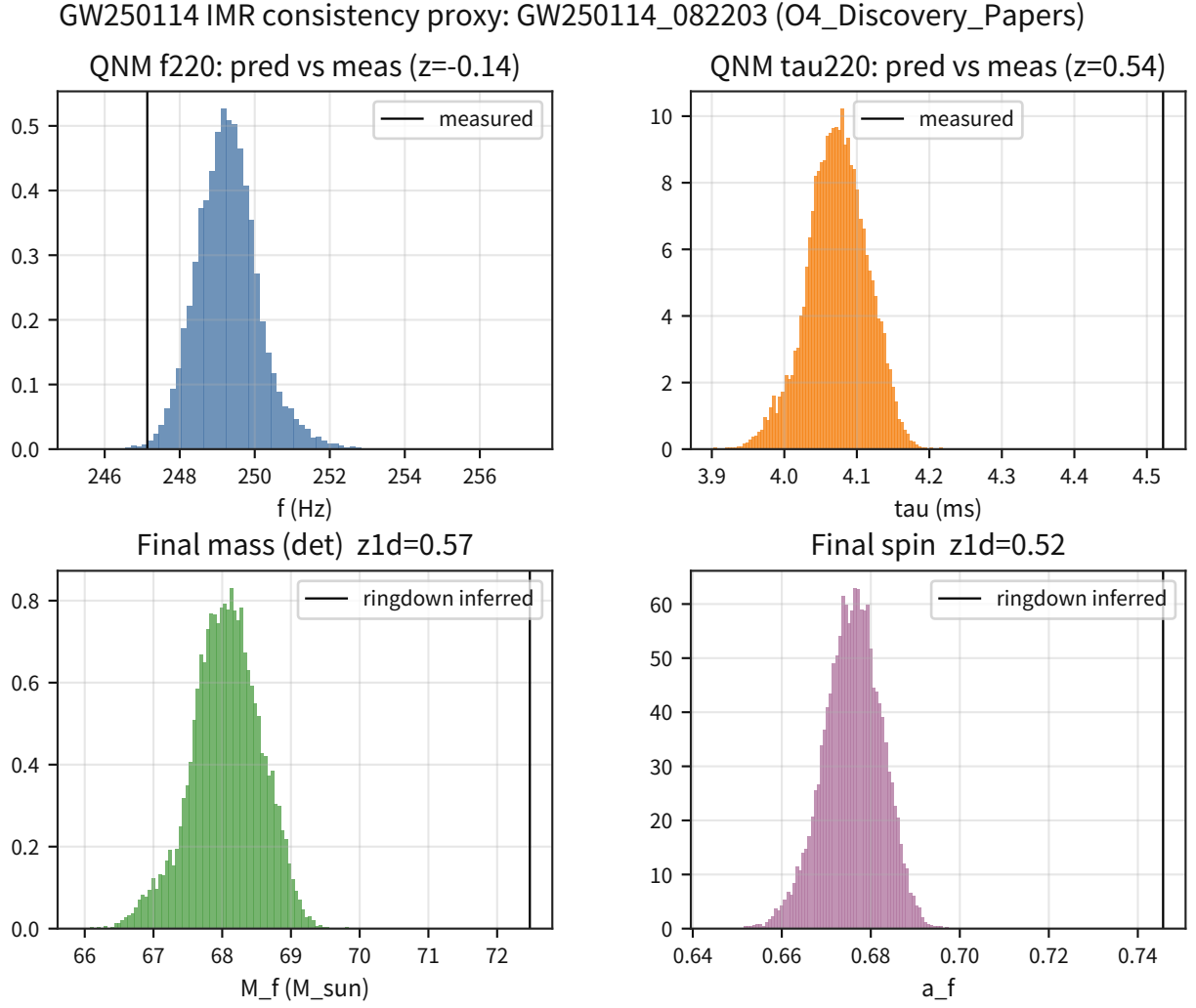


Figure 28: Sensitivity of the GW250114 IMR consistency proxy.

Supplement: because it depends on the public posterior and the reference-frame fit formula, the description of the reference framework is isolated under “Caution.”

4.11.1 Gravitational-Wave Polarization Modes (Audit of the P_μ Vector Wave)

Purpose: in addition to chirp-phase consistency, determine by an independent gate whether the P_μ extension introduced in Part I Section 2.8 is compatible with observational constraints on gravitational-wave polarization.

Input: use the H1/L1 amplitude-ratio audit for GW150914 and the polarization-degeneracy test in the H1/L1/V1 three-detector network.

Process: map the transverse spatial components of P_μ into a vector-polarization basis and compare the amplitude-ratio interval projected onto detector responses with the observed interval. The verdict is made in two stages:

1. H1/L1 hard gate (correlation, causal delay, amplitude-ratio overlap, tensor isomorphism)

2. H1/L1/V1 network gate (reduction/exclusion of scalar degeneracy, tensor-response consistency)

Equation: write the vector-polarization basis as

$$e_x^V = \hat{n} \otimes \hat{p} + \hat{p} \otimes \hat{n}, \quad e_y^V = \hat{n} \otimes \hat{q} + \hat{q} \otimes \hat{n}$$

and the detector response as

$$h_I(t) = D_I : (h_x e_x^V + h_y e_y^V)$$

This evaluates the interval in the H1/L1 observed amplitude ratio produced by transverse components in the plane orthogonal to the propagation direction.

Metric: gate_corr_pass, gate_lag_causal_pass, gate_vector_overlap_pass, gate_tensor_equiv_pass, and on the network side scalar_overlap_proxy (max/median), scalar_exclusion_pass, and tensor_consistency_pass. Here scalar_exclusion_pass indicates that scalar polarization has been excluded, and tensor_consistency_pass indicates that tensor polarization is consistent with the observations.

Result: for GW150914 in H1/L1, the hard gate passes, giving $|r|_{\max} = 0.749$, best lag -5.127 ms, and amplitude ratio median = 1.469. In the network analysis, however, n_usable_events=2 and scalar_overlap_proxy_max=0.500 with median = 0.417, so scalar_reduction_pass=true but scalar_exclusion_pass=false and tensor_consistency_pass=false. The overall verdict for this section is therefore fixed to Watch (scalar degeneracy reduced, but full exclusion not yet reached).

Discussion: with this revision, the polarization section moves from “declaring the scalar limit” to “auditing whether a P_μ vector wave can produce a signal isomorphic to the observation.” The current remaining task is the simultaneous achievement of scalar exclusion and tensor consistency in the three-detector network.

Caution: Watch here does not mean rejection, but failure so far to satisfy the adoption condition. Adoption requires not only a hard-gate Pass, but also scalar_exclusion_pass=true and tensor_consistency_pass=true on the network side.

4.11.2 Consistency Test of the Amplitude Ratio Between LIGO H1/L1 Detectors (GW150914)

Purpose: when the same event (GW150914) is observed by detectors with different geometry (H1/L1), to audit quantitatively whether the amplitude ratio remains stable within the time window after correcting for the arrival-time difference.

Input: GWOSC public strain (H1/L1, 32 s, 4 kHz).

Process: after a 30–350 Hz bandpass, perform a lag scan of ± 10 ms over the chirp window t in $[-0.12, -0.01]$ s, and align H1/L1 at the delay that maximizes the normalized correlation. After alignment, evaluate the envelope ratio $|H1|/|L1|$ as a time series and audit its median and scatter.

Equation: correlation at delay τ

$$r(\tau) = \text{corr}[h_{H1}(t + \tau), h_{L1}(t)]$$

envelope amplitude ratio

$$\mathcal{R}(t) = \frac{|h_{H1}(t)|}{|h_{L1}(t)|}$$

Metric: $|r|_{\max}$, best lag, $\text{median}(\mathcal{R})$, $p16/p84$, IQR/median , and in-window drift rate. The audit gates are $|r|_{\max} \geq 0.6$, $IQR/\text{median} \leq 0.35$, and drift rate ≤ 0.5 .

Result: for GW150914, the best lag is -5.13 ms, with $|r|_{\max} = 0.749$. The amplitude ratio is median = 1.469 ($p16 = 1.192$, $p84 = 1.824$), with $IQR/\text{median} = 0.314$ and in-window drift rate 0.257, so the consistency verdict is Pass.

Discussion: that the absolute amplitude ratio differs from 1 is itself permissible because of detector geometry (antenna-response) differences. What matters in this audit is that, after alignment at the best lag, the amplitude ratio does not diverge chaotically and remains within the consistency range for a single event.

Caution: this audit is a minimal consistency check between detectors and is not a substitute for polarization identification or full waveform estimation. Adoption/rejection is determined by combining independent gates for chirp phase, polarization, and the H1/L1 amplitude ratio.

4.11.3 Three-Detector Network: `corr_use_min` Stepwise Audit (Fixing Event Selection)

Purpose: to make explicit how the threshold setting `corr_use_min` affects the usable-event set in the three-detector (H1/L1/V1) polarization audit, and to determine the operational fixed value.

Input: public event sets to which the same I/F can be applied in a three-detector configuration (GW200115_042309, GW200129_065458, GW200311_115853), and stage thresholds `corr_use_min` in $\{0.05, 0.08, 0.10, 0.12, 0.15, 0.20, 0.25, 0.30\}$.

Process: sweep the threshold in stages and recalculate, at each stage, the number of usable events (those satisfying both geometric and ratio constraints). At the same time, fix the verdict (Reject/Watch/Pass) for the usable events and the evolution of the selected set.

Metric: `n_usable_events`, the usable-event set, and `overall_status` at each stage.

Result: for `corr_use_min=0.05`, one event is usable (GW200129_065458), with verdict `reject_tensor_response`. For `corr_use_min>=0.08`, the usable count drops to 0 and the audit moves to inconclusive. In addition, a local rescan (`corr_use_min=0.05..0.07`, `response_floor=0.05/0.08`, `min_ring=6/8`, `relax = 2/4`, 24 trials) also yields 0 Pass/Watch events, and the minimum `scalar_overlap_proxy` under the condition `n_usable_events>=2` remains fixed at 0.500. `scalar_overlap_proxy=0.0` appears only for `n_usable_events=1`, so with current public data, “exclusion (`proxy=0`) and multi-event coverage (`usable>=2`)” cannot be achieved simultaneously. The operational threshold for the present public data is therefore fixed to `corr_use_min=0.05`.

As a further coverage extension, the same local search was applied to event subsets anchored on GW200129_065458 (2–4 events, 7 subsets total). The target condition (`n_usable_events>=2` and `scalar_overlap_proxy<0.5`) was achieved in 0/7 subsets, and the best subset was fixed at GW200129_065458, GW200311_115853 with `n_usable_events=2`, `scalar_overlap_proxy=0.5`. `scalar_overlap_proxy=0` or 0.333 appears only for subsets with `n_usable_events=1`, and no setting has yet been found that keeps coverage (`usable >= 2`) while going below the Watch boundary (0.5). In addition, updates to the public three-detector event input are now recorded in the verification ledger as rerun candidates for the coverage audit, and are rerun explicitly with the same I/F only when needed. The operational state (input-hash monitoring, rerun counters, etc.) is managed in the Part IV verification ledger.

Discussion: with current data, tightening the threshold immediately loses statistical power, so an overly conservative threshold discards the information needed for polarization discrimination. At the same time, the fixed value 0.05 automatically separates events with weak inter-detector correlation into the inconclusive side.

Caution: this fixed value is an operational value for the current public dataset, not a universal constant. If additional high-S/N or strongly correlated three-detector public events appear, the rerun candidate is recorded in the verification ledger and the same stepwise audit is rerun explicitly with the same I/F only when needed.

4.12 Rotation (Frame Dragging: GP-B / LAGEOS)

Purpose: for the minimal extension of the rotation term, perform a reference comparison against published weak-field frame-dragging summary values (outside the primary verdict).

Input: the frame-dragging drift rate from GP-B and the agreement ratio for the Lense–Thirring effect in LAGEOS/LAGEOS II as stated in the published summaries. [17][18]

Process: under the covariant extension fixed in Part I Section 2.8,

$$P_\mu = (P_t, P_1, P_2, P_3), \quad \mathcal{L}_{\text{int}} = g_P P_\mu J^\mu, \quad \mathcal{L}_{\text{total}}^{\text{vec}} = \mathcal{L}_{\text{total}} + \mathcal{L}_{\text{int}} + \mathcal{L}_{\text{rot}}$$

the comparison ratio $\mathcal{R}_{\text{drag}}$ is used, with signs compared in absolute value because they depend on coordinate conventions. In the weak field,

$$J^i = 0 \Rightarrow P_i = 0, \quad P_t \equiv P$$

reduces to the existing scalar branch, and the frame-dragging component is evaluated only for rotating sources ($J^i \neq 0$). First, for the $\nu = \phi$ component, the quasistatic equation

$$\frac{1}{r^2} \partial_r (r^2 \partial_r P_\phi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta P_\phi) - \frac{P_\phi}{r^2 \sin^2 \theta} = -\frac{g_P}{c^2} J_\phi$$

is fixed with the external solution

$$P_\phi(r, \theta) = \frac{g_P J_*}{2c^2} \frac{\sin^2 \theta}{r^2} + \mathcal{O}(r^{-3}) \equiv \frac{\mu_{\text{drag}} \sin^2 \theta}{r^2}, \quad \mu_{\text{drag}} \equiv \frac{g_P J_*}{2c^2} \quad (13)$$

where J_* is the source angular momentum. Next, define the off-diagonal component of the effective metric by

$$g_{t\phi}^{(P)} \equiv -g_{tt}^{(P)} \beta_\phi^{(P)}, \quad \beta_\phi^{(P)} \equiv \xi_{\text{rot}} P_\phi$$

and reduce spin parallel transport,

$$U^\mu \nabla_\mu S^\nu = 0, \quad U_\mu S^\mu = 0$$

to the weak-field, low-velocity limit to obtain

$$\frac{d\mathbf{S}}{dt} = \mathbf{\Omega}_{\text{drag}}^{(P)} \times \mathbf{S}, \quad \mathbf{\Omega}_{\text{drag}}^{(P)} \equiv \frac{1}{2} \nabla \times \boldsymbol{\beta}^{(P)}$$

Substituting $\beta_\phi^{(P)} = \xi_{\text{rot}} P_\phi$ gives

$$(\nabla \times \boldsymbol{\beta}^{(P)})_r = \frac{3\xi_{\text{rot}} \mu_{\text{drag}} \sin \theta \cos \theta}{r^3}, \quad (\nabla \times \boldsymbol{\beta}^{(P)})_\theta = \frac{\xi_{\text{rot}} \mu_{\text{drag}} \sin^2 \theta}{r^3}$$

and on the equatorial plane ($\theta = \pi/2$),

$$\Omega_{\text{pred}}^{(P)} \equiv \left| \mathbf{\Omega}_{\text{drag}}^{(P)} \right| = \frac{|\xi_{\text{rot}} \mu_{\text{drag}}|}{2r^3} = \frac{|g_P \xi_{\text{rot}} J_*|}{4c^2 r^3} \quad (14)$$

GP-B follows an almost polar orbit ($\sim 90^\circ$ inclination), but because the published value is given as an

orbit-averaged precession rate, this paper normalizes the r^{-3} coefficient above to the orbit-averaged quantity and uses it for the $\mathcal{R}_{\text{drag}}$ comparison.

In other words, the chain $P_\phi \rightarrow g_{t\phi}^{(P)} \rightarrow \Omega_{\text{pred}}^{(P)}$ closes the precession prediction used in the observational comparison. The $\mathcal{R}_{\text{drag}}$ used in this section is defined with this $\Omega_{\text{pred}}^{(P)}$ in the denominator, yielding $\mathcal{R}_{\text{drag}} \approx 1$.

Equation: definition of the ratio

$$\mathcal{R}_{\text{drag}} \equiv \frac{|\Omega_{\text{obs}}|}{|\Omega_{\text{pred}}|}, \quad \Omega_{\text{pred}} \equiv \Omega_{\text{pred}}^{(P)}$$

Metric: $\mathcal{R}_{\text{drag}}$ and its σ , together with

$$z = \frac{\mathcal{R}_{\text{drag}} - 1}{\sigma}.$$

Result:

Experiment	$\mathcal{R}_{\text{drag}}$	z	Notes
GP-B	0.949±0.184	−0.28	
LAGEOS	0.99±0.05	−0.20	

Both remain consistent with $\mathcal{R}_{\text{drag}} = 1$ (reference comparison).

In addition, the integrated audit connecting the choice of background metric (flat background vs effective metric) and the axisymmetric boundary closure yields Reject for the flat background and Pass for the effective metric. The boundary diagnostic also satisfies the closure condition with $\text{max_flux_rel_std} = 1.25 \times 10^{-16}$, and the overall verdict for the rotation branch is fixed as `overall_status=Pass` (`deltaP_rot_required_under_effective_metric`).

4.12.1 GP-B Scalar-Limit Audit (Frame Dragging)

Purpose: to make explicit, using the GP-B observables, the limitation of the P-model when restricted to the pure-scalar limit (no rotational effect). [17]

Input: representative values of 6.6 arcsec/yr for geodetic precession and 0.039 arcsec/yr for frame dragging.

Process: compare a scalar-limit prediction in which geodetic precession is 6.6 arcsec/yr (consistent) but frame dragging is 0 arcsec/yr.

Metric:

$$z \equiv \frac{\Omega_{\text{obs}} - \Omega_{\text{scalar}}}{\sigma_{\text{obs}}}$$

(with audit gate $|z| > 3$ taken as Reject)

Result:

Channel	Observation [arcsec/yr]	Scalar prediction [arcsec/yr]	z	Verdict
Geodetic precession	6.6	6.6	0.00	Pass
Frame dragging	0.039	0.000	5.42	Reject

Discussion: Geodetic precession alone cannot exclude the scalar limit, but requiring frame dragging at the same time does exclude it. Accordingly, an extension carrying rotational effects (the minimal rotation term) is required.

A comparison between the observed $\mathcal{R}_{\text{drag}}$ and its error bars confirms that the rotational terms in GP-B and LAGEOS are consistent with the reference baseline ($\mathcal{R}_{\text{drag}} = 1$). This section is fixed as a Reference row (outside the primary verdict), and the rejection verdict is taken from the scalar-limit gates in 4.12.1–4.12.2.

Caution: differential predictions in the strong-field regime (near rotating black holes) remain a future task.

4.12.2 Integrated GP-B + LAGEOS Gate (Scalar Limit)

Purpose: not to rely on GP-B alone, but to judge the pure-scalar limit (no rotational effect) under a common gate across multiple weak-field experiments.

Input: GP-B (geodetic precession and frame dragging), and the agreement ratio for frame dragging in LAGEOS/LAGEOS II ($\mathcal{R}_{\text{drag}} = 0.99 \pm 0.05$).

Process: fix the scalar-limit prediction as

- geodetic precession: known value (control channel)
- frame dragging: 0

and judge both under the common z metric. For LAGEOS, use the published summary ratio (observation / reference-baseline prediction), and evaluate the scalar limit with $\mathcal{R}_{\text{drag,scalar}} = 0$.

Metric:

$$z \equiv \frac{X_{\text{obs}} - X_{\text{scalar}}}{\sigma_{\text{obs}}}, \quad |z| > 3 \Rightarrow \text{Reject}$$

Result:

Channel	Observed value	Scalar prediction	z	Verdict
GP-B geodetic precession (control)	6.6	6.6	0.00	Pass
GP-B frame dragging	0.039	0.000	5.42	Reject
LAGEOS frame dragging ($\mathcal{R}_{\text{drag}}$ ratio)	0.99	0.00	19.8	Reject

Discussion: the frame-dragging channel satisfies $|z| > 3$ in both GP-B and LAGEOS, so scalar_limit_rejected_by_frame_dragging is not dependent on a single experiment. The minimum condition needed to satisfy weak-field rotational observations is therefore the adoption of an extension with rotational effects.

Caution: for LAGEOS, the published summary value ($99 \pm 5\%$) is treated as an approximate 1σ scale for operational purposes, and this handling will be re-audited when primary-data updates become available.

4.13 XRISM (High-Resolution X-ray Spectra: Resolve)

Purpose: without relying on distance indicators (luminosity distance, Λ CDM), to fix directly from primary X-ray spectral data the “line shift” (velocity/redshift) and the “line width” (velocity dispersion), and thereby provide an entry point connecting to independent probes and high- γ velocity constraints.

Input:

- DARTS (ISAS/JAXA) XRISM public data (Resolve; rev3).
- NASA HEASARC (XRISM archive; mirror).
- For strong-field differences (Fe-K α broad line / ISCO): XMM-Newton / NuSTAR archives (XSA / HEASARC).

Process: from the Resolve PI spectrum and RMF, fit the lines in the Fe-K band (for example 5.5–7.5 keV) with a forward model and tabulate separately the statistical line-center energy E_{obs} and the procedure-dependent systematics. At minimum, the systematics are fixed through sweeps of:

- fit window (choice of band edges)
- gain (energy-scale uncertainty)
- rebinning dependence
- continuum-model dependence

Line detection is judged by the line-strength S/N ($\text{amp}/\sigma_{\text{amp}}$), and non-detections are excluded from the integrated verdict (“evaluate only detected lines” is fixed).

Equation: Energy ratio: $D_{\text{obs}} \equiv \frac{E_{\text{obs}}}{E_{\text{rest}}}$

Velocity (approximate): $\beta_{\text{obs}} \approx \frac{D_{\text{obs}} - 1}{D_{\text{obs}} + 1}$

Cluster X-ray redshift (line centroid): $z_{\text{xray}} \equiv D_{\text{obs}} - 1$

Systematic/statistical ratio (definition): $\text{sys/stat} \equiv \frac{\sigma_{\text{sys}}}{\sigma_{\text{stat}}}$

Metric:

- detection: $\text{amp}/\sigma_{\text{amp}} \geq 3$.
- systematic/statistical balance: sys/stat (combined systematic sweep divided by statistical uncertainty), together with its breakdown (gain/window/rebin/continuum, etc.).
- adoption gate (validation summary): for BH/AGN, detected obsids must satisfy $\text{sys/stat} \leq 10$; for clusters, this is combined with $\text{detected_obsids} \geq 2$.

Result:

Resolve (line centroids; BH/AGN and clusters)

Subsystem	Adoption gate	Current status	Verdict
BH/AGN	sys/stat $\leq$ 10 (detected obsids)	5/5 reached	Pass
Cluster	detected_obsids $\geq$ 2 and sys/stat $\leq$ 10	reached for seed targets	Pass

event-level QC (products vs event_cl)

Metric	Value	Notes
Fe-K band difference (L1)	0.7–3.4%	BH/AGN + cluster, 8 obsids
Mean energy difference	absolute value $\leq$ 0.8 eV	same

Fe-K α broad line / ISCO (GX 339-4; proxy)

Data	$r_{\text{in}} (r_g)$	stat	sys	sys/stat	Notes
NuSTAR 80001013002	5.50	0.73	4.10	5.6	
NuSTAR 80001013010	5.50	0.71	5.00	7.0	
NuSTAR 90702303001	5.50	0.74	4.00	5.4	

In addition, NuSTAR 80001013006 gives a proxy near $r_{\text{in}} \approx 2.5 r_g$, but with sys/stat $\approx$ 21 and thus systematics-dominated. NuSTAR 80001015001 and 80502325006/80702316002 are pinned to the lower bound ($r_{\text{in}} \approx 2.5 r_g$), and the two XMM obsids are pinned to the lower bound ($6 r_g$) in the minimal diskline model ($a_* = 0$) (r_in_bound=lower). A physical-model replacement (such as relxill) is therefore deferred to the next stage.

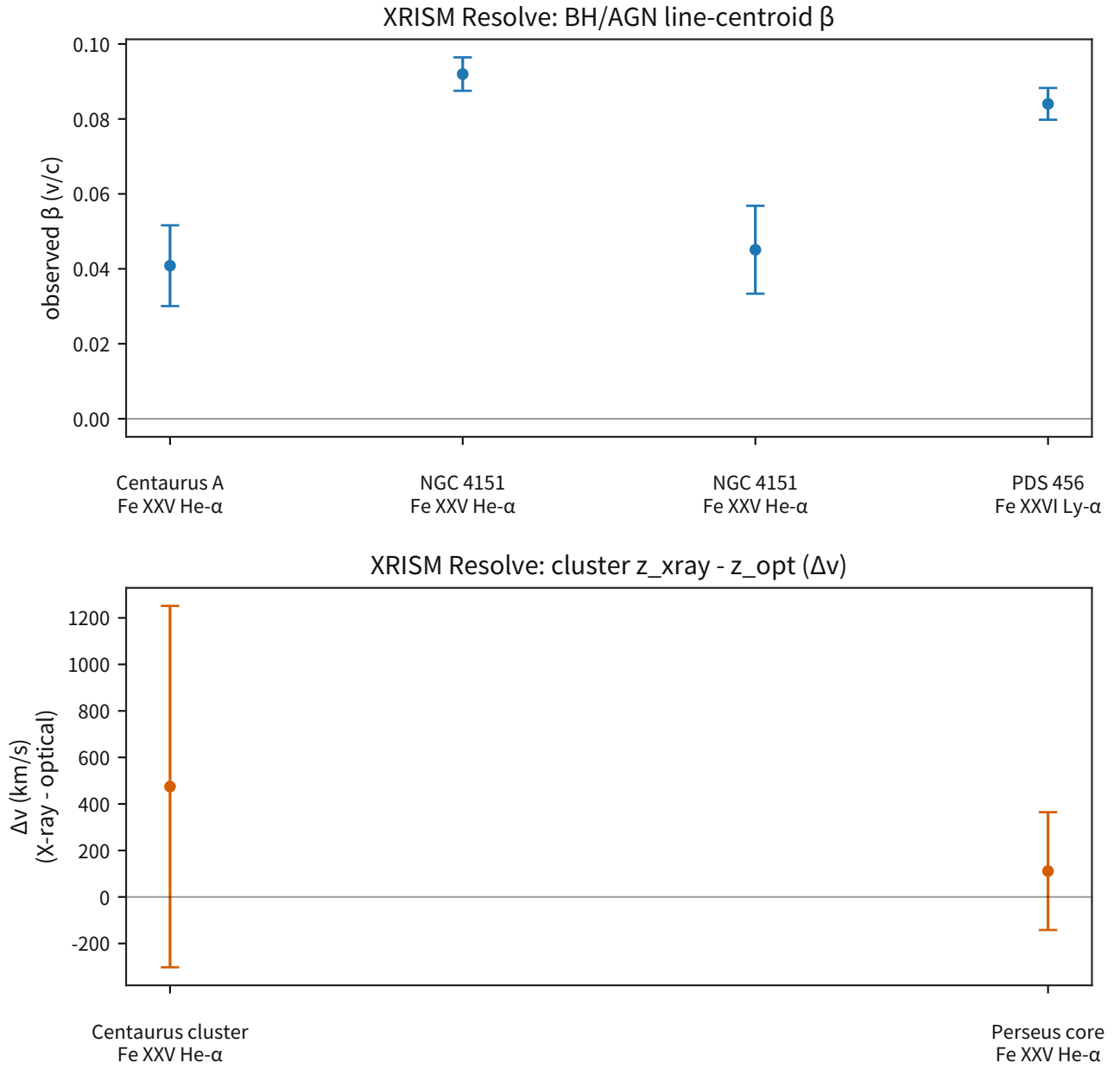


Figure 29: Overview of representative Resolve results for BH/AGN (β_{obs}) and clusters (a proxy for Δv).

Supplement: this fixes the grounds for adoption/non-adoption in a form that can be read directly from the figure.

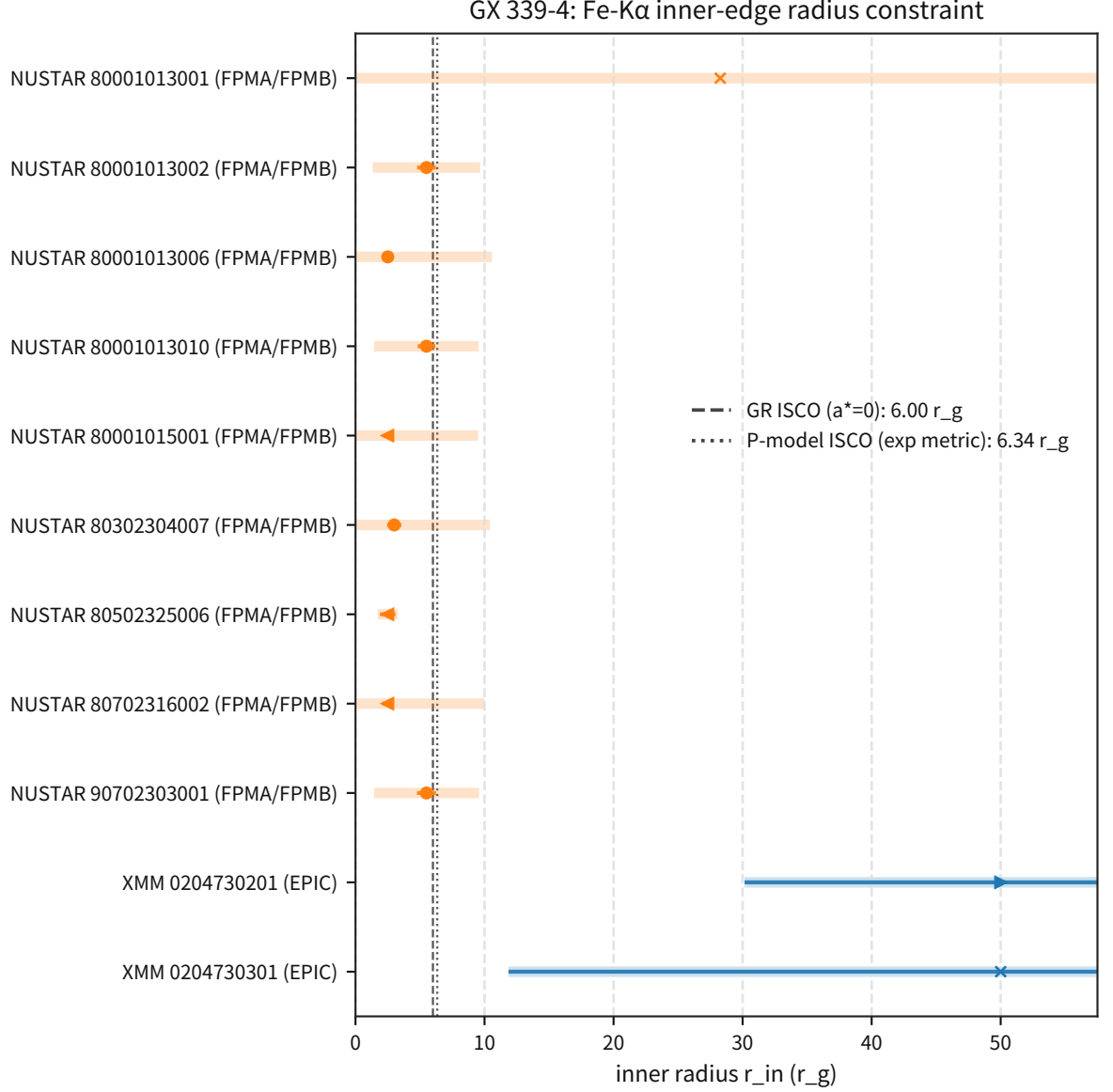


Figure 30: Cross-comparison of the inner radius r_{in} (ISCO proxy) inferred from the Fe-K α broad line in XMM/NuSTAR observations of GX 339-4.

Supplement: thin lines show statistical uncertainties, thick lines show statistical + systematic uncertainties, allowing boundary-pinned cases and systematics-dominated cases to be visualized in one figure. The vertical dashed line is the ISCO= $6 r_g$ of the reference framework (GR, $a_* = 0$), and the dotted line is the P-model candidate (exponential metric) at ISCO $\approx 6.34 r_g$. Systematics are tabulated as sweep contributions from knobs such as band, gain, rebinning, and region (σ_{sys}), and their dominance is evaluated through sys/stat ($\sigma_{sys}/\sigma_{stat}$).

Discussion: with new public obsids and calibration updates, the next stage is to increase the number of targets (multiple datasets), shrink the relative contribution of procedural systematics (sys/stat), carry out forward modeling of line identification and continuum components while making the degree-of-freedom ledger explicit, and separate BH and cluster branches by physical-model differences. This section therefore fixes a **reproducible analysis I/F** for future precision updates rather than claiming

a final conclusion.

Caution: the point of this section is not to declare “violation” or “support,” but to fix, for public primary data, how much the statistics move under procedural choices (window/gain/rebin/continuum) and to record that as the systematic component.

For strong-field differences (Fe-K α broad line / ISCO), the constraint is derived not from the line centroid but from the **shape** of the reflection spectrum. At present, the NuSTAR event_cl products are fixed as proxies (without RMF/ARF folding), while for XMM (PPS) an RMF-folded workflow using canned RMFs (SRSPEC 2400ch $\rightarrow$ RMF 800ch rebinning) has been added, together with XSPEC forward-folding (diskline), but these remain diskline proxies. A physical ISCO constraint and its comparison with the P-model ISCO prediction will be finalized in the next stage using reflection models with RMF/ARF folding and including cross-calibration and systematic decomposition.

4.14 Galaxy Rotation Curves (SPARC: V_{obs} / V_{bar} / V_P Audit)

Purpose: for SPARC rotation curves (175 galaxies), to quantify under a single degree of freedom M/L (Υ) whether the observed velocity $V_{\text{obs}}(R)$ can be explained by the baryonic term alone, or is improved by adding the P-model correction $V_P(R)$.

Input: SPARC Rotmod ($V_{\text{obs}}, \sigma_V, V_{\text{gas}}, V_{\text{disk}}, V_{\text{bulge}}$), together with the $H_0^{(P)}$ frozen through the redshift mapping. The derived scale is obtained from the late-time background law $P_{\text{bg}}(t) \propto \exp[-H_0^{(P)}(t - t_0)]$ through $\omega_{\text{bg}} = H_0^{(P)}$, $\lambda_{\text{bg}} = 2\pi c/H_0^{(P)}$, and $a_0 = c^2/\lambda_{\text{bg}} = cH_0^{(P)}/(2\pi)$.

Process: at each radius R , construct V_{bar} from gas+disk+bulge, and evaluate two comparison models: (i) baryon-only and (ii) P-model corrected. In both cases the only free parameter is Υ (with the bulge/disk ratio fixed), and a simultaneous all-point χ^2 minimization is carried out.

Equation:

$$V_{\text{bar}}^2(R) = V_{\text{gas}}^2 + \Upsilon V_{\text{disk}}^2 + (f_{\text{bulge}} \Upsilon) V_{\text{bulge}}^2$$

$$g_{\text{bar}} = \frac{V_{\text{bar}}^2}{R}, \quad g_P = \frac{g_{\text{bar}}}{1 - \exp\left(-\sqrt{g_{\text{bar}}/a_0}\right)}, \quad V_P = \sqrt{g_P R} \quad (15)$$

$$\chi^2 = \sum_i \left(\frac{V_{\text{obs},i} - V_{\text{model},i}}{\sigma_i} \right)^2, \quad \chi^2/\nu \quad (\nu = N - 1)$$

Here κ_a is not a fit coefficient dedicated to the SPARC section. It is fixed as a derived quantity from the late-time exponential law of the background wave, $P_{\text{bg}}(t) \propto \exp[-H_0^{(P)}(t - t_0)]$, via

$$\omega_{\text{bg}} = H_0^{(P)}, \quad \lambda_{\text{bg}} = \frac{2\pi c}{H_0^{(P)}}, \quad a_0 = \frac{c^2}{\lambda_{\text{bg}}} = \frac{cH_0^{(P)}}{2\pi}$$

which gives

$$\kappa_a \equiv \frac{a_0}{cH_0^{(P)}} = \frac{1}{2\pi}$$

The notation is kept distinct from the EHT κ (ring/shadow mapping) in Section 4.8.

Metric: global χ^2/ν , RMS residual, median galaxy-by-galaxy χ^2/ν , and $\Delta\chi^2 = \chi^2(\text{baryon}) - \chi^2(\text{P-model})$.

Result:

Metric	baryon-only	P-model correction
Best-fit Υ	0.94	0.54
Global χ^2/ν	106.45	21.34
RMS residual [km/s]	47.34	22.77
Median absolute residual [km/s]	31.53	11.53
Median galaxy-by-galaxy χ^2/ν	43.98	7.10

Additional summary (175 galaxies):

Metric	Value
$\Delta\chi^2(\text{baryon} - \text{P-model})$	2.885×10^5
Number of galaxies with $\Delta\chi^2 > 0$	150 / 175 (85.7%)

SPARC audit (single-Y fit)

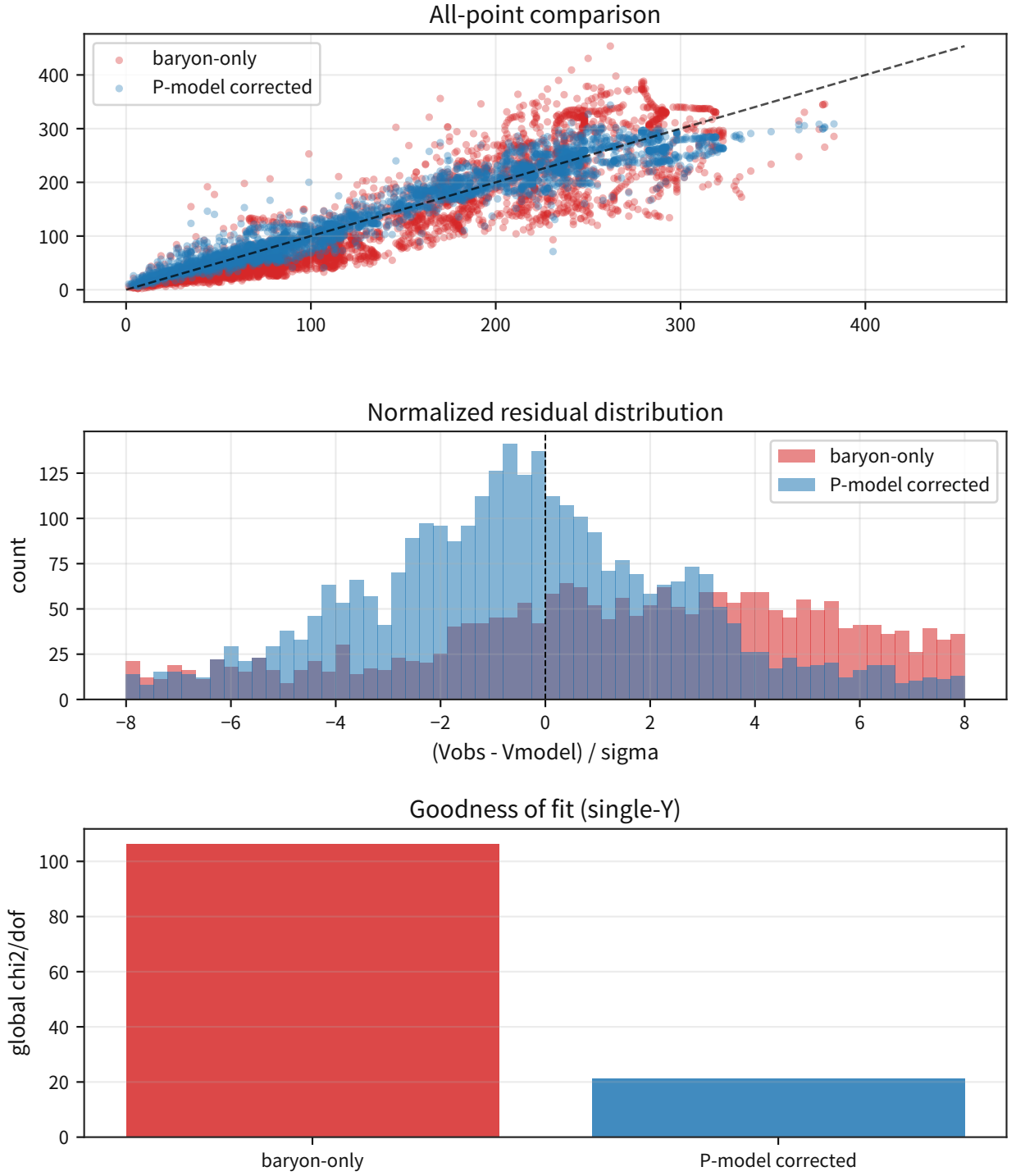


Figure 31: Upper panel: full-point comparison of V_{obs} and the model velocities (baryon-only / P-model correction). Middle panel: comparison of normalized residual (pull) distributions. Lower panel: comparison of global χ^2/ν for the single-Y fit.

Discussion: $\chi^2/\nu = 21.34$ is still large in the absolute sense and does not reach $\chi^2/\nu \approx 1$. However,

this audit imposes the strict condition of a single Υ shared across all 175 galaxies, so galaxy-specific systematics such as non-circular motions, inclination angles, and inner-beam smoothing readily accumulate into the residuals. Even under that restriction, the P-model correction yields an improvement of $\Delta\chi^2 = 2.885 \times 10^5$, with $\Delta\chi^2 > 0$ for 150/175 galaxies (85.7%), and the median galaxy-by-galaxy χ^2/ν decreases from 43.98 to 7.10. The present result therefore shows a significant improvement in explanatory power as a single- Υ baseline correction. Reaching an absolute $\chi^2/\nu < 1$ is left to the next stage, where galaxy-specific Υ values and systematic modeling will be released.

Caution: this section does not claim to decide the existence or non-existence of dark matter. It is an audit result comparing the explanatory power of V_{bar} and V_P on the SPARC observational I/F under the same number of degrees of freedom.

4.15 Cluster-Collision Offsets (Bullet-like Systems: Δx_{P-gas} / Δx_{P-lens})

Purpose: for the observed separation between the lensing centroid and the gas centroid in colliding galaxy clusters, compare on the same metric the baryon-only baseline (P peak = gas centroid) and the P -corrected branch (P peak leading), auditing both the direction (lead/lag) and the scale.

Input: the offset observables projected onto the collision axis (main/sub, two components), together with collision velocity templates (v_{pre}, v_{post}), the response time constant τ , the collision transition width, and the observation time. No coefficient borrowing (α) is used; the evaluation is restricted to the derived branch driven by J^μ .

Process: starting from the dynamical P equation in Part I Section 2.6, first derive the retarded potential for a uniformly moving point source. This confirms that a single uniform-velocity source does not produce a center shift at $\mathcal{O}(v/c)$. For a collision system, the offset Δx is then evaluated through the combination of “velocity transition + finite response + retarded kernel.” Specifically, the first-moment equation for the P -peak centroid X_P is integrated in time, and ξ is fitted as the only free degree of freedom. τ and ξ are fixed by the Part I derivation chain ($\tau_{free}/\tau_{int}/\tau_{damp} \rightarrow$ kernel first moment $\rightarrow \tau_{eff} \rightarrow \xi$), with no external hand-tuned additions.

Equation:

$$\begin{aligned}
 \left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 \right) u &= \frac{4\pi G}{c^2} \rho, & \rho(\mathbf{r}, t) &= M \delta^3(\mathbf{r} - \mathbf{v}t) \\
 u(\mathbf{r}, t) &= \frac{GM/c^2}{(1 - \mathbf{n}_r \cdot \boldsymbol{\beta}_v) R_r} \Big|_{t_r}, & \mathbf{a} &= c^2 \nabla u \\
 \mathbf{a} &= -GM \frac{\mathbf{s}}{|\mathbf{s}|^3} + \mathcal{O}\left(\frac{v^2}{c^2}\right), & \mathbf{s} &\equiv \mathbf{r} - \mathbf{v}t \\
 \frac{1}{c_w^2} \partial_{tt} u_L - \partial_{xx} u_L &= S_L, & \tau_{free} \partial_{tt} u_L + \partial_t u_L &= D \partial_{xx} u_L + c_w^2 S_L, \quad D \equiv c_w^2 \tau_{free} \\
 \varepsilon_{slow} &\equiv \frac{\tau_{free}}{T_{slow}}, & T_{slow} &\equiv \min(\tau_{int}, \tau_{damp}), \quad \varepsilon_{slow} \ll 1 \Rightarrow \partial_t u_L = D \partial_{xx} u_L + c_w^2 S_L \\
 \partial_t P_L + \partial_x q &= -\frac{P_L - J^0}{\tau_{free}}, & q &= -D \partial_x P_L - \xi[(1 - \eta)J^x + \eta J^x(t - \Delta t)] \\
 \tau \frac{dX_P}{dt} + X_P &= X_g + \xi \tau [(1 - \eta)\chi(t)v(t) + \eta R_{delay}(t)] \\
 R_{delay}(t) &= \int_0^t K_{\tau_k}(t - s) \chi(s - \Delta t) v(s - \Delta t) ds, & \Delta x_{P-gas} &= X_P - X_g \\
 \tau_{free} &= \frac{L_{path}}{c_w}, & \tau_{int} &= \sqrt{\frac{\langle (\chi v)^2 \rangle}{\langle (d(\chi v)/dt)^2 \rangle}}, \quad \tau_{damp} = \sqrt{\frac{\langle v^2 \rangle}{\langle (dv/dt)^2 \rangle}} \\
 \eta &= \frac{\tau_{free}}{\tau_{free} + \tau_{int}}, & \tau_{eff} &= \tau_{free} + \eta \tau_{kernel}, \quad \tau_{kernel} = w_f \tau_f + w_s \tau_s, \quad w_i \propto \frac{1}{\tau_i}
 \end{aligned} \tag{16}$$

$$\xi = \frac{\tau_{\text{int}} + \tau_{\text{damp}}}{\tau_{\text{eff}}}$$

$$\Gamma_{\text{path}} \equiv \tau_{\text{free}}^{-1} = \frac{g_P^2}{\chi P} \int_0^\infty \langle \delta J_x(t) \delta J_x(0) \rangle dt = \Gamma_{\text{adv}} + \Gamma_{\text{coll}} \quad (17)$$

$$\Gamma_{\text{adv}} \simeq \frac{v}{L_{\text{corr}}}, \quad \Gamma_{\text{coll}} \simeq A_{\text{col}} \rho T^{-3/2}, \quad L_{\text{path}} = \frac{c_w}{\frac{v}{L_{\text{corr}}} + A_{\text{col}} \rho T^{-3/2}}$$

$$\frac{L_{\text{path}}}{L_{\text{path},0}} = \frac{1 + \Pi_0}{r_v + \Pi_0 r_\rho r_T^{-3/2}}, \quad \Pi_0 = \frac{\tau_{\text{int},0}}{\tau_{\text{free},0}} - 1$$

Below, the τ in the first-order delay equation is fixed to mean τ_{eff} .

$$\chi^2 = \sum_i \left(\frac{\Delta x_{P\text{-}gas}^{(i)} - \Delta x_{\text{obs}}^{(i)}}{\sigma_{\text{obs}}^{(i)}} \right)^2$$

Metric: χ^2/ν , $\max|z(\Delta x_P - lens)|$, `tau_reconstruction_rel_error`, `lw_mean_peak_rel_error`, and `ad_hoc_parameter_count` (audit of the prohibition on ad hoc additions).

Result:

Metric	Value
<code>overall_status</code>	Watch
χ^2/ν	3.142
low-DoF boundary p_{upper} ($\nu_{\text{eff}} = 1$, χ_{red}^2 proxy)	0.076
$\max z(\Delta x_P - lens) $	2.034
<code>tau_reconstruction_rel_error</code>	0.124
$\epsilon_{\text{slow}} = \tau_{\text{free}}/\min(\tau_{\text{int}}, \tau_{\text{damp}})$	0.207
<code>tau_origin_mode</code>	<code>derived_from_pmu_kernel</code>
<code>lw_mean_lag_rel_error</code>	0.333
<code>lw_mean_peak_rel_error</code>	0.128
<code>retarded_kernel_mode</code>	<code>double_exp_derived</code>
<code>xi_mode / ξ</code>	<code>derived / 6.5466</code>
$\Pi_0 = \tau_{\text{int}}/\tau_{\text{free}} - 1$	5.540
$L_{\text{path}}(2\rho)/L_{\text{path}}(\rho)$	0.541
$L_{\text{path}}(2v)/L_{\text{path}}(v)$	0.867
$L_{\text{path}}(2T)/L_{\text{path}}(T)$	2.210
Bullet reevaluation χ^2/ν	3.142
Bullet reevaluation $\max z $	2.034
<code>ad_hoc_parameter_count</code>	0
<code>background_metric_choice</code>	<i>effective</i> $g_{\mu\nu}(P)$
<code>metric_caseA / caseB</code>	Reject / Pass

Cluster-by-cluster decomposition of contributions ($\Delta x_{\text{peak}} \approx |(\xi \tau \chi_{\text{obs}} v_{\text{obs}}) + s_{\text{lag}} \Delta x_{\text{lag}}|$, $s_{\text{lag}} = -\text{sign}(v_{\text{obs}})$):

cluster	current term (kpc)	lag term (kpc)	est. abs. offset (kpc)	sim. abs. offset (kpc)	residual rel. err.
bullet_main	139.154	-12.264	126.890	139.149	0.088
bullet_sub	-171.402	42.176	129.226	110.697	0.167

Supplement: the JSON-key correspondence is fixed as `lw_current_term_signed_kpc / lw_lag_term_signed_kpc / lw_est_abs_offset_kpc / sim_abs_offset_kpc / residual_rel_error`.

The updated estimate satisfies the Watch condition $\text{lw_mean_peak_rel_error} \leq 0.50$, while maintaining the strict derivation (`ad_hoc_parameter_count=0`), and at the low-degree-of-freedom boundary (two Bullet components), conservatively downgrades $\chi^2/\nu > 3$ to a Watch verdict.

Discussion: the retarded potential of a uniformly moving source alone cannot produce a first-order offset. What dominates here is the signed combination of the “current term” generated by the post-collision J^μ transition and the “lag term” from the retarded response. The current term is the principal component and the lag term is the correction component. The derivation-chain audit fixes `tau_origin_mode=derived_from_pmu_kernel`, `xi_mode=derived`, and `ad_hoc_parameter_count=0` under the same gate so that the audit values can be traced directly from the equations in the text. Moreover, because $\Pi_0 > 0$ (the collisional branch is significant), L_{path} decreases monotonically with ρ and v , and increases monotonically with T . This ratio scaling can be extrapolated to non-Bullet systems (for example El Gordo) by replacing only (r_ρ, r_v, r_T) . In the transfer reevaluation carried out under the same I/F, the two Bullet components give $\chi^2/\nu = 3.142$ and $\max|z|=2.034$, which do not reach the 3σ rejection domain, but because of the low-DoF boundary ($p_{\text{upper}} \approx 0.076$ under the conservative estimate $\nu_{\text{eff}} = 1$), the verdict is Watch. Also, $\epsilon_{\text{slow}} = 0.207$ lies near the boundary of the asymptotic condition $\epsilon_{\text{slow}} \ll 1$, so the contribution of higher-order corrections (at least $\mathcal{O}(\epsilon_{\text{slow}}^2)$) must be evaluated at the next stage. Non-Bullet systems remain Watch because the observed offsets are not yet registered, so they remain at the input-preparation stage.

Caution: this section does not declare a final theoretical solution for Bullet systems. It presents an operational audit with the derived double-exponential kernel fixed as a reproducible I/F. The background-metric adoption has already been fixed by the metric-comparison audit as caseA (flat $\eta_{\mu\nu}$)=Reject and caseB (effective metric $g_{\mu\nu}(P)$)=Pass, so the present separation analysis formally adopts the effective metric (caseB). Extrapolation to non-Bullet systems (e.g. El Gordo), as well as the recording of rerun candidates associated with future EHT data updates (input-hash monitoring, event counters, etc.), is managed entirely in the Part IV verification ledger (Registry) and rerun explicitly with the same I/F only when needed. It is not described in the main text here.

4.16 Macroscopic Emergence of a DM-Replacement Mechanism and ISW / Weak-Lensing Predictions

Purpose: to unify the “dark-matter-like behavior” observed in SPARC and Bullet systems not by assuming dark-matter particles, but as a macroscopic emergence of the dynamical P field, while making explicit both the observable predictions in ISW / weak lensing and the present limits.

Input: the SPARC audit in 4.14 ($V_{\text{obs}}, V_{\text{bar}}, V_P$), the colliding-cluster offset audit in 4.15 ($\Delta x_{P\text{-gas}}, \Delta x_{P\text{-lens}}$), and the background mapping $H_{\text{eff}}(z)$.

Process: apply the same P branch to the galaxy-scale quasi-stationary regime and to nonstationary cluster collisions, connecting two limiting pictures: “in the stationary regime it acts as additional gravity,” and “in the collision regime it produces centroid separation through response delay.”

Equation:

$$g_{\text{eff}}(R) = g_{\text{bar}}(R) + g_P(R), \quad V_{\text{obs}}^2(R) = R g_{\text{eff}}(R)$$

$$\nabla^2 \phi_P = 4\pi G \rho_P^{\text{eff}}, \quad \phi_{\text{eff}} = \phi_{\text{bar}} + \phi_P$$

$$\kappa_{\text{eff}}(\boldsymbol{\theta}) \propto \nabla_{\perp}^2 \int d\chi \frac{2\phi_{\text{eff}}}{c^2}$$

$$\left. \frac{\Delta T}{T} \right|_{\text{ISW}} = -2 \int d\eta \partial_{\eta} \left(\frac{\phi_{\text{eff}} + \psi_{\text{eff}}}{2c^2} \right)$$

Metric: SPARC $\Delta\chi^2$ and median χ^2/ν , Bullet χ^2/ν and $\max|z|$, and for future verdicts `isw_sign_pass` and `lensing_peak_alignment_pass`.

Result:

Item	Fixed content	Current verdict
Galaxy rotation curves (SPARC)	g_P -inclusive g_{eff} improves over baryon-only in 150 of 175 galaxies ($\Delta\chi^2 > 0$)	Pass
Colliding clusters (Bullet-like systems)	reproduces a leading P peak through current term + lag term and explains $\Delta x_{P\text{-gas}}$ under the same I/F	Watch
ISW prediction	in collisions and non-equilibrium systems, the axial dipole pattern in $\partial_{\eta}(\phi_{\text{eff}} + \psi_{\text{eff}})$ is enhanced	Watch (qualitatively fixed)
Weak-lensing prediction	in mergers, the lensing peak shifts toward the P peak rather than the gas peak; in relaxed systems the difference shrinks	Watch (qualitatively fixed)

Discussion: the DM-replacement mechanism in this paper appears as an effective gravity enhance-

ment through g_P on galaxy scales, and as centroid separation through delayed response via τ_{eff} in collision systems. The key difference from an “additional particle” hypothesis is that both are described by the same equation system.

Caution: ISW and weak lensing are only at the prediction-fixing stage in this paper, and a full simultaneous fit to all-sky ISW correlations or cosmic-shear two-point statistics has not yet been performed.

4.17 Joint Strong-Field Higher-Order Constraint (EHT + GW + Pulsar + Fe-K α ISCO)

Purpose: in the strong-field regime beyond the weak-field approximation, fix the higher-order coefficient λ_H with one degree of freedom and audit whether a simultaneous constraint across EHT, GW, Pulsar, and Fe-K α ISCO can be maintained.

Input: the EHT shadow-diameter constraints (M87* / Sgr A*), GW IMR consistency for GW250114 (final mass / final spin), the first-strain similarity indicators for GW150914 (H1/L1 correlation), the pulsar orbital-decay channel ratios, and the Fe-K α ISCO substitute indicator.

Process: start from the nonlinear steady axisymmetric PDE for the strong-field branch fixed in Part I 2.8.8, treat the higher-order contribution as a one-parameter extension of the shared mapping, and then audit whether the same λ_H can satisfy all channels without violating the hard gates on channel count, single-coefficient consistency, and excessive free-parameter expansion.

Equation:

$$\begin{aligned} \frac{1}{\sqrt{|g(P)|}} \partial_\mu \left(\sqrt{|g(P)|} F_{(P)}^{\mu\nu} \right) &= 0, \quad F_{(P)}^{\mu\nu} = g^{\mu\alpha}(P) g^{\nu\beta}(P) (\partial_\alpha P_\beta - \partial_\beta P_\alpha) \\ \frac{1}{r^2} \partial_r (r^2 e^{-2u} \partial_r P_t) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta e^{-2u} \partial_\theta P_t) &= \mathcal{N}_0^{(2)}[u, P_t, P_\phi] \\ \frac{1}{r^2} \partial_r (r^2 e^{-2u} \partial_r P_\phi) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta e^{-2u} \partial_\theta P_\phi) - \frac{e^{-2u} P_\phi}{r^2 \sin^2 \theta} &= \mathcal{N}_\phi^{(2)}[u, P_t, P_\phi] \end{aligned}$$

For $u = \epsilon\psi$ ($\epsilon \ll 1$), expand $e^{-2u} = 1 - 2u + 2u^2 + O(u^3)$ up to $O(\epsilon^2)$ and discard $O(\epsilon^3)$ and above. The time component is then split as the linear branch plus an $O(\epsilon^2)$ perturbation.

$$\begin{aligned} \mathcal{N}_0^{(2)} &= \mathcal{N}_{\text{metric}}^{(2)} + \mathcal{N}_\phi^{(2)} \\ \mathcal{N}_{\text{metric}}^{(2)} &= \epsilon_{\text{nl}}^2 2 \left[(\partial_r u)(\partial_r P_t) + \frac{1}{r^2} (\partial_\theta u)(\partial_\theta P_t) \right] \\ \mathcal{N}_\phi^{(2)} &= \epsilon_{\text{nl}}^2 e^{-2u} \left[\frac{(\partial_r P_\phi)^2}{r^2 \sin^2 \theta} + \frac{(\partial_\theta P_\phi)^2}{r^4 \sin^2 \theta} \right] \end{aligned}$$

Fix the minimal rotation branch as $P_\phi(r, \theta) = \mu_{\text{drag}} \sin^2 \theta / r^2$ and write

$$\begin{aligned} P_t &= P_t^{(\text{lin})} + \delta P_t, \quad \bar{\mathcal{N}}_0(r) = \frac{1}{2} \int_0^\pi \mathcal{N}_0^{(2)}(r, \theta) \sin \theta d\theta \\ \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d \delta P_t}{dr} \right) &= \bar{\mathcal{N}}_0(r), \quad \delta P_t = - \int_r^{r_{\text{out}}} d\xi \xi^{-2} \int_\xi^{r_{\text{out}}} ds s^2 \bar{\mathcal{N}}_0(s) \end{aligned}$$

To express the higher-order contribution as a shift in the fitted coefficient, define

$$\Delta C_{\text{core}} \equiv C_{\text{linear}} - C_{\text{ref}}, \quad \Delta C_{N0} \equiv C_{\text{full}} - C_{\text{linear}}$$

$$\lambda_H^{(N0)} \equiv \frac{\Delta C_{N0}}{\Delta C_{\text{core}}} = \frac{\text{n0_contribution_pct}}{\text{core_gap_pct}}$$

$$\lambda_H^{(N0)} = \epsilon_{\text{nl}}^2 \Xi_{N0}$$

The fixed audit values are $\text{core_gap_pct}=4.6278\%$, $\text{n0_contribution_pct}=0.1806\%$, and $\epsilon_{\text{nl}}^2 = 0.0064$, which imply $\lambda_H^{(N0)} = 0.0390$ and $\Xi_{N0} = 6.10$. The $\lambda_H(\text{joint})$ estimated below is treated as an **effective coefficient** in the integrated observational channels (diagnostic fit) and is kept distinct from the physical definition of $\lambda_H^{(N0)}$.

The background-metric adoption is already fixed by the weak-field coefficient audit as caseA (flat $\eta_{\mu\nu}$)=Reject and caseB (effective metric $g_{\mu\nu}(P)$)=Pass, so the present section adopts caseB. The boundary conditions are fixed as $r \rightarrow \infty$ ($P_t/P_{\text{ref}} \rightarrow 1 + 1/r + \lambda_2/r^2$, $P_\phi \rightarrow 0$), $\theta = 0, \pi$ ($P_\phi = 0$, $\partial_\theta P_t = 0$), and $r \rightarrow r_H^+$ (P_t, P_ϕ finite), and the closure audit satisfies $\text{boundary_closure_pass}=\text{true}$, $\text{max_flux_rel_std}=1.2527\text{e-}16$ (threshold $1.0\text{e-}3$), and $\text{max_asymptotic_rel_error}=0.0$. Accordingly, this section connects to the simultaneous λ_H constraint under the same boundary conditions as Section 4.12.

For the external branch, write $P_t/P_{\text{ref}} = 1 + a_1/r + a_2/r^2 + \dots$ and $P_\phi = (\mu_{\text{drag}} \sin^2 \theta)/r^2 + \dots$, and map the same branch to EHT ring diameter / asymmetry and to rotation-related GW observables with $\Omega_{\text{drag}} \propto r^{-3}$.

Metric: $\lambda_H(\text{EHT})$, $\lambda_H(\text{GW})$, $\lambda_H(\text{Pulsar})$, $\lambda_H(\text{X-ray})$, $\lambda_H(\text{joint})$, χ^2/dof (baseline vs higher-order term), ΔAIC , the EHT κ precision ratios (compressed / required), the first-strain similarity metric for GW150914 (abs_best_corr), the multi-event similarity metric for GW ($\text{median_abs_corr_usable}$, n_usable_events), the GW area-theorem significance ($\text{sigma_gaussian_combined}$), the GW/Pulsar consistency gate $|\lambda_{\text{GW}} - \lambda_{\text{PULSAR}}|/\sigma_{\text{combined}}$, the GW/X-ray consistency gate $|\lambda_{\text{GW}} - \lambda_{\text{XRAY}}|/\sigma_{\text{combined}}$, and the independent X-ray channel count.

Result:

Metric	Value	Verdict
$\lambda_H(\text{EHT})$	-0.071 ± 0.085	—
$\lambda_H(\text{GW})$	$+0.074 \pm 0.099$	—
$\lambda_H(\text{Pulsar})$	$-4.96e - 5 \pm 6.63e - 5$	—
$\lambda_H(\text{X-ray})$	-0.155 ± 0.362	—
$ \lambda_{\text{EHT}} - \lambda_{\text{GW}} /\sigma_{\text{combined}}$	1.12	Hard Pass
$ \lambda_{\text{GW}} - \lambda_{\text{PULSAR}} /\sigma_{\text{combined}}$	0.751	Pass
$ \lambda_{\text{GW}} - \lambda_{\text{XRAY}} /\sigma_{\text{combined}}$	0.611	Pass
Strong-field baseline coefficient gap ($c_{\text{core}}/c_{\text{ref}} - 1$)	4.6278%	fixed
$\mathcal{N}_0^{(2)}$ contribution (additive shift to the coefficient gap)	+0.1806%	Pass
Composite coefficient gap (baseline + $\mathcal{N}_0^{(2)}$)	4.8084%	fixed
$\mathcal{N}_0^{(2)}$ contribution ratio (relative to baseline)	3.90%	Reference
$\lambda_H^{(N0)} = \Delta C_{N0}/\Delta C_{\text{core}}$	0.0390	physical anchor
$\Xi_{N0} = \lambda_H^{(N0)}/\epsilon_{\text{nl}}^2$	6.10	fixed

Continued on next page

Metric	Value	Verdict
$\max \delta P_t $ (perturbative solution)	2.03×10^{-3}	finite Pass
δu_{eff} (effective ring-radius shift)	8.65×10^{-4}	finite Pass
Background-metric adoption	$effective g_{\mu\nu}(P)$	fixed
Strong-field axisymmetric boundary closure	boundary_closure_pass=true (max_flux_rel_std=1.2527e-16, max_asymptotic_rel_error=0.0)	Pass
$\lambda_H(joint)$	$-4.97e-5 \pm 6.63e-5$	—
χ^2/dof ($\lambda_H = 0$)	0.294	—
χ^2/dof (joint fit)	0.272	—
ΔAIC (baseline–fit)	-1.44	support not reached (Watch)
inverse-variance contribution ratios (joint)	Pulsar 99.99989% / EHT 0.000061% / GW 0.000045% / X-ray 0.000003%	dominated by Pulsar under $\propto 1/\sigma^2$ weighting
$\lambda_H(\text{joint}; \text{ no Pulsar})$	$-1.395 \times 10^{-2} \pm 6.349 \times 10^{-2}$	reference (sensitivity diagnosis)
EHT κ precision ratio (compressed / required)	5.21	Watch
EHT κ precision ratio (conservative / required)	12.26	Reference
EHT κ required- σ gap (compressed minus required)	0.0416	Watch
EHT κ required- σ reduction rate (compressed baseline)	80.8%	Watch
EHT κ required independent-information multiplier ($\sigma \approx 1/\sqrt{N}$)	27.1x	Watch
GW150914 first-strain similarity (abs_best_corr)	0.749	Pass
GW H1/L1 multi-event similarity (median_abs_corr_usable)	0.701 ($n_{usable} = 2$)	Pass
GW250114 area theorem (sigma_gaussian_combined)	4.447	Pass
Pulsar independent channel count	4	Pass
X-ray ISCO independent channel count	5	Pass
Best AIC scenario (subset)	$\Delta AIC = -1.29$ (EHT only)	Watch
Can the current 1-DoF + fixed-channel setup reach $\Delta AIC \geq +2$?	no	Watch
Best ansatz-extended model (AIC)	shared_lambda ($\Delta AIC = -1.44$)	Watch

Continued on next page

Metric	Value	Verdict
Theoretical upper bound of AIC support ($k = 1$, $\chi^2_{\text{fit}} \rightarrow 0$)	$\Delta\text{AIC}_{\text{max}} = +1.82$	support impossible
Required χ^2_{fit} to reach AIC support ($k = 1$)	< 0 (-0.179)	impossible (with current data)
Missing $\Delta\chi^2$ gain to reach AIC support ($k = 1$)	3.44 (against target 4.0)	Watch
Required $ z $ from one new constraint under additive- χ^2 approximation	1.85 (fit residual 0σ) / 1.92 (0.5σ) / 2.11 (1σ)	planning metric
Required number of new constraints under the same assumption (all new constraints well-fitted)	$ z = 1.0 \rightarrow 4$, $1.5 \rightarrow 2$, $2.0 \rightarrow 1$	planning metric

The integrated verdict is **Watch**. The hard gates (number of input channels, single-coefficient consistency in EHT/GW, and immediate rejection of excessive free-parameter freedom) are passed. On the GW side, both the single-event gate (GW150914) and the multi-event gate (H1/L1) pass the similarity checks, and the Pulsar and X-ray sides also pass the independent-channel-count and GW-consistency gates. The remaining Watch items are only the $\Delta\text{AIC} \geq +2$ support gate and the EHT κ precision. Also, the main reason that $\lambda_H(\text{joint})$ nearly matches $\lambda_H(\text{Pulsar})$ is that the $1/\sigma^2$ weighting makes the Pulsar contribution overwhelmingly dominant.

Discussion: at present, one cannot conclude that the higher-order term is required, but one also cannot conclude that it is rejected. The practical bottleneck is the EHT κ systematics: even after compression, about 5.2 times the required precision remains. In σ space this means an additional compression of 0.0416 (an 80.8% reduction), which corresponds to increasing the independent information volume by about 27.1 times under the approximation $\sigma \approx 1/\sqrt{N}$. Even in the AIC scenario matrix including subsets, the best case stays at $\Delta\text{AIC} = -1.29$, so the current 1-DoF ansatz for λ_H and the fixed channel set alone do not reach the support gate ($\geq +2$). Furthermore, even in the ansatz-extension matrix ($k = 1/2/4$), the best model is shared_lambda with $\Delta\text{AIC} = -1.44$, and even for $k = 1$ the theoretical upper bound is only +1.82 under the $\chi^2_{\text{fit}} \geq 0$ constraint. That means the support condition cannot be satisfied structurally with the present data. Detailed injection studies and candidate management for GW polarization and cross-domain extra constraints are therefore kept out of the main text here, leaving only the physical interpretation of the key metrics (ΔAIC , required $\Delta\chi^2$, required $|z|$).

Caution: this section does not provide a final decision on the higher-order term. It fixes the extended I/F for a simultaneous constraint across *EHT+GW+Pulsar+Fe-K α ISCO*. The final accept/reject decision will be revisited after updates to the first-strain similarity audit and the EHT systematics compression. Also, extrapolation to non-Bullet systems (such as El Gordo) and the registration of rerun candidates associated with future EHT data updates (input-hash monitoring, event counters, and so on) are all managed in the Part IV verification ledger (Registry) and rerun explicitly under the same I/F only when needed; they are not described in the main text here.

5 Differential Predictions (Falsifiability)

In this chapter, we will confirm the regions where the P-model and the reference frame can match, and then quantify the observables that show differences in the form of "required precision" and "rejection conditions". In other words, which experiment can be the deciding point in future data updates is determined by determining the threshold value (e.g. 3σ) and error budget (stat/sys).

5.1 EHT: Difference in the Shadow-Diameter Coefficient

Purpose: Using the ring diameter of EHT, quantify the difference in "shadow diameter coefficient" under β freezing as a differential prediction, the accuracy required for discrimination (the falsification condition) is fixed.

Input: EHT ring diameter (M87*, Sgr A*) and mass/distance constraints (Section 4.8). [25][29][34]

Process: Separate the ring/shadow mapping κ as an external system and perform the processing necessary to determine the difference in C (P-model vs. reference coefficient) by 3σ . Estimate the total uncertainty (contribution of θ_{unit} , ring σ , κ).

Equation: Ring/shadow mapping and angular scale Ring/shadow mapping: $\theta_{\text{ring}} = \kappa \theta_{\text{sh}}$

$$\theta_{\text{sh}} = C \frac{r_s}{D}, \quad r_s \equiv \frac{2GM}{c^2}$$

Coefficient (β fixed; $\beta=1$ for reference)

$$C_P = 2e\beta \quad (\beta = 1 \Rightarrow C_P = 2e \simeq 5.437), \quad C_{\text{ref}} = \sqrt{27} \simeq 5.196 \quad (18)$$

(If the $r_g \equiv GM/c^2$ standard is used within the implementation, C will be doubled, but the difference rate from C_P/C_{ref} will remain unchanged.)

Metric: Relative difference $|C_P - C_{\text{ref}}|/C_{\text{ref}}$ and total 1σ uncertainty required for 3σ discrimination (breakdown: θ_{unit} / ring σ / $\kappa\sigma$).

Result:

Item	Value	Remarks
Coefficient difference $ C_P - C_{\text{ref}} /C_{\text{ref}}$	4.8084%	Includes the $\mathcal{N}_0^{(2)}$ correction discussed in Section 4.17
Total 1σ required for 3σ discrimination	Approximately 1.6% or less	$\lesssim$ difference/3
rate-limiting term	M87*: θ_{unit} , Sgr A*: ring σ and κ systematics	The breakdown is fixed in the diagram.

The coefficient difference of 4.8084% used in this section is the final composite value obtained by adding the correction for the strong field nonlinear term determined in Section 4.17 (+0.1806%) to the baseline coefficient difference (approximately 4.63%).

The current κ systematic uncertainty is about $\sigma(\kappa) \approx 0.12$ (including the model-dependent lower bound; error budget in Section 4.8). This far exceeds the $\sigma(\kappa) \approx 0.011$ required to distinguish the coefficient difference of 4.8084% by 3σ .

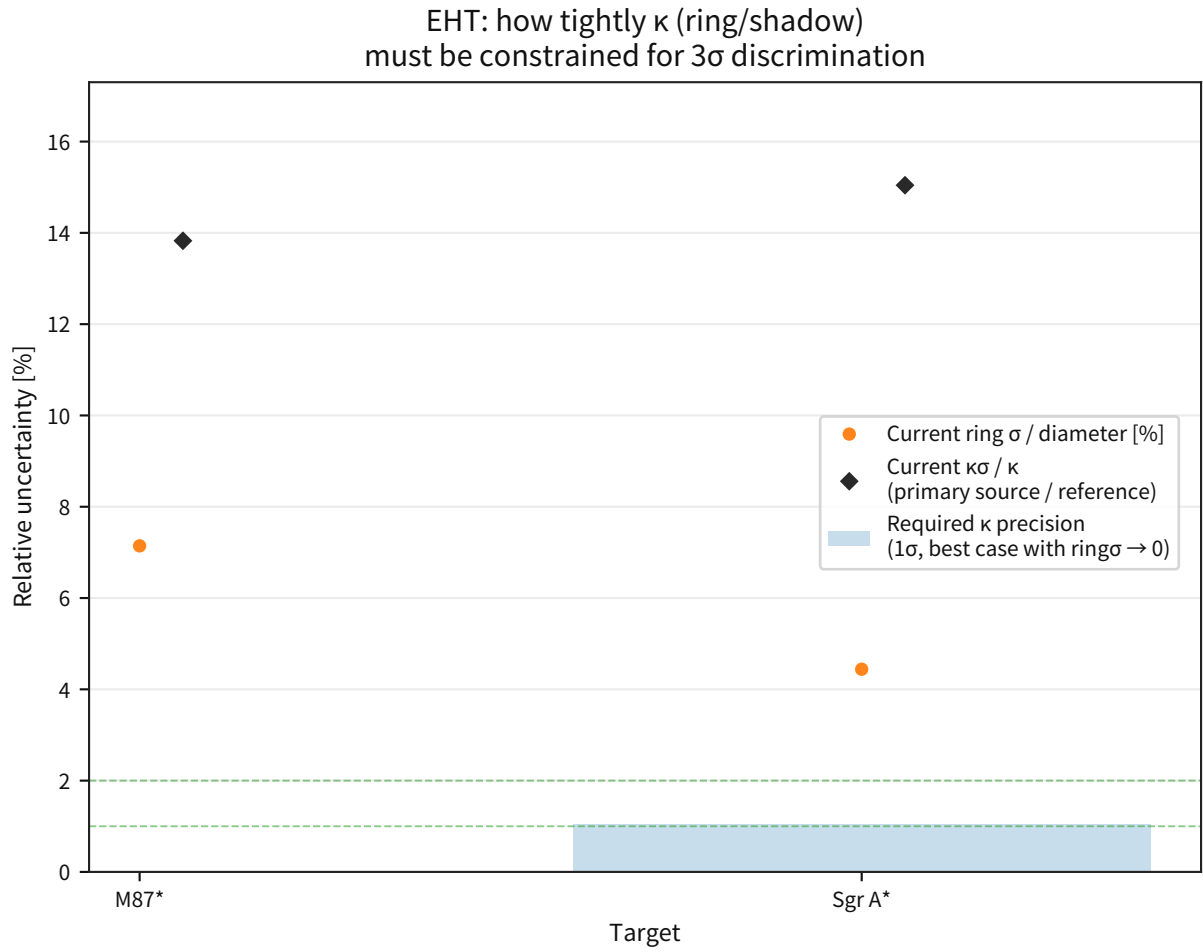


Figure 32: For M87*/Sgr A*, the current accuracy and "accuracy required for 3σ discrimination" are shown side by side.

Supplement: Visualize which uncertainty ($\theta_{\text{unit}}/\text{ring}/\kappa$) is the rate-limiting term.

EHT: κ precision tradeoff (3σ)

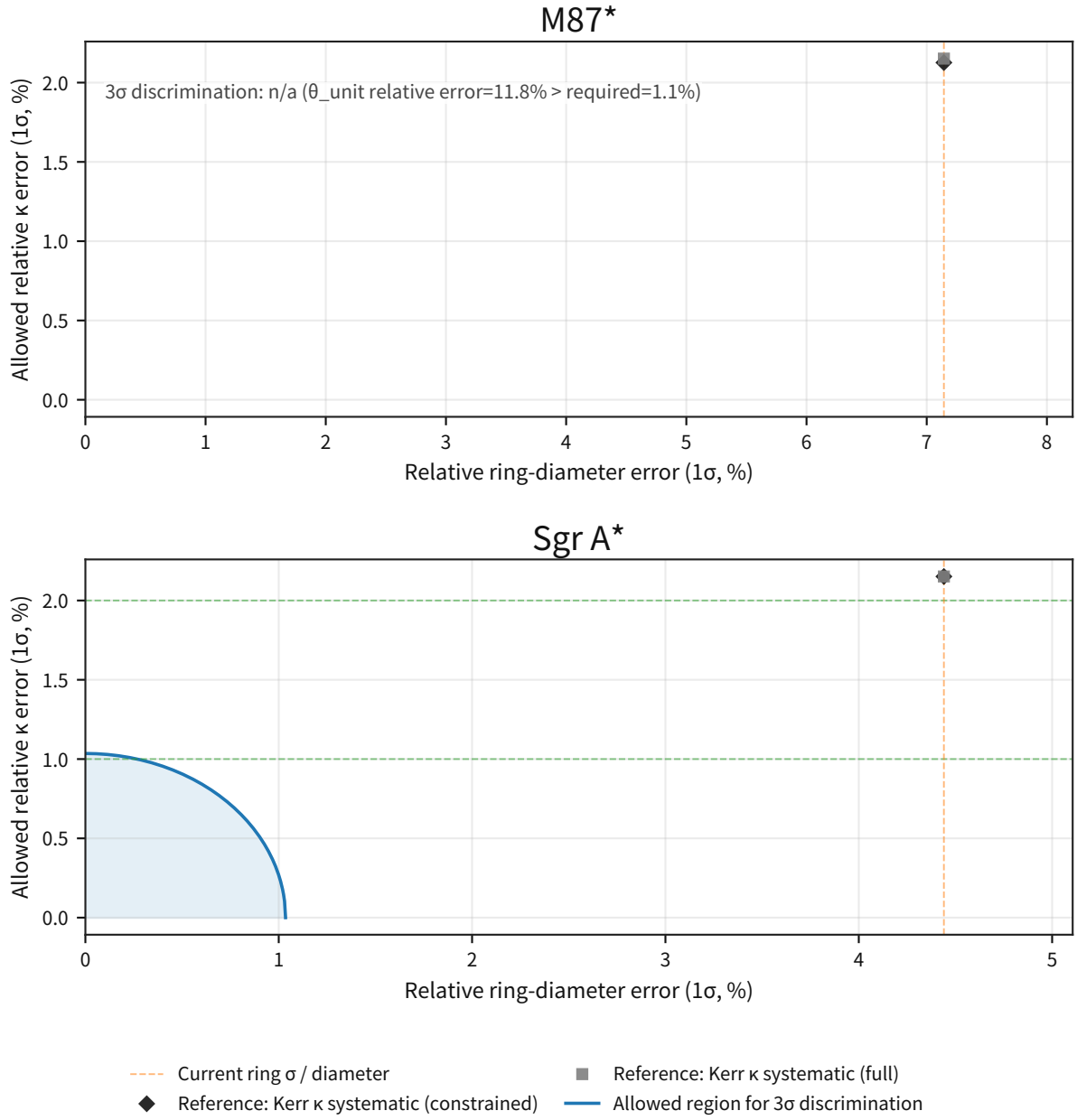


Figure 33: Indicates the trade-off between κ , ring σ , and θ_{unit} , and gives priority for improvement.

Supplement: Fix conditional branches such as “M/D is the rate-limiting term before packing κ ” as a diagram.

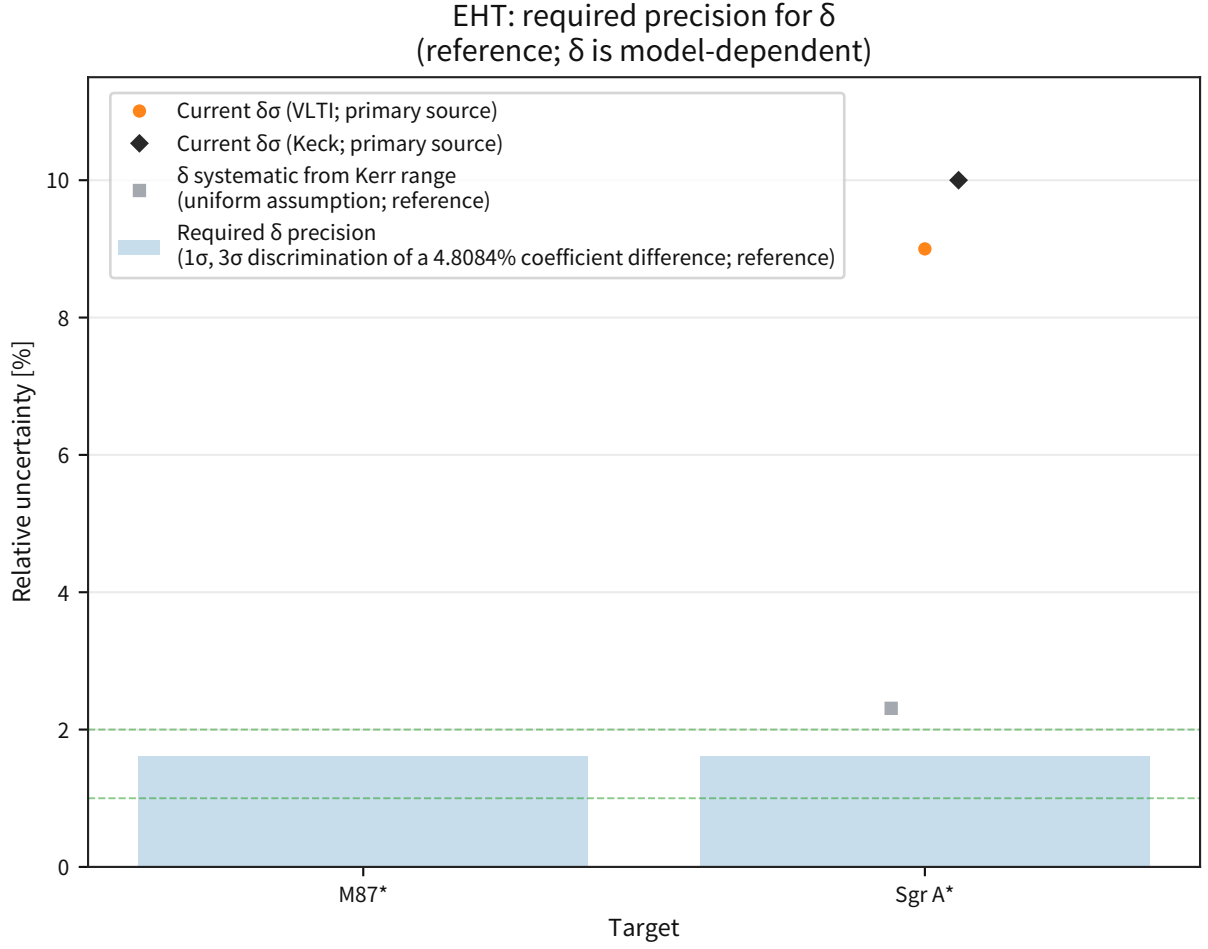


Figure 34: The coefficient difference of the differential prediction is converted to the required accuracy (1 σ) and shown.

Supplement: Similar to Section 5.2 (upper limit of high γ), quantify “with what precision it is rejected” .

Discussion: Difference detection is possible, but since κ and strong field geometry (rotation/tilt/definition difference) are currently the dominant systems, it is first necessary to fix the system ledger and cross-check.

Caution: C_{ref} is the coefficient of the reference frame (Schwarzschild approximation), and the rotation (spin/tilt) and “diameter definition” are treated as systematics (see the figures in Section 4.8).

5.2 Velocity Saturation δ : Upper Limit from High γ

Purpose: Fix the upper bound on the velocity-saturation parameter δ_0 (an extended hypothesis not included in the core mapping of Part I) required to remain consistent with known high- γ observations.

Input: Using a typical example of high γ (accelerator/cosmic ray/high energy ν), map to the upper limit of δ_0 using $\gamma \approx E/(mc^2)$ as a guide (Input is a list of published representative values; primary sources are summarized in the appendix).

Process: Calculate $\delta_0 \lesssim 1/\gamma_{\text{obs}}^2$ for observed γ_{obs} from the relationship of $\gamma_{\text{max}} \simeq 1/\sqrt{\delta_0}$ (example).

Equation: Saturation

$$\gamma_{\text{max}} \simeq \frac{1}{\sqrt{\delta_0}} \quad (19)$$

Upper limit (example)

$$\delta_0 \lesssim \frac{1}{\gamma_{\text{obs}}^2}$$

Metric: δ_0 upper limit (function of γ_{obs}) for each representative example.

Result:

Example	γ_{obs} (approximate)	δ_0 Upper limit (approximate)	remarks
LHC (proton 7 TeV)	7.46×10^3	$\lesssim 1.8 \times 10^{-8}$	
UHECR (Proton 10^{20} eV)	1.07×10^{11}	$\lesssim 8.8 \times 10^{-23}$	
High energy ν (10 PeV)	10^{17}	$\lesssim 10^{-34}$	$m = 0.1$ eV assumption (mass dependent)

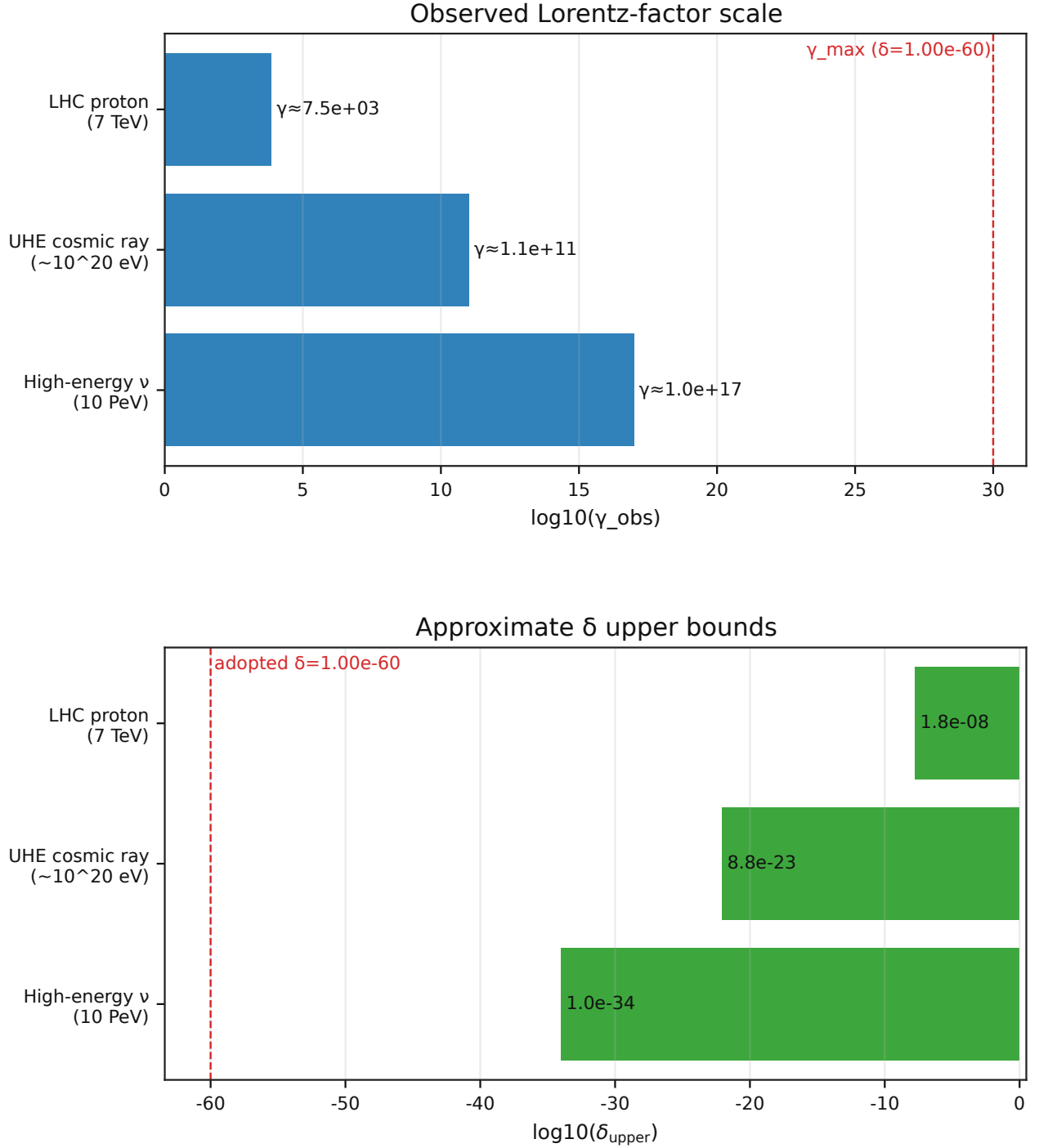


Figure 35: Compare the δ_0 upper limit derived from representative high γ observations.

Supplement: δ_0 is not a “measured value” at this stage, but is used for the purpose of fixing at least a range that does not contradict known observations.

Caution: δ_0 is a phenomenological ansatz, and this paper does not fit and align δ_0 afterwards. (Adoption will be decided based on whether primary data and threshold values that allow differences to be identified are available). Since the upper limit of ν strongly depends on the mass assumption, it is treated here as a “order guideline” .

5.3 Cosmology: Distance Duality (Prerequisite Audit: η and $\eta^{(P)}$)

The thermal history and primordial light-element synthesis of the early universe in the P-model are handled on the quantum side of the program. In Part III, the weak-interaction freeze-out branch together with the minimal He-4 chain is fixed only as a raw Watch/Pass entry after derivation-side error propagation. By contrast, the overall multi-isotope consistency across D/H, He-3/He-4, and Li-7 remains at Watch, and this section takes those boundary conditions as given. The present subsection therefore moves to the later universe and audits gravitational large-scale structure formation ($f\sigma_8$) together with light-propagation observables such as BAO.

5.3.1 Direct Comparison of Pantheon+ $\mu(z)$ and Cosmic Chronometers $H(z)$

Purpose: As a basic requirement for an alternative cosmology, fix a common comparison interface between the Pantheon+ distance modulus $\mu(z)$ and the Cosmic Chronometers measurement $H(z)$, and record both χ^2 and ΔAIC (sign convention: $AIC_{\text{baseline}} - AIC_{\text{P-model}}$).

Input: Pantheon+ distance table (redshift, distance modulus, error) and Cosmic Chronometers public table (z , $H(z)$, σ_H).

Process: Evaluate flat Λ CDM and the P-model in parallel, using the complete H_{eff}^2 expression on the P-model side. Fix H_0 by linear least squares on the $H(z)$ side and Δ_μ by linear least squares on the supernova side. For the baseline, minimize χ_{total}^2 on a one-dimensional grid in Ω_m . For the P-model, search the minimum with bounded L-BFGS-B optimization over $(\alpha_r, \alpha_m, \alpha_\Lambda)$. Implementation details and fixed outputs follow the public snapshot/audit ledger in Part IV.

Equation:

$$H_{\Lambda\text{CDM}}(z) = H_0 \sqrt{\Omega_m(1+z)^3 + 1 - \Omega_m}$$

$$D_L^{(\Lambda\text{CDM})}(z) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + 1 - \Omega_m}}$$

$$H_P(z) = H_0 E_P(z), \quad E_P^2(z) = 1 + \alpha_r [(1+z)^4 - 1] + \alpha_m [(1+z)^3 - 1] - \alpha_\Lambda \ln(1+z)$$

$$D_L^{(P)}(z) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{E_P(z')}$$

$$\alpha_r \equiv \frac{\Omega_r^2}{2H_0^2}, \quad \alpha_m \equiv \frac{2\Omega_m^2}{3H_0^2}, \quad \alpha_\Lambda \equiv \frac{2\Omega_\Lambda^2}{H_0^2}$$

$$\mu_{\text{obs}}(z) \approx \mu_{\text{model}}(z) + \Delta\mu, \quad \Delta AIC = AIC_{\text{baseline}} - AIC_{\text{P-model}}$$

Metric: n_{SN} , n_{CC} , best parameters (Ω_m or $\alpha_r, \alpha_m, \alpha_\Lambda$, H_0 , Δ_μ), χ_{SN}^2 , χ_{CC}^2 , χ_{total}^2 , AIC , ΔAIC .

Result:

model	n_{SN}	n_{CC}	shape parameter	H_0	Δ_μ	χ_{SN}^2	χ_{CC}^2	χ_{total}^2	AIC
flat Λ CDM	1588	32	$\Omega_m =$ 0.3693	65.98	-0.206	709.88	15.36	725.25	731.25
P-model (complete formula)	1588	32	$\alpha_r = \alpha_\Lambda =$ 0, $\alpha_m =$ 0.3694	65.98	-0.206	709.88	15.37	725.25	735.25

$\Delta\text{AIC} = \text{AIC}_{\text{baseline}} - \text{AIC}_{\text{P-model}} = -4.00$ (baseline advantage due to negative value). Pass condition (P-model superiority in $\Delta\text{AIC} (= \text{AIC}_{\text{baseline}} - \text{AIC}_{\text{P-model}}) > 2$) is not satisfied.

Direct comparison between Pantheon+ $\mu(z)$ and Cosmic Chronometers $H(z)$ (Vector PDF original). Note: PNG for compatibility display also exists under the same name.

Discussion: In the complete-equation refit, the best solution converges to $\alpha_r \rightarrow 0$, $\alpha_\Lambda \rightarrow 0$, and $\alpha_m \approx 0.369$. Within the current observation band ($z \lesssim 2$), the fit is therefore effectively reduced to a matter-dominated form close to ΛCDM . The χ^2 value improves to the same level as the baseline, but that gain is not enough to overcome the AIC penalty associated with the additional degrees of freedom ($k = 5$), so the cosmology judgment remains Watch / Reject rather than Pass.

Caution: This comparison uses the diagonal error (MU_SH0ES_ERR_DIAG) of Pantheon+ and is fixed at the first stage. The strict version that uses full covariance will be subject to the next stage of auditing.

5.3.2 Full Time-Evolution Equation for the Background Field $P_{\text{bg}}(t)$

Purpose: Build a single I/F for the complete time evolution of $P_{\text{bg}}(t)$ that includes not only the radiation-dominant branch but also the matter-dominant branch and the late-time neighborhood, so that it can be linked directly to the distance / $H(z)$ comparison in Section 5.3.

Input: Background value $P_{\text{bg}}(t_0)$ at reference time t_0 , effective density $\rho_{r,0}, \rho_{m,0}, \rho_{\Lambda,0}$, and $u_{\text{bg}}(t) \equiv \ln(P_{\text{bg}}(t)/P_{\text{bg}}(t_0))$.

Process: Use the continuous-component relation

$$\dot{\rho}_i + 3(1 + w_i)H_{\text{eff}}\rho_i = 0, \quad H_{\text{eff}} \equiv -\dot{u}_{\text{bg}}$$

to rewrite each component as $\rho_i(u_{\text{bg}})$ and thereby close the background equation as a one-dimensional ODE.

Equation:

$$\ddot{u}_{\text{bg}} = \Omega_r^2 e^{4u_{\text{bg}}} + \Omega_m^2 e^{3u_{\text{bg}}} - \Omega_\Lambda^2,$$

$$\Omega_r^2 \equiv 8\pi G\rho_{r,0}, \quad \Omega_m^2 \equiv 4\pi G\rho_{m,0}, \quad \Omega_\Lambda^2 \equiv 8\pi G\rho_{\Lambda,0}.$$

First integral:

$$\dot{u}_{\text{bg}}^2 = H_0^2 + \frac{\Omega_r^2}{2}(e^{4u_{\text{bg}}} - 1) + \frac{2\Omega_m^2}{3}(e^{3u_{\text{bg}}} - 1) - 2\Omega_\Lambda^2 u_{\text{bg}}.$$

Four factorization (complete solution):

$$t - t_0 = \int_0^{u_{\text{bg}}(t)} \frac{d\tilde{u}}{\sqrt{H_0^2 + \frac{\Omega_r^2}{2}(e^{4\tilde{u}}} - 1) + \frac{2\Omega_m^2}{3}(e^{3\tilde{u}}} - 1) - 2\Omega_\Lambda^2 \tilde{u}}.$$

Observation map ($1 + z = e^{u_{\text{bg}}}$):

$$H_{\text{eff}}^2(z) = H_0^2 + \frac{\Omega_r^2}{2}[(1+z)^4 - 1] + \frac{2\Omega_m^2}{3}[(1+z)^3 - 1] - 2\Omega_\Lambda^2 \ln(1+z).$$

Single component closed form ($w > -1$):

$$P_{\text{bg}}^{(w)}(t) = P_* \left[1 + \frac{3}{2}(1+w)H_*(t-t_*) \right]^{-\frac{2}{3(1+w)}}, \quad q(w) = \frac{2}{3(1+w)}.$$

Metric: $q_r = 1/2$, $q_m = 2/3$, and $F(z) \equiv H_{\text{eff}}^2(z) \geq 0$ (redshift region to be analyzed) Fixed as an operational gate.

Result:

branch	formula	Fixed index/condition
radiation dominant	$P_{\text{bg}} \propto t^{-1/2}$	$q_r = 1/2$
material dominance	$P_{\text{bg}} \propto t^{-2/3}$	$q_m = 2/3$
late-time neighborhood	$u_{\text{bg}} \approx -H_\Lambda \Delta t - \frac{\Omega_\Lambda^2}{2} \Delta t^2$	C^1 continuous at connection point
full time	Four-branch decomposition of the above ODE	Evaluate $H_{\text{eff}}(z)$ on the same I/F

Discussion: Re-running the direct fit of Section 5.3 with the complete expression shows that, within the current observation band, α_r and α_Λ stay close to zero, while the contracted branch in which only α_m remains active provides the best fit. At this stage, the priority is therefore not to add more terms, but to improve the simultaneous fit including the high- z band (CMB / BAO first-order statistics) so that the discriminability of the $\ln(1+z)$ term is fixed on the same interface.

Caution: This section fixes not only the definition of the background equation but also the direct Pantheon+ / CC comparison. However, the full-covariance SN constraints and the simultaneous CMB / BAO constraints are not yet integrated. The final judgment therefore remains under continuous audit (Watch / Reject).

Purpose: Treat distance duality (DDR) not as an immediate rejection/acceptance statement, but as a **prerequisite audit** that asks which $(1+z)$ factor is embedded in the distance-estimation I/F, and fix the indicator $\eta^{(P)}$ and the falsification condition (required accuracy) to be used on the P-model side.

Input: A list that integrates the representative constraints of DDR (e.g. ϵ_0 of SNIa+BAO) and publication constraints of distance-indicator categories (cluster/strong lens, etc.). Here, it is treated as a reinterpretation of publication constraints rather than a “reanalysis of primary data.” [50]

Process:

- Adopt the publication constraint format $D_L = (1+z)^{(2+\epsilon_0)} D_A$ and read ϵ_0 as a common parameter for comparison.
- Under the same published constraints, use the P-model-specific indicator

$$\eta^{(P)}(z) \equiv \frac{D_L}{(1+z) D_A} \quad (20)$$

i.e. the quantity that divides out only the flux-side factor $(1+z)$ from η , and evaluate it on the reference scale $z_{\text{ref}} = 1$.

- The representative constraints are (i) the minimum σ (tightest) when uses-BAO=true, and (ii) the minimum $|z|$ (least rejecting) when uses-BAO=false.

- Falsification conditions (including rejection thresholds and application conditions) are fixed in Part IV (verification materials) (GitHub: [EnterOgawa/p-model](#), snapshot policy: base [v0.1.2](#) + strong-field bridge [v0.1.3](#)). This section treats as a prerequisite audit the requirement that the distance-estimation I/F must not embed the FRW expansion assumption. [1]

Equation:

- Distance duality (standard indicator):

$$\eta(z) \equiv \frac{D_L}{(1+z)^2 D_A} \quad (21)$$

- Parameterization of publication constraints:

$$D_L(z) = (1+z)^{2+\epsilon_0} D_A(z) \quad (\eta = (1+z)^{\epsilon_0})$$

- Publication-constraint format on the same P-model-specific indicator:

$$\eta^{(P)}(z) \equiv \frac{D_L}{(1+z) D_A} = (1+z)\eta(z) = (1+z)^{1+\epsilon_0} \quad (22)$$

- Model predictions (compared with the same $\eta^{(P)}$ metric):
- FRW+photon conservation: $\eta = 1$ ($\epsilon_0 = 0$) so $\eta^{(P)} = (1+z)$ (exponent $p = 1$).
- P-model minimum (static geometry + background P redshift + time stretching derived from Part I 2.5.1 $p_t = 1$; in the publication-constraint " ϵ_0 format," this corresponds to $\epsilon_0 = -1$):

$$D_L^{(P)} = (1+z) D_A^{(P)} \quad (23)$$

Therefore, $\eta^{(P)} = 1$ (exponent $p = 0$), which corresponds to $\epsilon_0 = -1$ in the publication-constraint format.

- Internal consistency (independent approximation):

$$z_{ij} = \frac{\epsilon_i - \epsilon_j}{\sqrt{\sigma_i^2 + \sigma_j^2}}$$

Metrics: ϵ_0 (publication constraint), $p \equiv 1 + \epsilon_0$ ($\eta^{(P)} = (1+z)^p$), $\eta^{(P)}(z_{\text{ref}} = 1)$, and z-score (FRW: $p = 1$ vs P-model minimum: $p = 0$).

Result:

Publication constraints	uses BAO	ϵ_0 (1σ)	$p = 1 + \epsilon_0$ (1σ)	$\eta^{(P)}(1)$	z_{FRW}	$z_{\text{P,min}}$
SN Ia + BAO (2021)	1	0.013±0.029	1.013±0.029	2.018±0.041	-0.45	-25.10
BAO + SN Ia (2013)	1	-0.159±0.097	0.841±0.097	1.791±0.120	+1.73	-6.57
SN Ia + H(z) (2010)	0	-0.040±0.037	0.960±0.037	1.945±0.051	+1.08	-18.70
JLA + H(z) (2015)	0	0.081±0.166	1.081±0.166	2.115±0.243	-0.47	-4.58
Union2.1 + H(z) (2014)	0	0.025±0.079	1.025±0.079	2.034±0.112	-0.31	-9.24
H(z) + JLA (2017, no H0 prior)	0	0.018±0.084	1.018±0.084	2.025±0.118	-0.21	-8.69
Clusters + SN Ia (2012)	0	-0.474±0.433	0.526±0.433	1.440±0.433	+1.29	-1.02

Continued on next page

Publication constraints	uses BAO	ϵ_0 (1σ)	$p = 1 + \epsilon_0$ (1σ)	$\eta^{(P)}(1)$	z_{FRW}	$z_{\text{P,min}}$
Clusters + SNIa (2012, 0 non-iso)		-0.713 $\pm$ 0.263	0.287 $\pm$ 0.263	1.220 $\pm$ 0.222	+3.51	-0.99
Clusters + SNIa (2011) 0		-0.184 $\pm$ 0.282	0.816 $\pm$ 0.282	1.760 $\pm$ 0.344	+0.70	-2.21
Clusters + SNIa (2011, 0 Bonamente)		-0.415 $\pm$ 0.194	0.585 $\pm$ 0.194	1.500 $\pm$ 0.201	+2.49	-2.49
Clusters + SNIa (2011, 0 $\eta_0/(1+z)$)		-0.144 $\pm$ 0.192	0.856 $\pm$ 0.192	1.810 $\pm$ 0.241	+0.79	-3.36
Clusters + SNIa (2011, 0 Bonamente, $\eta_0/(1+z)$)		-0.313 $\pm$ 0.135	0.687 $\pm$ 0.135	1.610 $\pm$ 0.150	+2.59	-4.05
SGL + SNIa + GRB (2016, PLAW)	0	-0.160 $\pm$ 0.188	0.840 $\pm$ 0.188	1.790 $\pm$ 0.233	+0.90	-3.40
SGL + SNIa + GRB (2016, SIS)	0	0.270 $\pm$ 0.150	1.270 $\pm$ 0.150	2.412 $\pm$ 0.251	-1.64	-5.63
SNe + radio (2017, $\eta_0 z$)	0	0.089 $\pm$ 0.077	1.089 $\pm$ 0.077	2.128 $\pm$ 0.113	-1.13	-9.95
SNe + radio (2017, $\eta_0 z/(1+z)$)	0	0.136 $\pm$ 0.127	1.136 $\pm$ 0.127	2.198 $\pm$ 0.194	-1.02	-6.18

We perform a side-by-side comparison of the constraints on the $\eta^{(P)}$ index for (i) $\eta^{(P)} = 1$ (P-model minimum) and (ii) $\eta^{(P)} = (1+z)$ (FRW + photon preservation), and show that the representative SNIa+BAO example (2021), with $p = 1.013 \pm 0.029$ ($\eta^{(P)} = 2.018 \pm 0.041$ at $z = 1$), is significantly different from $p = 0$.

Supplement: In the representative uses-BAO=true example, $p \approx 1$ is strongly supported, while $p = 0$ is significantly excluded on the $z = 1$ scale. However, since this depends on "which distance estimation assumptions are included in the publication constraints," an I/F audit is required in the next step.

This is a diagnostic diagram that color-codes 16 published constraints (ϵ_0) according to BAO inclusion/non-inclusion, and compares the distance from both standards $\epsilon_0 = -1$ (P-model minimum mapping) and $\epsilon_0 = 0$ (FRW).

Supplement: Large tension such as "35 σ " is treated as a diagnosis of premise dependence that appears when the "expansion side" ($1+z$) is embedded in the distance-estimation I/F. (The fixed reference in the figure is the public snapshot ledger in Part IV.)

In addition to the 16 published constraints, we fix a cross-sectional audit of where the "expansion-side ($1+z$)" factor can enter the distance-estimation I/F, including the independence of the SNe Ia standardization process (SALT-family fit). Here, the "expansion-side ($1+z$)" corresponds to the FRW relation $D_A = D_M/(1+z)$ and must be distinguished from the flux-side ($1+z$) that comes from photon energy and arrival rate.

Publication constraints	Main configuration of I/F	Embedding location on expansion side ($1+z$) (audit point)	Shortest time to return to primary quantity
SNe Ia standardization (cross-sectional audit)	SALT2/SALT3 Light Curve Fit (m_B,x_1,c,M_B)	Standard cosmological assumptions and empirical rules may be implicitly mixed into the determination of color correction, width correction, and absolute magnitude anchor (including host mass step).	Re-standardize from primary photometry and fix correction term and anchor determination in a separate layer as "cosmology-independent I/F"

Continued on next page

Publication constraints	Main configuration of I/F	Embedding location on expansion side (1+z) (audit point)	Shortest time to return to primary quantity
SNIa+BAO (2021)	SNe Ia + BAO compression distance	$D_A(z) = D_M(z)/(1+z)$ can be implicit on the BAO side. In addition, procedural dependencies of fiducial/recon/cov remain.	BAO treats $D_M(z)/r_d$, $D_H(z)/r_d$ as primary and makes $D_A(z)$ transformation explicit in a separate layer
BAO+SNIa (2013)	SNe Ia + BAO compression distance	Same as above (the definition on the D_A side remains even in the opacity interpretation)	Same as above
SNIa+H(z) (2010)	SNe Ia + cosmic chronometer $H(z)$	An expansion assumption is entered in the definition stage of $H(z) = -(1/(1+z))dz/dt$. Distance reconstruction can rely on FRW integrals	Return $H(z)$ input to $\Delta t, \Delta z$ and separate where (1+z) enters
JLA+H(z) (2015)	SNe Ia + $H(z)$	Same as above	Same as above
Union2.1+H(z) (2014)	SNe Ia + $H(z)$	Same as above	Same as above
H(z)+JLA (2017, no H0 prior)	SNe Ia + $H(z)$	Same as above	Same as above
Clusters+SNIa (2012)	SZ+X-ray D_A + SNe Ia D_L	(1+z) may be included in X-ray surface brightness and SZ calibration, and geometric model selection may dominate	Decompose the (1+z) factor of surface brightness scaling and check consistency with independent constraints such as $T(z)$
Clusters+SNIa (2012, non-iso)	Same as above	Same as above (differences in geometric models directly enter the results)	Same as above
Clusters+SNIa (2011)	Same as above	Same as above	Same as above
Clusters+SNIa (2011, Bonamente)	Same as above	Same as above	Same as above
Clusters+SNIa (2011, $\eta_0/(1+z)$)	Same as above	Same as above (parameterization difference is difference in expression)	Same as above
Clusters+SNIa (2011, Bonamente, $\eta_0/(1+z)$)	Same as above	Same as above	Same as above
SGL+SNIa+GRB (2016, PLAW)	Strong lens distance ratio + SNe Ia + GRB extrapolation	Geometric assumptions (FRW) such as distance sum rule can be fixed. $(1+z_l)$ is entered for time-delay	Make the three distance parameters D_d, D_s, D_{ds} independent and clearly indicate whether or not the sum rule is fixed.
SGL+SNIa+GRB (2016, SIS)	Same as above	Same as above (the lens mass model systems also overlap)	Same as above
SNe+radio (2017, $\eta_0 z$)	SNe Ia + Ultra-compact radio source (angular diameter distance)	Distance assumptions can be included in standard ruler and calibration (population evolution/size model)	Specify the estimation formula for D_A (e.g. l/θ) and calibration I/F, and fix only the z mapping

Continued on next page

Publication constraints	Main configuration of I/F	Embedding location on expansion side (1+z) (audit point)	Shortest time to return to primary quantity
SNe+radio (2017, $\eta_0 z/(1+z)$)	Same as above	Same as above (parameterization difference is difference in expression)	Same as above

Discussion: $D_L = (1+z)^2 D_A$ of FRW includes (a) $(1+z)$ of flux attenuation (photon energy x arrival rate); (b) $(1+z)$ of $D_A = D_M/(1+z)$ is included due to spatial expansion (physical scale of radiation time). The P-model minimum maintains (a) but does not adopt (b), so it becomes the following formula.

$$D_L^{(P)} = (1+z) D_A^{(P)}$$

Therefore, DDR is based on (b), unless distance estimation is established (estimation I/F audit). It cannot be treated as a “physical rejection” and is positioned here as the entry point for a prerequisite audit (where discrepancies occur).

5.3.3 $(1+z)$ hand-calculated trace of DDR audit (SNe Ia SALT2 / BAO)

Purpose: Expand the distance-estimation chains for SNe Ia (SALT2) and BAO line by line, and identify in a machine-determinable manner the stage at which the implicit relation $D_A = D_M/(1+z)$ is injected.

Input: SALT2 standardized formula; the published BAO distance ratios; the derived distances $D_A(z)$ and $D_V(z)$; and the DDR definition η .

Process:

- The SNe Ia side expands the chain of $m_B^*, x_1, c \rightarrow \mu \rightarrow D_L$ and classifies the place where $(1+z)$ enters.
- The BAO side expands the chain mapping from $D_M(z)/r_d$, $D_H(z)/r_d$ to published values $D_A(z)/D_V(z)$.
- For each row, audit whether $(1+z)$ enters on (i) the flux/time-stretching side or (ii) the geometric side through $D_A = D_M/(1+z)$.
- Substitute into the DDR combination expression $\eta = D_L/[(1+z)^2 D_A]$ and identify the injection position of the geometric side $(1+z)$.

For the P-model-specific indicator $\eta^{(P)}(z) \equiv D_L/[(1+z) D_A]$ used in this section, evaluate errors on the reference scale $z_{\text{ref}} = 1$ after dividing out only the flux-side $(1+z)$ factor.

Equation:

$$\mu_{\text{SN}} = m_B^* - M_B + \alpha x_1 - \beta c + \Delta_M + \Delta_B, \quad D_L = 10^{(\mu_{\text{SN}} - 25)/5} \text{ Mpc}$$

$$\alpha_{\perp} = \frac{D_M(z)/r_d}{D_{M,\text{fid}}(z)/r_{d,\text{fid}}}, \quad \alpha_{\parallel} = \frac{D_H(z)/r_d}{D_{H,\text{fid}}(z)/r_{d,\text{fid}}}$$

$$D_A = \frac{D_M}{1+z}, \quad D_V = \left[(1+z)^2 D_A^2 \frac{cz}{H(z)} \right]^{1/3} = \left[D_M^2 \frac{cz}{H(z)} \right]^{1/3}$$

$$\eta(z) = \frac{D_L(z)}{(1+z)^2 D_A(z)}, \quad D_A(z) = \frac{D_M(z)}{1+z} \Rightarrow \eta(z) = \frac{D_L(z)}{(1+z) D_M(z)}$$

1 case full trace (SNIa+BAO 2021; [50]): The primary quantities of BAO are the angular width and redshift width, and the lateral and line-of-sight directions are

$$\Delta\theta = \frac{r_d}{D_M(z)}, \quad \Delta z = \frac{r_d}{D_H(z)}, \quad D_H(z) = \frac{c}{H(z)}$$

(published in the functional form of $D_M(z)/r_d$, $D_H(z)/r_d$). In BOSS DR12

$$D_M(z) = (1+z)D_A(z)$$

is adopted as the definition of traversal distance (Introduction definition formula; [53]). Based on this, Martinelli+2021 is Eq.(12)

$$d_A(z) = \frac{c}{1+z} \int_0^z \frac{dz'}{H(z')}$$

is adopted and evaluated by combining $\eta = d_L/[d_A(1+z)^2]$ in Eq. (4) with SNe Ia (D_L obtained from SALT2). Therefore, when $d_A(z) = d_M(z)/(1+z)$ ($D_A(z) = D_M(z)/(1+z)$) is assigned to the equivalent value,

$$\eta(z) = \frac{D_L(z)}{(1+z)^2 D_A(z)} = \frac{D_L(z)}{(1+z) D_M(z)}$$

Therefore, the injection point of $(1+z)$ on the geometry side can be specified at the expression level. This is the "pointing point" of this audit, and DDR claims that do not go through the estimated I/F audit will be affected by the inclusion of assumptions.

Metric: rows_with_1pz, rows_geometry_1pz, geometry_injection_detected, and injection row list.

Result:

audit chain	Representative line	judgement
SNe Ia(SALT2)	$t_{\text{rest}} = t_{\text{obs}}/(1+z)$	$(1+z)$ is detected on the time-stretching side, not on the geometric side
BAO distance mapping	$D_A(z) = D_M(z)/(1+z)$	Detected as injection point on geometric side $(1+z)$
BAO compression distance	$D_V(z) = \left[(1+z)^2 D_A(z)^2 cz/H(z) \right]^{1/3}$	Inherit the same geometric side transformation
DDR combination expression	Assignment of D_A to $\eta = D_L/[(1+z)^2 D_A]$	Specify that one $(1+z)$ on the geometric side is injected

The audit hard gate (geometry_side_1pz_injection_trace_complete) passes and fixes the stage where $D_A(z) = D_M(z)/(1+z)$ enters the distance-estimation I/F with a line number.

Discussion: This manual calculation trace allows us to avoid immediately interpreting DDR mis-

matches as physical rejections. The operational rule is therefore fixed: first separate and evaluate the stage of the distance definition at which the geometric-side $(1+z)$ factor is used.

Caution:

- $\eta^{(P)}$ has meaning as a falsification indicator on the P-model side when the “distance estimation I/F that does not automatically assume the FRW expansion factor” is set up.

If that audit is omitted, the conclusion is likely to be recovered to the distance definition (premises).

- This section has a lot of content, so the details will be organized as follows: ((1+z) hand-calculated trace), (BAO first-order statistics), (distance-indicator and independent verification), (distance-indicator dependence and BAO assumption).

The falsification condition pack ($\eta^{(P)}$) adopts the following as the main points of fixed JSON (3σ standard).

```
{
  "definition": "eta_P(z) = DL / ((1+z) DA) = (1+z)^(1+epsilon0)",
  "model_predictions": {
    "FRW_photon_conservation": "eta_P=(1+z) (p=1)",
    "P_model_minimal": "eta_P=1 (p=0)"
  },
  "operational_rule": [
    "Evaluate eta_P after auditing that the distance estimation I/F does not embed expansion assumptions such as 'DA=DM/(1+z)'",
    "If the deviation from eta_P=1 consistently exceeds 3sigma in multiple datasets with the same I/F fixed, the P-model minimum (eta_P=1) is rejected."
  ],
  "precision_targets_examples": {
    "sigma_p_required_for_3sigma_separation": 0.33,
    "sigma_eta_required_for_3sigma_separation_at_z_ref_1": 0.33
  }
}
```

5.3.4 Detailed Audit of BAO Primary Statistics

Purpose: BAO’s compressed output (e.g. $D_M(z)/r_d$, $D_H(z)/r_d$) can be driven by assumptions (fiducial/reconstruction/covariance/broadband degrees of freedom). Therefore, we fix the “entrance” for re-deriving the AP warping index $\hat{\epsilon}$ from the primary statistics (ξ_ℓ/P_ℓ). For the P-model distance mapping (pbg/lcdm), record how much $\hat{\epsilon}$ moves with the procedure knob (sys) and connect it to DDR reconnection (5.3.3).

Input:

- BOSS DR12: post-recon ξ_0/ξ_2 public package (Ross et al.) [54]
- BOSS DR12: pre-recon ξ_0/ξ_2 public package (Satpathy et al.) [55]
- BOSS DR12v5 LSS: Galaxy+random (pre-recon; recalculate $\xi(s, \mu)$ and project to ξ_ℓ)
- DESI DR1: Self-generate $dv = [\xi_0, \xi_2] + \text{cov}$ from LSS / VAC ($\text{Ly}\alpha$) (promotion check)

Process:

- ξ_ℓ peakfit (minimal model): $r = 50\text{--}150$ (Mpc/h), $\mu\text{-bin}=80$, smooth basis 1, $1/r$, $1/r^2$ + Gaussian peak (template centered $r_0 = 105$, $\sigma = 10$ (Mpc/h)).
- The fit estimates α (isotropic scale) and $\hat{\epsilon}$ (AP warping) and compares (i) $\hat{\epsilon}$ free, (ii) $\hat{\epsilon} = 0$, (iii) $\hat{\epsilon}$ fixed to P_{bg} prediction (calculated from F_{AP} ratio; r_d canceled).
- BOSS will check the consistency of the entrance using screening (Stage A; diag error), and DESI will perform promotion (Stage B) based on stability using dv+cov.
- Detailed artifacts and reproduction leads refer to the public snapshot ledger in Part IV.

Equation:

$$\xi_\ell(s) = \frac{2\ell+1}{2} \int_{-1}^1 \xi(s, \mu) P_\ell(\mu) d\mu$$

$$\alpha = (\alpha_\perp^2 \alpha_\parallel)^{1/3}, \quad \varepsilon = (\alpha_\parallel / \alpha_\perp)^{1/3} - 1$$

Metric: $\hat{\epsilon}$ and CI (screening), and promotion condition ($|z_{\text{score,combined}}| \geq 3$ without sign reversal and $\text{tracer} \geq 2$).

Result:

dataset	Results (representative)	positioning
BOSS DR12 (Ross post-recon; diag)	$z = 0.38 : \hat{\epsilon} = +0.008$, $z = 0.51 : \hat{\epsilon} = -0.002$, $z = 0.61 : \hat{\epsilon} = -0.010$	screening (Stage A)
DESI DR1 (pbg; promotion check)	LRG3+ELG1: $z_{\text{score,combined}} \approx -11.13 \cdots + 0.70$ (with sign inversion), Ly α QSO: $\approx +0.68 \cdots + 1.35$ ($ z < 3$)	Promotion not achieved with passing tracer 0/2 (Stage B is Watch)
BOSS+DESI cross-integration	Stage A (BOSS): $\max z = 1.063$ (pass), Stage B (DESI): promoted=false / passing=0	overall=Watch (no need to update the total number of Part IV judgment table)

Apply minimum peakfit to BOSS DR12 (post-recon) ξ_0/ξ_2 and extract AP warping $\hat{\epsilon}$.

Supplement: In this article, the starting point is fixed as the way $\hat{\epsilon}$ is derived from the primary statistics, not as a compressed-output comparison.

Recalculate $\xi(s, \mu) \rightarrow \xi_\ell$ from BOSS DR12v5 catalogs (galaxy+random) and compare it with $\hat{\epsilon}$ derived from public multipoles.

Supplement: Visualize the movement range of $\hat{\epsilon}$ as sys during reconstruction/coordinate conversion (pbg/lcdm).

Aggregate $z_score_combined$ of DESI DR1 using covariance estimation method (jack-knife/RascalC/VAC) and λ -shrinkage sweep.

Caution: The current snapshot does not meet the confirmation (Stage B) promotion condition of “2 or more tracers that satisfy $|z| \geq 3$ without inverting the sign.”

Integrate BOSS (Stage A) and DESI (Stage B) into the same judgment I/F, and audit whether the overall judgment and Part IV judgment table need to be updated.

Supplement: overall=watch, part4_scoreboard_update.recommend_update=false are fixed, and the total number of judgment tables is unchanged this turn. (The enumeration of figures, JSON/CSV, and reproduction commands is based on the public ledger in Part IV.)

Caution: The detailed log below (recon gap, weight/random specification sensitivity, setting sensitivity, additional figures) will be retained as an internal reference, and the main points of the main text of the paper will be fixed in the bullet points above.

5.3.5 Cosmology I: distance-independent probes

Purpose: Confirm through primary observations independent of the distance indicator that DDR tension cannot be absorbed solely by the time-dilation index p_t or by the photon-energy side (CMB temperature scaling).

Input:

- SNe Ia time dilation (p_t): Blondin et al. 2008 [51]
- CMB temperature scaling (β_T): Avgoustidis et al. 2011 [52]
- AP (F_{AP}): BOSS DR12 consensus (D_M , H , correlation) [53]

Equation:

$$\Delta t_{\text{obs}} = (1+z)^{p_t} \Delta t_{\text{em}}$$

$$T(z) = T_0(1+z)^{1-\beta_T}$$

$$F_{AP}(z) \equiv \frac{D_M(z) H(z)}{c}$$

Result:

theme	Primary constraint (representative)	implication
SNe Ia time dilation (p_t)	0.97 ± 0.10 (1σ)	Consistent with the P-model minimum prediction ($p_t = 1$) derived in Part I 2.5.1. Strongly excludes $p_t = 0$ (no time dilation).
CMB temperature scaling (β_T)	0.004 ± 0.016 (1σ ; up to $z \approx 3$)	Consistent with the P-model minimum prediction ($\beta_T = 0$) [$T \propto (1+z)$].
AP (F_{AP} ; P_{bg} prediction)	$z \approx 2.56/1.59/1.22$ ($z = 0.38/0.51/0.61$)	AP alone does not lead to a definitive rejection.

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theme	Primary constraint (representative)	implication
Structure formation growth rate (RSD $f\sigma_8$)	$\tau_{\text{eff}} = 0.0556$ Gyr, $\chi^2/\text{dof} = 0.153/2$, $\max z_{\text{score}} = 0.308$, overall=Pass	The minimum map containing the effective friction term $\Gamma_{\text{eff}}\delta'$ derived from the continuity equation + Euler + delayed response is consistent at three low- z points
CMB acoustic peak (ℓ_1 - ℓ_3 reverse identification, ℓ_4 - ℓ_6 holdout)	$\ell_A^{(P)} = 299.76$, $\phi_P = 0.249$, $u_\Psi = 0.0585$, $c_{s,P}/c = 0.5242$, $\lambda_{\text{sc}}/r_s = 1.746$, $R_P = 0.213$, $\delta_P = 0.286$, $\kappa_D = 5.2$; core: $\Delta\ell_4 = -0.24$, $\Delta A_4/A_4 = -21.6\%$; extended: $\max \Delta\ell_{4-6} = 0.68$, $\max \Delta A_{4-6}/A_{4-6} = 32.2\%$ (ℓ_6); No DM attenuation: $A_3/A_1 = 0.435 < 1$	The mode constant is uniquely determined by closed-form inverse solution from the 1st to 3rd peaks, and the 3rd peak attenuation is fixed as a theorem in the DM-free branch.
CMB polarization phase (TE/EE; Planck binned)	$\Delta\phi_{EE} =$ 0.5667 ± 0.0334 , $\Delta\phi_{TE} =$ 0.2675 ± 0.0556 ; $\max \Delta\ell_{EE} =$ 30.13 , $\max \Delta\ell_{TE} = 45.51$	Consistent with the half-phase / quarter-phase shift predicted by the Stokes+Thomson extension of Part I 2.8.9 (audit gate pass).

Here, $\Delta\phi_{EE} \approx 0.5$ and $\Delta\phi_{TE} \approx 0.25$ are inevitable values derived from $\Theta \propto \cos(kr_s)$ and $\Pi, E \propto \sin(kr_s)$, which shows that Planck's TT/TE/EE phase shift can be reproduced only by the fluid + scattering equation on the P-model side.

Process (Added: Minimum-mode audit of CMB acoustic peaks): From the peak position/height of $n=1,2,3$, after fixing the P side acoustic mode parameters ($\ell_A^{(P)}$, ϕ_P), R_P, δ_P is derived from the first-principles closure of "gravitational potential well u_Ψ + sound speed c_s, P + scattering length λ_{sc} ". Using the derived R_P, δ_P , compare $n=4,5,6$ as a holdout prediction. Here, instead of "simultaneously fitting all components of the spherical likelihood", we fix the feasibility judgment limited to the peak structure.

Process (Added: CMB polarization-phase audit): Part I Using Stokes radiative transport ($S = (I, Q, U)^T$) fixed at 2.8.9 and Thomson phase matrix, First, fix $\ell_1.. \ell_3$ to ℓ_A, ϕ of TT, and then extract the extreme value positions of EE (max) and TE (max/min alternating) from Planck binned. Map the extracted ℓ to $\ell_n^{EE} \approx (n - \phi + 1/2)\ell_A$ and $\ell_m^{TE,ext} \approx (m - \phi + 1/4)\ell_A$, Determine the phase difference $\Delta\phi_{EE}, \Delta\phi_{TE}$ and the peak position residual $\Delta\ell$ using the same gate.

Process (Added: Structure-formation $f\sigma_8$ audit): Part I Using the minimum mapping (continuity formula + Euler + delayed response potential) of 2.6.1 as is, τ_{eff} freezes the derived values of the collision system and fixes only μ_{ref} in a single scan (no readjustment at each redshift). The background map uses $a_{\text{eff}} = 1/(1+z)$, calculates $f = d\ln D/d\ln a_{\text{eff}}$ and $f\sigma_8 = f \cdot \sigma_8$ from $D = \delta/\delta_{\text{ref}}$, and compares them with the three published points of RSD.

Equation (Added):

$$\Theta'_k = -\frac{k}{3}v_{pb,k}$$

$$v'_{pb,k} + \Gamma_P v_{pb,k} = 3c_{s,P}^2 k (\Theta_k + \Psi_P)$$

$$\Theta''_k + \Gamma_P \Theta'_k + c_{s,P}^2 k^2 \Theta_k = -k^2 \Psi_P \quad (24)$$

$$\tilde{\Theta}_k \equiv \Theta_k + \frac{\Psi_P}{c_{s,P}^2}, \quad \tilde{\Theta}''_k + c_{s,P}^2 k^2 \tilde{\Theta}_k = 0$$

$$r_s(s) \equiv \int_0^s c_{s,P}(s') ds', \quad \tilde{\Theta}_k(r_s) = A_k \cos(kr_s) + B_k \sin(kr_s)$$

$$\Theta_k(r_s) = \Theta_{eq,k} + [\Theta_k(0) - \Theta_{eq,k}] \cos(kr_s) + \frac{\Theta'_k(0)}{k c_{s,P}} \sin(kr_s), \quad \Theta_{eq,k} \equiv -\frac{\Psi_P}{c_{s,P}^2}$$

$$\Theta'_k(0) = 0 \text{ (adiabatic branch)} \Rightarrow \Theta_k(r_s) = \Theta_{eq,k} + [\Theta_k(0) - \Theta_{eq,k}] \cos(kr_s)$$

$$\text{acoustic extrema: } kr_s = m\pi \quad (m = 1, 2, 3, \dots)$$

$$\ell_n^{(P)} \approx (n - \phi_P) \ell_A^{(P)}, \quad A_n^{(P)} = A_0 e^{-\delta_P(n-1)} [1 + (-1)^{n+1} R_P] D_n, \quad D_n = \exp \left[-\left(\frac{\ell_n}{\ell_D} \right)^2 \right] \quad (25)$$

Here, if κ_D ($\ell_D = \kappa_D \ell_A$) is fixed first with an independent audit of the damping tail, then From $(\ell_1, \ell_2, \ell_3, A_1, A_2, A_3)$, the parameters can be inversely identified in closed form.

$$S_i = \sum_{n=1}^3 n, \quad S_{ii} = \sum_{n=1}^3 n^2, \quad S_\ell = \sum_{n=1}^3 \ell_n, \quad S_{i\ell} = \sum_{n=1}^3 n \ell_n$$

$$\ell_A^{(P)} = \frac{3S_{i\ell} - S_i S_\ell}{3S_{ii} - S_i^2}, \quad b = \frac{S_\ell - \ell_A^{(P)} S_i}{3}, \quad \phi_P = -\frac{b}{\ell_A^{(P)}}$$

$$\delta_P = -\frac{1}{2} \ln \left[\frac{A_3}{A_1} \frac{D_1}{D_3} \right], \quad q = \frac{A_2}{A_1} e^{\delta_P} \frac{D_1}{D_2}, \quad R_P = \frac{1-q}{1+q}, \quad A_0 = \frac{A_1}{(1+R_P)D_1}$$

The above equation gives a unique solution under the physical conditions of $A_n > 0$, $D_n > 0$. Therefore, the identification in this section is It is fixed as an algebraic inverse solution from the observed peak, rather than a curve fit in the optimizer.

$$\frac{c_{s,P}^2}{c^2} = \frac{1}{3(1+R_P)}, \quad R_P = \frac{u_\Psi}{(c_{s,P}/c)^2}, \quad u_\Psi \equiv \frac{|\Psi_P|}{c^2}$$

$$\Rightarrow u_\Psi = \frac{R_P}{3(1+R_P)}, \quad R_P = \frac{3u_\Psi}{1-3u_\Psi}$$

$$\Gamma_{\text{sc}} \simeq \frac{c_{s,P}}{\lambda_{\text{sc}}}, \quad k_A = \frac{\pi}{r_s}, \quad \delta_P = \frac{\Gamma_{\text{sc}}}{2} \frac{\pi}{k_A c_{s,P}} = \frac{r_s}{2\lambda_{\text{sc}}}$$

$$\frac{A_3}{A_1} = \exp(-2\delta_P) \frac{D_3}{D_1} < 1 \quad (\delta_P > 0, D_3 \leq D_1) \quad (26)$$

$$k^\mu \nabla_\mu^{(g(P))} \tilde{\mathbf{S}}_\nu = \tilde{\mathbf{J}}_\nu - \tilde{\mathbf{A}}_\nu \tilde{\mathbf{S}}_\nu + \dot{\tau}_T \mathbf{M}(\mu, \varphi) \tilde{\mathbf{S}}_\nu, \quad \tilde{\mathbf{S}}_\nu = (\tilde{I}_\nu, \tilde{Q}_\nu, \tilde{U}_\nu)^T$$

$$\Theta'_k = -\frac{k}{3} v_{pb,k}, \quad \Theta_k(r_s) = \Theta_{eq,k} + \Delta \Theta_k \cos(kr_s) \Rightarrow v_{pb,k}(r_s) = 3c_{s,P} \Delta \Theta_k \sin(kr_s)$$

$$\nabla \cdot v_{pb,k} = ik v_{pb,k} \propto \sin(kr_s)$$

$$\Pi'_k + \dot{\tau}_T \Pi_k = \frac{8}{15} k v_{pb,k}$$

$$\dot{\tau}_T \gg \partial_s \Rightarrow \Pi_k \simeq \frac{8}{15} \frac{k}{\dot{\tau}_T} v_{pb,k} \simeq \frac{8}{15} \frac{1}{\dot{\tau}_T} \nabla \cdot v_{pb,k}$$

$$k^\mu \nabla_\mu^{(g(P))} (\tilde{Q}_\nu \pm i \tilde{U}_\nu) \supset \dot{\tau}_T \frac{3}{4} (1 - \mu^2) \Pi_k e^{\pm 2i\varphi}$$

$$C_\ell^{TT} \propto \cos^2 x, \quad C_\ell^{EE} \propto \sin^2 x, \quad C_\ell^{TE} \propto \frac{1}{2} \sin(2x), \quad x \equiv kr_s$$

Therefore, the extreme position is

$$x_n^{TT} = n\pi, \quad x_n^{EE} = \left(n + \frac{1}{2}\right)\pi, \quad x_m^{TE, \text{ext}} = \left(m + \frac{1}{4}\right)\pi$$

So, the phase difference is

$$\Delta x_{EE-TT} = \frac{\pi}{2}, \quad \Delta x_{TE-TT} = \frac{\pi}{4} \quad (27)$$

is fixed mathematically. In other words, EE ($\sin^2$) is shifted by half a wavelength with respect to TT ($\cos^2$), and TE ($\sin \cdot \cos$) is in the middle phase.

$$\ell_n^{EE} \approx (n - \phi + \frac{1}{2})\ell_A, \quad \ell_m^{TE, \text{ext}} \approx (m - \phi + \frac{1}{4})\ell_A$$

$$\tau_{\text{eff}} \delta^{(3)} + \delta^{(2)} + \tau_{\text{eff}} c_s^2 k^2 \delta^{(1)} + (c_s^2 k^2 - 4\pi G \bar{\rho}) \delta = 0$$

$$\delta^{(2)} + \Gamma_{\text{eff}} \delta^{(1)} + (c_s^2 k^2 - 4\pi G \bar{\rho}) \delta = 0, \quad \Gamma_{\text{eff}} = 4\pi G \bar{\rho} \tau_{\text{eff}} \quad (28)$$

$$a_{\text{eff}} = \frac{1}{1+z}, \quad f = \frac{d \ln D}{d \ln a_{\text{eff}}}, \quad f \sigma_8(z) = f(z) \sigma_8(z)$$

Metric (Added): Evaluate $\max |\Delta \ell_n|$ and $\max |\Delta A_n/A_n|$ at the 1st to 3rd peaks, and $|\Delta \ell_n|$ and

$|\Delta A_n/A_n|$ at the 4th to 6th holdout peaks. The core gates are $\max |\Delta \ell_{1-3}| \leq 15$, $\max |\Delta A_{1-3}/A_{1-3}| \leq 0.15$, $|\Delta \ell_4| \leq 50$, $|\Delta A_4/A_4| \leq 0.25$. Extended diagnosis (Watch/Pass) was set to $\max |\Delta \ell_{4-6}| \leq 80$ and $\max |\Delta A_{4-6}/A_{4-6}| \leq 0.35$. The polarization-phase hard gate is defined by $|\Delta \phi_{EE} - 0.5| \leq 0.12$, $|\Delta \phi_{TE} - 0.25| \leq 0.12$, and the alternating TE-extremum sign pattern (+,-,+, -, ...). If any of these conditions fails, the judgment is fixed to Reject. Only after this phase hard gate is satisfied are $\max |\Delta \ell_{EE}| \leq 50$ and $\max |\Delta \ell_{TE}| \leq 55$ used for the Watch / Pass decision. Structure formation gates are `growth_mapping_defined=true`, `fsigma8_gate_defined=true`, `friction_term_positive=true`. The Pass condition is fixed by satisfying $\max |z_{\text{score}}| \leq 2.5$ and `delay_vs_instant_nonworse=true` at the same time.

In conclusion, the P-model minimum predictions ($p_t = 1$, $\beta_T = 0$) derived in Part I 2.5.1 are maintained within the range of primary observations (p_t , β_T) that are independent of distance estimation.

In the peak structure of CMB TT (1st to 3rd inverse identification + 4th to 6th holdout), both core/extended gates are filled with the revised model including Silk attenuation correction ($D_n = \exp(-(\ell_n/\ell_D)^2)$, $\ell_D = \kappa_D \ell_A$, $\kappa_D = 5.2$). However, the amplitude holdout margin is thin, $|\Delta A_4/A_4| = 21.6\%$ (threshold 25%, margin 3.4pt) and $|\Delta A_6/A_6| = 32.2\%$ (threshold 35%, margin 2.8pt), so the audit of threshold setting sensitivity will be continued in the next stage. First-principles closure also restores $R_P = 0.213$, $\delta_P = 0.286$ from $u_\Psi = 0.0585$, $c_{s,P}/c = 0.5242$, $\lambda_{\text{sc}}/r_s = 1.746$, and the third peak is fixed on the decay side ($A_3/A_1 = 0.435$) by the decay theorem $A_3/A_1 = \exp(-2\delta_P)(D_3/D_1) < 1$ of the DM-free branch.

On the polarized side, $\Delta \phi_{EE} = 0.5667 \pm 0.0334$ and $\Delta \phi_{TE} = 0.2675 \pm 0.0556$ are obtained from Planck binned TE/EE, and the difference from the theoretical prediction (0.5, 0.25) is within 0.0667 and 0.0175, respectively. All gates of phase, position, and sign were satisfied at the same time, and CMB polarization audit was fixed at Pass.

Result (added: CMB TT full C_ℓ fit; $\ell \leq 2500$):

index	value	judgement
Number of bins	83	
$\chi^2_{\Lambda\text{CDM}} / \chi^2_{\text{P}}$	65.122 / 64.139	
$\text{AIC}_{\Lambda\text{CDM}} / \text{AIC}_{\text{P}}$	65.122 / 72.139	baseline advantage
$\Delta \text{AIC} = \text{AIC}_{\text{baseline}} - \text{AIC}_{\text{P}}$	-7.018	Reject
P-model remap coefficient	$\alpha = 1.001$, $\Delta \ell =$ -0.5, gain = 0.9997, offset = 1.1246	

(Figures, metrics, and summaries of TT full C_ℓ fit are aggregated in the public ledger in Part IV, and individual file names are not listed in this article.)

Caution: In the TT full-range operational fit (template remap), χ^2 is close, but the AIC including degree of freedom penalty cannot exceed the baseline. Therefore, the TT independent judgment for $\ell \leq 2500$ is currently Reject.

Results (added: CMB TT/TE/EE simultaneous full C_ℓ fit; $\ell \leq 2500$):

index	value	judgement
Number of bins (TT/TE/EE)	83/66/66 (total 215)	

Continued on next page

index	value	judgement
$\chi^2_{\Lambda\text{CDM}} / \chi^2_{\text{P}}$ (simultaneously)	200.765 / 198.340	
$\text{AIC}_{\Lambda\text{CDM}} / \text{AIC}_{\text{P}}$	200.765 / 214.340	baseline advantage
(simultaneous)		
$\Delta\text{AIC} = \text{AIC}_{\text{baseline}} - \text{AIC}_{\text{P}}$	-13.575	Reject
Shape contribution /	0.000 / +2.425 / -16.000	Reject Maintain
Channel linear contribution /		
AIC penalty		
shared remap factor	$\alpha = 1.0000, \Delta\ell = 0.0$	

(Figures, metrics, and summaries of TT/TE/EE simultaneous full C_ℓ fit are aggregated in the public ledger in Part IV, and individual file names are not listed in this article.)

Caution: In simultaneous fit, the entire amount of χ^2 improvement comes from channel linear adjustment, and the amount of improvement for shared shape remap is 0. Therefore, the bottleneck at the current stage is not that TE/EE is not implemented, but that it is not possible to obtain a shape information gain that exceeds the AIC penalty (16.0).

On the structure formation side, the delayed branch that freezes $\tau_{\text{eff}} = 0.0556$ Gyr becomes $\chi^2/\text{dof} = 0.153/2$, $\max |z_{\text{score}}| = 0.308$, which satisfies Pass for the RSD $f\sigma_8(z)$ of low- z three points. Since the difference between the delayed and immediate branches is small ($\Delta\chi^2 \approx 2.7 \times 10^{-9}$) in the current data, an extended audit on the high-precision, high-redshift side is required to determine the presence or absence of $\Gamma_{\text{eff}}\delta'$. Since the residuals on the higher-order peak side (especially ℓ_6) still remain significant, the next step is to further improve the extrapolation stability by improving the driving term approximation. Therefore, the tension in the DDR should not be "absorbed by changing p_t/β_T " but should be evaluated by connecting to the prerequisite audit of the distance estimation I/F (Section 5.3).

We confirm that the explanation that p_t alone absorbs DDR tension (e.g. $p_t \approx 3$) does not hold due to the first-order constraint on the time dilation index p_t obtained from the aging rate of the SNe Ia spectrum.

Due to the first-order constraint on the CMB temperature scaling (β_T) obtained from SZ etc., we confirm that the explanation for restoring DDR only by the exponent on the photon energy side does not hold true.

Read Planck binned TT/TE/EE with the same I/F and audit the phase shift of EE/TE with respect to ℓ_A, ϕ fixed from TT.

Reconstruct $f\sigma_8(z)$ with the minimum growth equation obtained from the continuity equation + Euler + delayed response and compare it with the BOSS DR12 consensus 3 points.

Shows the difference between F_{AP} and P_{bg} predictions calculated from BOSS DR12.

Supplement: This group of figures shows that (1) DDR tension cannot be resolved by changing p_t or β_T alone, (2) polarization phase audit directly verifies the prediction of static background wave P_{bg} with primary data, (3) the growth rate is low- z , within the gate, and non-worse even in the delayed branch including $\Gamma_{\text{eff}}\delta'$, and (4) the AP test is weak when used alone, so an integrated judgment with BAO primary statistics / distance indicators is required.

Caution: The CMB handled in this section is limited to late-time audits of "temperature scaling" and "acoustic peak structure." TT full C_ℓ ($\ell \leq 2500$) and TT/TE/EE simultaneous binned operational fit have been added in this section, but replacement of Planck full likelihood (covariance/foreground simultaneous estimation) has not been implemented. BBN (light element synthesis ratio such as

H/He/Li) is explained in Part III (Quantum Physics Edition). The minimal He-4 chain is fixed as raw watch and pass after derived error propagation. In this paper, we refer to the results as boundary conditions. The overall consistency of multi-isotope (D/H, He-3/He-4, Li-7) is currently being audited as Watch. The derivation of the origin of the CMB blackbody spectrum will continue to be investigated as a remaining issue in the early thermal history. In addition, for the growth rate of structure formation ($f\sigma_8$ of RSD), the minimum mapping for three low- z points is fixed in this section. However, the growth history, nonlinear branches, and dependence on initial conditions on the high redshift side have not yet been separated, and will be handled separately in the next audit. Therefore, Pass/Watch/Reject in this section is not interpreted as a final acceptance or rejection of the early universe model, but as a judgment of consistency with the observation I/F of the late universe. The detailed logs below (JWST/MAST x1d acquisition, individual target $z_{\text{confirmed}}$, etc.) will be retained as internal references, and the main points of the paper will be fixed in the bullet points above.

5.3.6 Cosmology II: Distance-indicator dependence and BAO assumptions

Purpose: Decompose, for each distance-indicator premise, the tension between the static-background P minimum (P_{bg} , $\epsilon_0 = -1$) and the observational DDR constraint ($\epsilon_{0,\text{obs}} \pm \sigma$), fix the form of rejection, and quantify the correction required to compensate for the tension (opacity α , light-source evolution s_L , ruler evolution s_R) together with the reachable limit z_{limit} from the error budget.

Input:

- DDR Constraint Catalog (integrated list; see appendix “Data Sources (Primary Sources)”)
- SNe error budget: Pantheon / Pantheon+ (distance modulus error and systematic covariance)
- BAO error budget: BOSS DR12 consensus (D_M , H , correlation) [53]

Equation:

$$\eta(z) \equiv \frac{D_L}{(1+z)^2 D_A} = (1+z)^{\epsilon_0}$$

$$\Delta\mu(z) = 5 \log_{10}((1+z)^{\Delta\epsilon}), \quad \tau(z) = 2\alpha \ln(1+z) \quad (29)$$

Result:

index	value	remarks
Representative (SNIa+BAO; 2021)	$\epsilon_{0,\text{obs}} = 0.013 \pm 0.029 \rightarrow \Delta\epsilon_{\text{needed}} = 1.013$	$z = 1$ with extra D_L factor = 2.018, $\Delta\mu = 1.525$ mag, $\tau = 1.404$
Error budget envelope (opt_total; 3σ)	$z_{\text{limit}} \approx 0.048$ (with BAO), $z_{\text{limit}} \approx 0.201$ (without BAO)	A guideline for “ z at which the necessary correction can be hidden in error”

The correction necessary to shift the static background P minimum ($\epsilon_0 = -1$) to the observed ϵ_0 is shown by converting it into $\Delta\mu(z)$ and $\tau(z)$.

How to read: Intuitively fix “how many mags are needed for $z=1$ ” and “ z where the correction can be hidden by error” .

The error budget (statistics/system) of SNe (Pantheon/Pantheon+) and BAO (BOSS) is shown mapped to the equivalent $\Delta\mu$.

Confirmation point: Confirm that the necessary correction cannot be hidden in the current error budget (z_{limit} is small).

Visualize the “reconnection conditions” that restore ϵ_0 with the combination of opacity α / light source evolution s_L / ruler evolution s_R .

Position: Clarify that the place where the tension is absorbed is not unique (the degree of freedom in deciding which premise to push into). (The diagrams, ledgers, and reproduction lines in this section are based on the public snapshot in Part IV, and individual file names are not listed in this article.)

Discussion: DDR tension ($z \approx -25.10$ in the typical example) is a **serious issue** for the current P-model minimum. This paper does not hide this discrepancy, but clearly discloses the reconnection conditions (required amount) and reach limits. Preliminary auditing is not a means of “avoiding rejection,” but is positioned as a diagnosis to isolate which distance estimation I/F stage tension is amplified.

Fair comparison: When introducing corrections to the I/F side in the preliminary audit, it is essential to compare the same corrections applied to the Λ CDM side. If the same correction improves Λ CDM to the same extent, that improvement is not a basis specific to P-model.

Result (minimal fair-comparison check):

Comparison axis (SNIa+BAO 2021)	setting	$z(\Lambda\text{CDM})$	$z(\text{P-model min})$
baseline	Same as published value	-0.45	-25.10
Same condition I/F correction	Observation: Apply the same shift δ_{IF} to both models	-0.45 (unchanged)	-25.10 (unchanged)

The same shift

$$z_i = \frac{[(\eta_{\text{obs}} + \delta_{\text{IF}}) - (\eta_i + \delta_{\text{IF}})]}{\sigma_\eta} = \frac{\eta_{\text{obs}} - \eta_i}{\sigma_\eta}$$

shows that the relative difference between the models does not change. Therefore, the effect of an audit-side correction cannot be judged by whether only the P-model appears to improve; the difference from Λ CDM must be evaluated under the same corrected conditions.

Caution: Please keep the detailed log below (Tolman, static-infinite pack, tension attribution, etc.) as an internal reference, and fix the main points of the paper in the bullet points above.

6 Discussion

In this chapter, we will organize the scope and limits reached so far (dominant systems and unfixed assumptions), and connect the work that needs to be completed next to a roadmap. The conclusion of the main text is closed on the P-model side, and comparisons with reference frames (GR/ Λ CDM, etc.) are isolated to “Caution”.

6.1 Current Scope of Achievement

As an essential first-edition set in this repository, we prepared a configuration in which weak-field tests (LLR/Cassini/Viking/Mercury/GPS) and representative strong-field examples (EHT) can be compared within the same framework (P and frozen β). However, each theme has an explicit definition of observables and a matching systematic-error model, and EHT in particular still depends on radiation models, scattering, and spin. The significance evaluation of the differential predictions therefore remains to be refined.

Regarding EHT, the current comparison is based on the approximation that “ring diameter $\approx$ shadow diameter” ($\kappa = 1$). Under this assumption, while both the P-model and GR do not significantly contradict observations, it is not at the stage where it is possible to claim statistically definitive superiority (evidence of substitutability). The consistency check index (z score) is shown in the figure below.

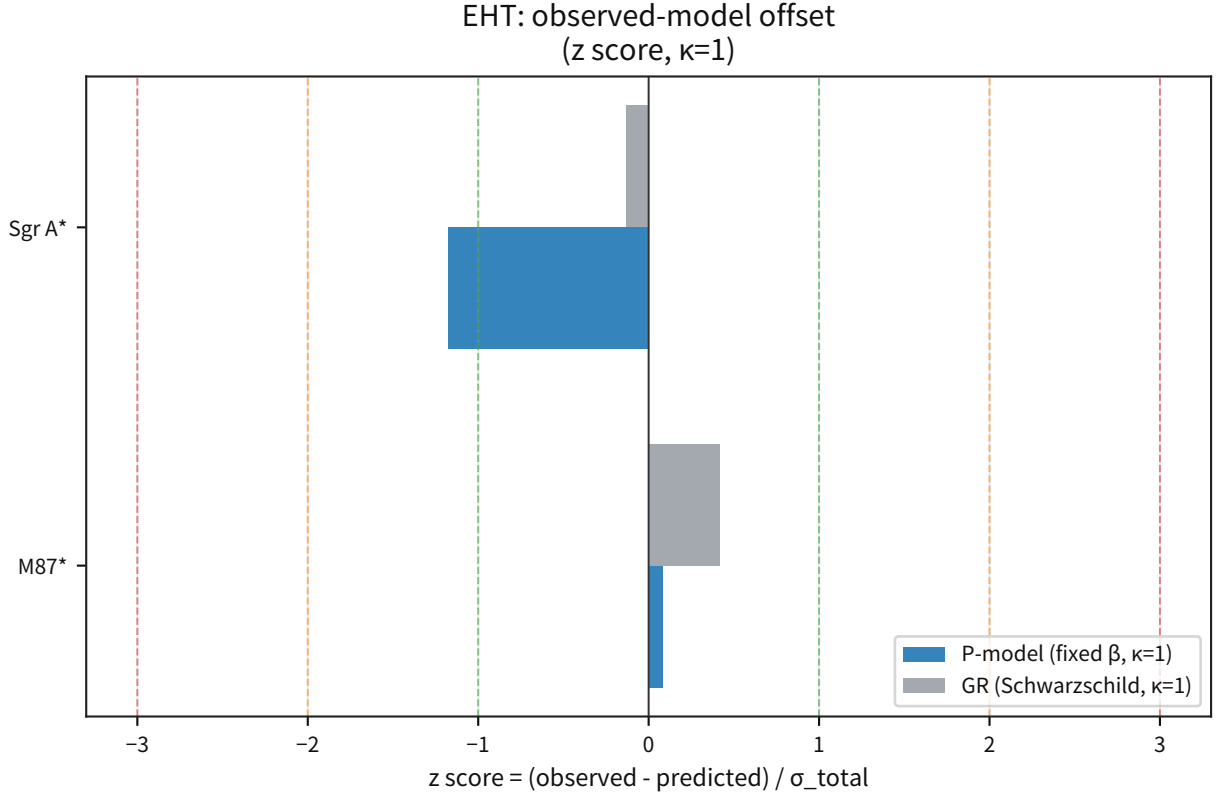


Figure 36: For M87* and Sgr A*, the deviation z between the observed ring diameter and the model shadow diameter under the assumption of $\kappa = 1$ is juxtaposed, and the sign and error width are compared.

Supplement: $z = 0$ is a match, and $|z| < 1$ is a guideline for 1σ match.

Therefore, the current position is (i) "understanding the ring/shadow system scale" and rough consistency check through κ_{fit} , and (ii) predicting the difference in the "coefficient difference (several percent)" that remains after fixing β (corresponding diagrams are summarized in Part IV).

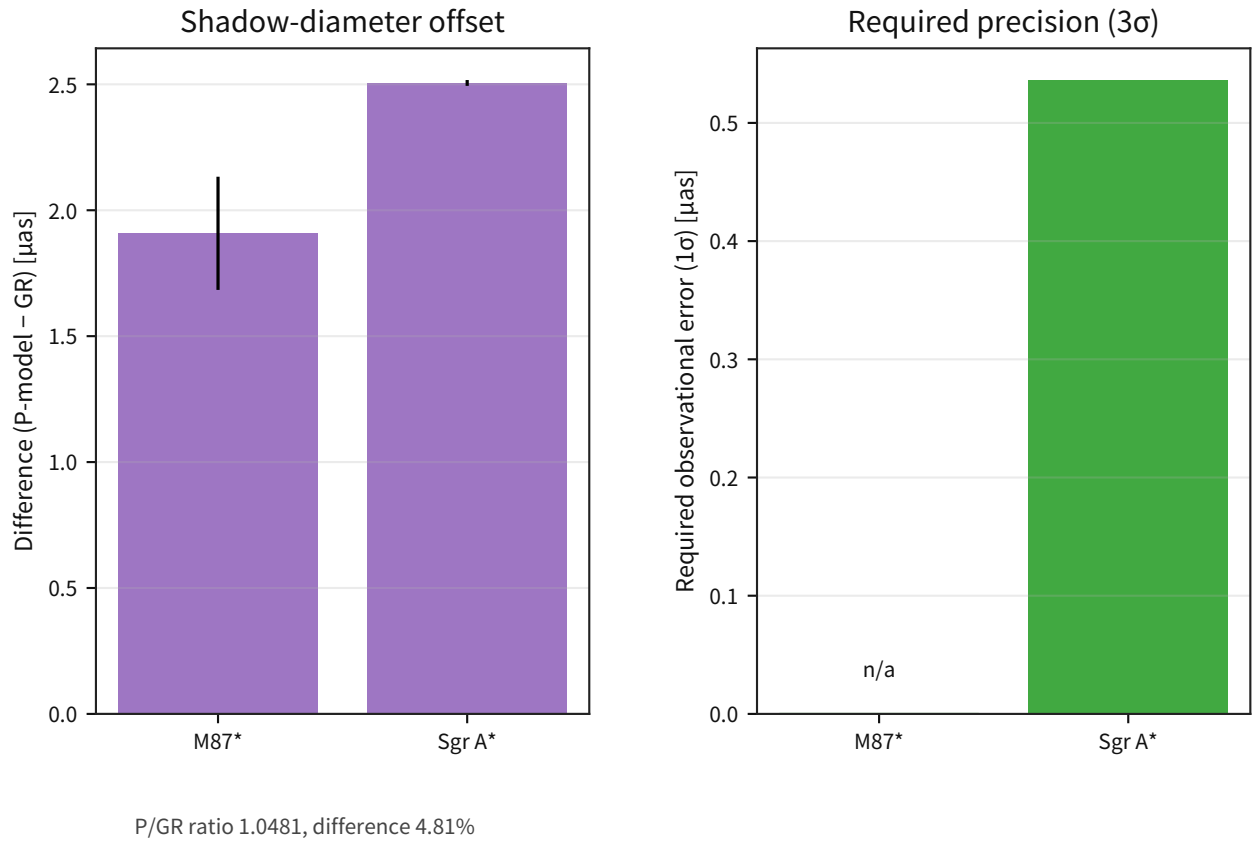


Figure 37: Differential prediction of coefficient differences (β freezing).

6.2 Limitations and Next Steps

- LLR: Outliers are traced to the source line (*sourcefile : lineno*). This was separated by auditing time-tag sensitivity and target contamination sensitivity (figures are summarized at the end of this section). The largest spikes have a high probability of “target contamination” (reflector mix-up), so suspected contamination is excluded from the standard statistics and confirmed in the primary-data row (no automatic reassignment).
- Cassini: Continue to track TDF observable definitions (differences from predictions, plasma correction, missing data imputation) with primary sources.
- Viking: In this paper, we reproduced the peak round-trip Shapiro delay from HORIZONS and compared it with typical values in the literature. Proceeding to the first “time-series” verification requires re-analyzing the observation data system of that period (range/Doppler, applied corrections, and estimated parameters).
- Mercury: The perihelion precession is reproduced with a reduced numerical model. For rigorous verification, the framework must be expanded to include n-body perturbations, solar J_2 , coordinate-time conventions, ephemeris alignment, and systematic comparison against independent references (modern ephemerides and the literature).
- GPS: At present, the comparison is based on the residual RMS against IGS Final, and the result varies depending on the satellite, period, and correction system. Stronger claims require statistics over multiple days and multiple arcs together with explicit handling of the observation models (ionosphere / troposphere / relativity corrections). The design must also make clear which quantities are estimated and which are held fixed.
- Cosmology (DDR): A large tension (e.g. “35 σ ”) based on the published ϵ_0 constraint can arise when the distance-estimation I/F embeds the “expansion side” $(1+z)$ directly into the distance definition, for example by defining the angular-diameter distance as the comoving distance divided by $(1+z)$. In this paper, that tension is treated as a **serious issue** for the P-model minimum rather than being hidden from the conclusion. This paper uses $\eta^{(P)} = D_L/((1+z)D_A)$ as an index, but the falsification conditions apply only to **distance estimates that have passed the distance-estimation I/F audit**. The preliminary I/F correction must also be compared under the same conditions on the Λ CDM side. The next issues are (i) fixing the BAO entry point that uses first-order statistics (ξ_ℓ , etc.) rather than compressed distances, (ii) standardizing and calibrating distance indicators (standard candles / standard rulers), and (iii) checking whether the rejection threshold of $\eta^{(P)}$ can then be applied consistently across multiple datasets.
- DM alternative branch (SPARC/collision system/ISW/weak lens): The emergent mechanism of the g_P correction and delayed response, as shown for SPARC and the Bullet systems, is effective, but the full simultaneous fit to the ISW correlation and weak-lensing two-point statistics (including covariance) remains incomplete. In addition to the TT full C_ℓ audit ($\ell \leq 2500$), the CMB side has been extended to the simultaneous binned TT/TE/EE audit, but the replacement of the Planck full likelihood (simultaneous covariance / foreground estimation) has not yet been performed. At this stage, the overall verdict for the DM alternative branch remains Watch.
- Growth of Structure: For the RSD-derived growth rate index ($f\sigma_8$), the minimum mapping at three low-redshift points was integrated into this paper. On the other hand, the contribution of $\Gamma_{\text{eff}}/H_{\text{eff}}$ is small in the current data, and the discrimination power of the delayed branch is insufficient. In the next stage, we will conduct homogeneous audits with higher precision over a wider redshift range using the primary statistics of DESI/BOSS, and continue to prioritize the identification of delay terms.

- GW polarization mode (most important limitation): The biggest limitation at present is that, in the 3-detector (H1/L1/V1) polarization determination, complete exclusion of the scalar limit `scalar_exclusion_pass=true` has not yet been achieved (currently false). Even in the local re-search (24 trials), the minimum `scalar_overlap_proxy=0.500` is maintained at $Pass/Watch = 0$ with `n_usable_events` ≥ 2 , and `scalar_overlap_proxy=0.0` is restricted to the `n_usable_events=1` condition. The same boundary (`best_u2=0.500`) is reproduced in the subset expansion (7 subsets), and operational monitoring of public 3-detector events is continuing, while the detailed logs are separated into Part IV. The main bottleneck at this stage is the number of configurable highly correlated 3-detector event coverages. Therefore, the bottleneck at this stage is not the theoretical branch itself, but the highly correlated event coverage that can be used simultaneously by three detectors. Here `scalar_exclusion_pass` indicates that scalar polarization has been excluded, and `tensor_consistency_pass` indicates that tensor polarization is consistent with the observations. The introduction of the P_μ vector wave is fundamental for describing the tensor-like nature of gravitational waves and strong-field rotation (frame dragging). Simultaneously satisfying `scalar_exclusion_pass=true` and `tensor_consistency_pass=true` under improved network S/N is fixed as an absolute survival condition for the P-model to surpass the pure scalar theory.
- EHT: κ (ring/shadow) and Kerr spin dependence are treated as systematic uncertainties. The corresponding diagram in Part IV organizes (i) the allowable range of κ , (ii) the coefficient difference that remains after fixing β , and (iii) the observational accuracy required for discrimination. (The figures in Section 5.1, `eht_kappa_tradeoff.pdf` / `eht_kappa_precision_required.pdf`, and figure `eht_shadow_systematics.pdf` in Section 4.8 are referred to without being reproduced.) The remaining issue is whether the physics of κ (radiation model/scattering/definition difference) can be constrained to the 1-2% level.
- Quantum (premise verification) is outside the scope of this paper and will be summarized in Part III (Quantum Physics Edition).
- Velocity saturation δ_0 (extended hypothesis): The saturation of the velocity term is a phenomenological ansatz, and δ_0 itself has not yet been “measured.” The upper limit consistent with known high- γ observations has already been organized in the correspondence diagram in Part IV (refer to `delta_saturation_constraints.pdf` in Section 5.2 without reprinting it). The present role of that figure is to arrange the upper limit from high- γ observations, that is, the tolerable range of δ_0 . The next step is to specify which observations and which precision would determine a direct detection or an updated upper limit.

LLR audit diagram referenced in Section 6.2:

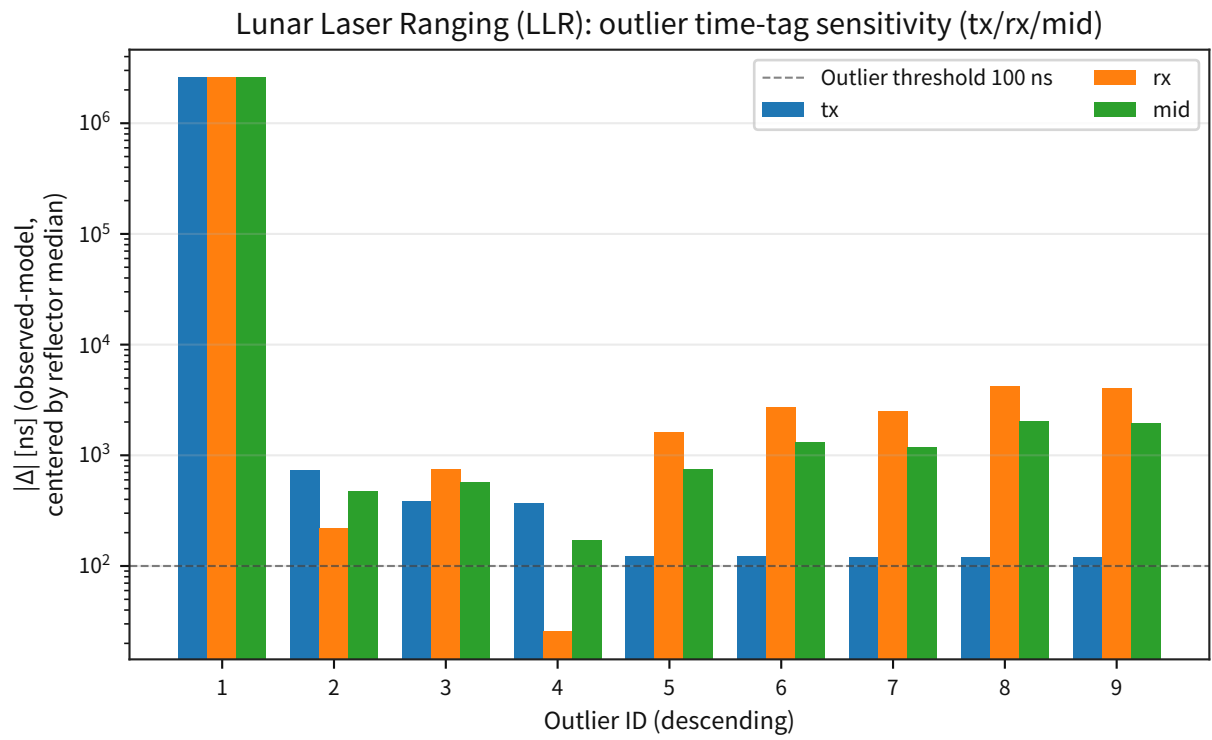


Figure 38: Audit diagram of time-tag sensitivity.

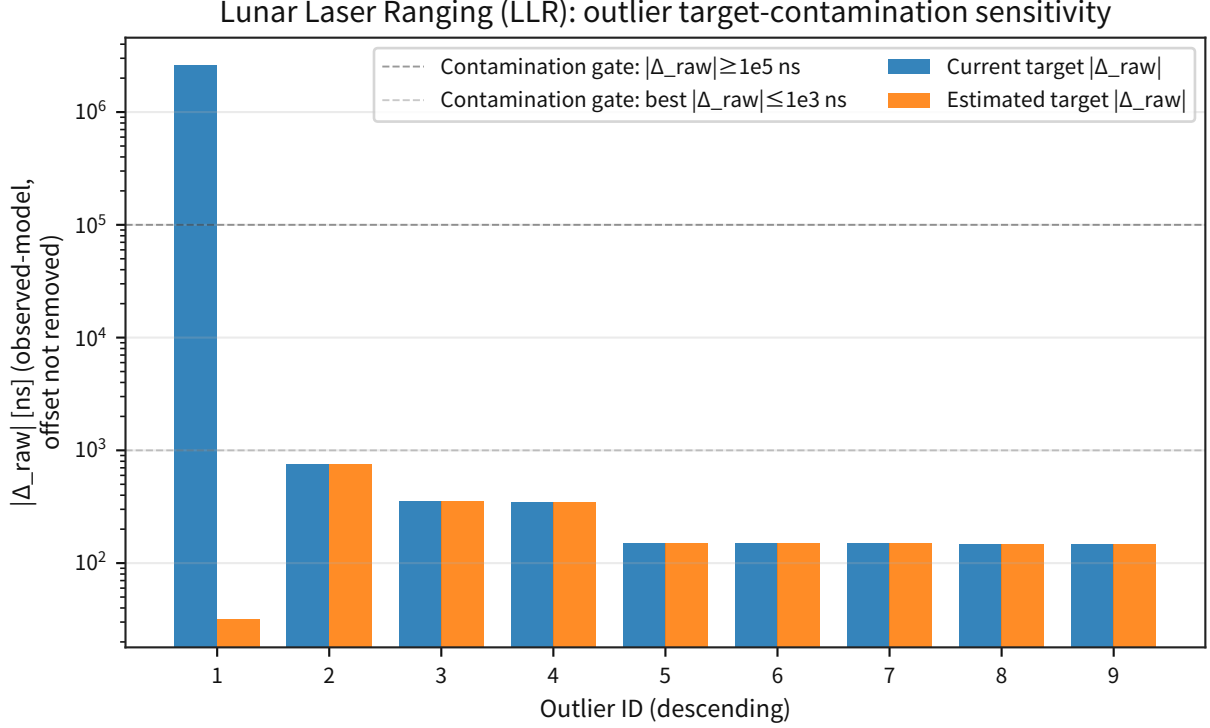


Figure 39: Audit diagram of target contamination sensitivity.

6.3 Discrimination Conditions (Form of Falsifiability)

- EHT: After fixing β through solar-system verification (Cassini etc.), κ and the Kerr lineage (spin/tilt/scattering) must be constrained at the 1-2% level. If the statistical error of the ring diameter can be reduced to the same level, significant discrimination of the coefficient difference (approximately 4.8%) comes into view. M87* is rate-limited by θ_{unit} (mass/distance), and Sgr A* is rate-limited by ring σ and κ systematics (see the correspondence diagram in Section 5.1 for details). Because this accuracy range is directly connected to the design goals of ngEHT and space-VLBI (BHEX, etc.), this paper fixes the differential prediction as “required accuracy” in a form that can be updated mechanically with future data.
- Velocity saturation δ_0 (extended hypothesis): Even when introducing δ_0 , the known maximum γ (high-energy particles/cosmic neutrinos, etc.) will update the upper limit of δ_0 . To directly measure the “difference” due to δ_0 , it is necessary to observe the region where γ approaches $\gamma_{\text{max}} \approx 1/\sqrt{\delta_0}$.

The main points of required precision (EHT) and rejection conditions (EHT/ δ_0 candidates) are summarized in the following figure as a falsification condition pack.

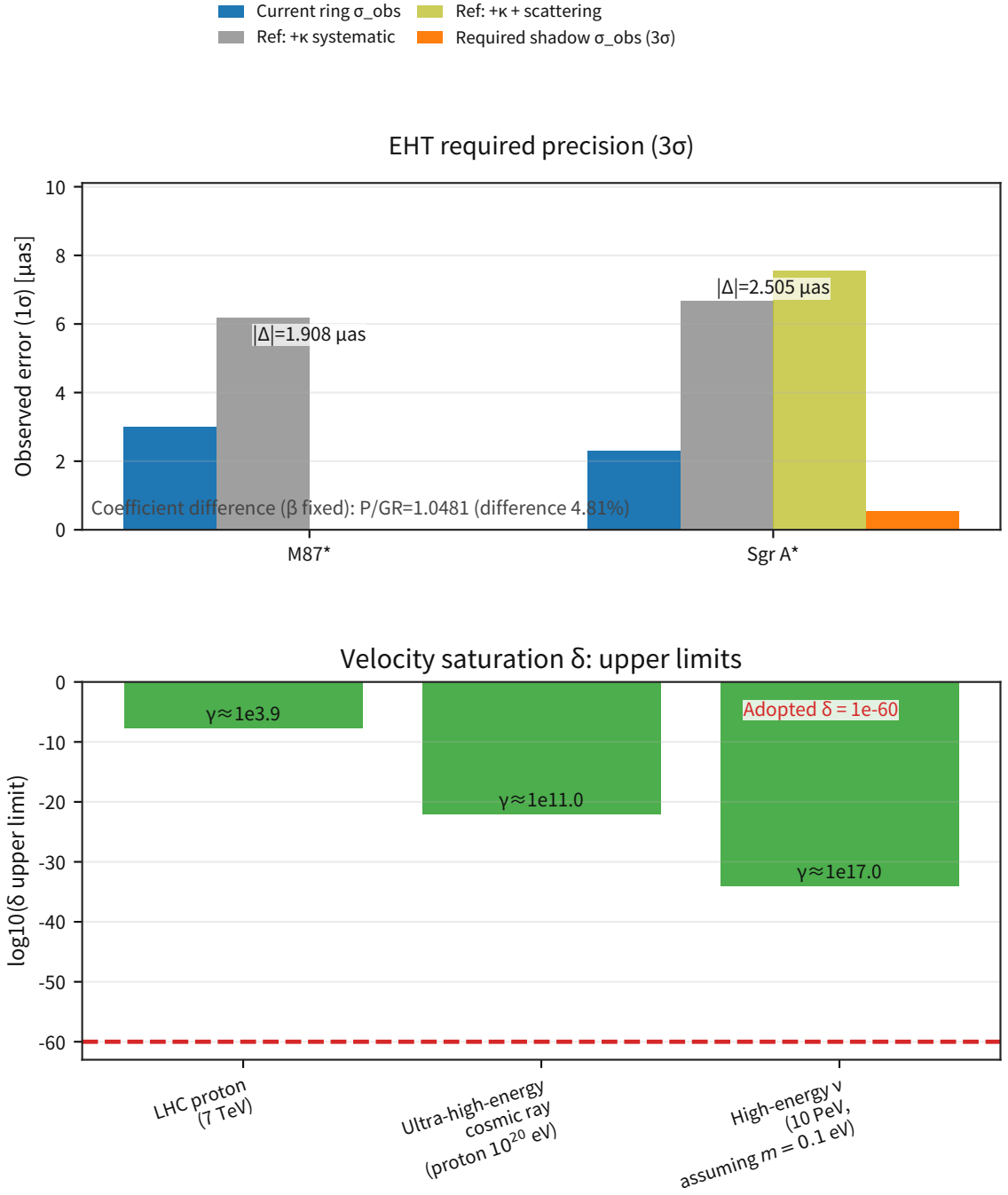


Figure 40: The main gates of weak fields (LLR/Cassini/Viking/GPS/Mercury) and strong fields (EHT/ δ_0 candidates) are integrated into one sheet, and the required accuracy, current statistics, and Pass/Watch/Reject boundaries are fixed as a falsification condition pack that can be determined by machine.

7 Conclusion

P-model is a framework that uniformly expresses gravity, clocks, and light propagation through the time-wave density P . In this paper, we established a reproducible validation pipeline for representative cases ranging from weak fields to strong fields and cosmological scales, and presented the entry points for both the differential prediction (the EHT coefficient difference) and the upper-bound constraint on the extension hypothesis (velocity saturation δ_0).

In cosmology, by generating $\xi_0/\xi_2 + \text{cov}$ directly from the DESI DR1 raw products and performing the BAO ϵ fit, we confirmed that tension across multiple tracers appears stably in pbg coordinates. However, this paper does not immediately interpret that result as a rejection of the P-model. Instead, it quantifies the result as a reconnection condition based on the premise structure of the distance indicators ($\alpha, s_L, s_R; \Delta\epsilon$), and fixes it as a falsifiable prediction.

Regarding DDR (distance duality), this paper explicitly treats the large discrepancy with the P-model minimum as a **serious issue**, and organizes it as a preliminary audit of which $(1+z)$ factor is embedded in the distance-estimation I/F. Furthermore, the correction introduced in the preliminary audit must be applied under the same conditions to the Λ CDM side as well, so that the comparison remains fair. On that basis, we adopt the P-model-specific index $\eta^{(P)}$ together with the rejection condition of 3σ across multiple datasets.

For the DM alternative branch, we fixed a mechanism that explains both the rotational support of galaxies and the center-of-mass separation in colliding systems by the same equation, through the SPARC g_P correction and the delayed response of the Bullet-type systems. On the other hand, the ISW / weak-lensing side remains at the prediction-fixed stage, and the simultaneous fit to the CMB full C_ℓ and weak-lensing statistics has not yet been completed. Therefore, the overall verdict for the DM alternative branch remains Watch.

For the derivation of β in the weak-field regime, the cross-channel audit in Part IV is treated as the current official verdict. In the LLR decontaminated branch, $\beta = 1.000674 \pm 0.000961$ is consistent with the theoretical prediction $\beta = 1$, with $\text{abs}(z(\beta - 1)) = 0.702$. By contrast, the cross consistency with VLBI fails at $\text{abs}(z(\beta_{\text{vlbi}} - \beta_{\text{llr}})) = 13.392$, so the VLBI comparator eligibility remains fixed at false. However, because the LLR bias / promotion gate and the messenger pair gate have been cleared and the promotion conditions of policy_D have been confirmed, the active policy has already been switched to policy_D_exclude_plus_messenger_fusion. Accordingly, the terminal verdict is fixed at beta_terminal=Pass.

From here onward, the priority is to isolate the conditions under which differential predictions become observationally discriminable, that is, the factors that dominate the uncertainty, and to reduce the strong-field and cosmological falsification routes to more direct observational proposals. The cosmological conclusion of this paper focuses on the observational audit of the late universe, while the BBN side has already fixed the minimal He-4 chain ($Y_p \approx 0.25$) in Part III (the quantum volume) as **raw watch** and, after propagating the derivation uncertainty, pass. However, the overall multi-isotope consistency remains Watch, and the final acceptance or rejection is deferred to continued auditing. This paper does not reproduce that derivation, but adopts it as a boundary condition and retains the audit of the origin of the CMB blackbody spectrum as an open task.

Likewise, the growth-rate audit for structure formation (RSD-derived $f\sigma_8$) is presently fixed only at the minimum mapping of the three low- z points. The next priority is to extend the audit to a wider redshift range, using an isomorphic procedure based on the primary statistics, and to secure sufficient S/N to discriminate the delay term. For the quantum domain, Part III develops the validation procedure under the same I/F (primary data $\rightarrow$ fixed procedure $\rightarrow$ metric $\rightarrow$ falsification threshold), and evaluates whether the trilogy maintains the same mapping, the same freezing rule, and the same

falsification form across scales.

8 Reproducibility / Availability

The data used in this paper are based on publicly available primary sources, and the sources are summarized in the appendix. The analysis code and processing conditions are organized so that the same input $\rightarrow$ the same result can be reproduced. Primary sources (URLs/reference dates) are summarized in the appendix "Data sources (primary sources)," and processing conditions (windows/outlier rules/frozen values) are specified in the main text.

In this paper (Part II), the lists of reproduction artifacts, published outputs, and original reproduction commands are consolidated in Part IV (verification materials). Input identifiers mentioned in the text are kept consistent with those in Part IV, while the I/F key is retained as the minimal reference for reproduction conductors. The fixed artifact lists, reproduction commands, and audit gates are therefore governed by Part IV (verification materials) (GitHub: [EnterOgawa/p-model](#), snapshot policy: base [v0.1.2](#) + strong-field bridge [v0.1.3](#)). [1]

9 Appendix A: Data Sources (Primary Sources)

This appendix is a window for collecting the references of the primary sources (observations themselves/officially published data) referenced in the main text. In the main text of Part II, only the input conditions and audit gates for each observation theme are fixed, giving priority to the readability of the reproduction procedure, and the complete URL/reference date/obtainment meta information of the primary source is assumed to be the ledger in Part IV (verification materials). The key in square brackets (e.g. [Will2014]) in the text is used as a reference key for the Part IV ledger, and Part IV is the only reference for the URL/reference date/acquisition history.

The authoritative operational sources of truth are:

- Primary source ledger (URL/reference date/obtained meta information): Part IV (verification material) ledger/manifest
- Freezing conditions (windows, outlier rules, thresholds): Part II main text (each section 4.x / 5.x)
- Fixed artifacts (diagram/CSV/JSON support): List of fixed names in Part IV (verification materials)

Note: This appendix is intentionally minimal. Its only role is to close the references cited from the main text, while updates to the primary-source list are managed centrally on the Part IV side.

References

- [1] P-model Verification Materials (supplementary material for reproduction commands, fixed artifact lists, audit gates, hashes, and manifests). Web: [EnterOgawa/p-model](#) (Releases: [GitHub Releases](#)). Accessed: 2026-02-15.
- [2] Will, C. M. “The Confrontation between General Relativity and Experiment.” *Living Reviews in Relativity* **17**, 4 (2014). DOI: 10.12942/lrr-2014-4. Accessed: 2026-01-03.
- [3] Ashby, N. “Relativity in the Global Positioning System.” *Living Reviews in Relativity* **6**, 1 (2003). DOI: 10.12942/lrr-2003-1. Accessed: 2026-01-03.
- [4] Lebach, D. E., et al. “Measurement of the Solar Gravitational Deflection of Radio Waves Using Very-Long-Baseline Interferometry.” *Physical Review Letters* **75**, 1439 (1995). DOI: 10.1103/PhysRevLett.75.1439.
- [5] Shapiro, I. I., et al. “Measurement of the Solar Gravitational Deflection of Radio Waves Using Geodetic Very-Long-Baseline Interferometry Data, 1979–1999.” *Physical Review Letters* **92**, 121101 (2004). DOI: 10.1103/PhysRevLett.92.121101. Accessed: 2026-01-03.
- [6] Fomalont, E. B., Kopeikin, S. M., Lanyi, G. E., and Benson, J. M. “Progress in Measurements of the Gravitational Bending of Radio Waves Using the VLBA.” *The Astrophysical Journal* **699**(2), 1395-1402 (2009). DOI: 10.1088/0004-637X/699/2/1395. arXiv:0904.3992. Accessed: 2026-01-03.
- [7] Bertotti, B., Iess, L., and Tortora, P. “A test of general relativity using radio links with the Cassini spacecraft.” *Nature* **425**, 374-376 (2003). DOI: 10.1038/nature01997. Accessed: 2026-01-03.
- [8] NASA Planetary Data System (PDS). “CASSINI RSS RAW DATA SET - SCE1 V1.0” (CO-SS-RSS-1-SCE1-V1.0). DOI: 10.17189/ske1-av41. Landing page: <https://pds.nasa.gov/ds-view/pds/viewDataset.jsp?dsid=CO-SS-RSS-1-SCE1-V1.0>. Accessed: 2026-01-03.
- [9] Shapiro, I. I., et al. “Fourth test of general relativity: New radar results.” *Journal of Geophysical Research* **82**(28), 4329-4334 (1977). DOI: 10.1029/JS082i028p04329.
- [10] Reasenberg, R. D., et al. “Viking relativity experiment: Verification of signal retardation by solar gravity.” *The Astrophysical Journal Letters* **234**, L219-L221 (1979). DOI: 10.1086/183144.
- [11] NASA/JPL SSD. “HORIZONS system.” Web: <https://ssd.jpl.nasa.gov/horizons/> (API: <https://ssd.jpl.nasa.gov/api/horizons.api>). Accessed: 2026-01-03.
- [12] Nobili, A. M., and Will, C. M. “The real value of the perihelion advance of Mercury.” *Nature* **320**, 39-41 (1986). DOI: 10.1038/320039a0.
- [13] Dow, J. M., Neilan, R. E., and Rizos, C. “The International GNSS Service in a changing landscape of Global Navigation Satellite Systems.” *Journal of Geodesy* **83**, 191-198 (2009). DOI: 10.1007/s00190-008-0300-3.
- [14] IGS Public Data (BKG mirror) “root_ftp/IGS” (products/BRDC/CLK/SP3). Web: https://igs.bkg.bund.de/root_ftp/IGS/. Accessed: 2026-01-03.
- [15] Vessot, R. F. C., et al. “Test of Relativistic Gravitation with a Space-Borne Hydrogen Maser.” *Physical Review Letters* **45**, 2081 (1980). DOI: 10.1103/PhysRevLett.45.2081. Accessed: 2026-01-04.

- [16] Delva, P., et al. “A gravitational redshift test using eccentric Galileo satellites.” *Physical Review Letters* **121**, 231101 (2018). DOI: 10.1103/PhysRevLett.121.231101. arXiv:1810.12191. Accessed: 2026-01-04.
- [17] Everitt, C. W. F., et al. “Gravity Probe B: Final Results of a Space Experiment to Test General Relativity.” *Physical Review Letters* **106**, 221101 (2011). DOI: 10.1103/PhysRevLett.106.221101. arXiv:1105.3456. Accessed: 2026-01-08.
- [18] Ciufolini, I., and Pavlis, E. C. “A confirmation of the general relativistic prediction of the Lense-Thirring effect.” *Nature* **431**, 958-960 (2004). DOI: 10.1038/nature03007. Accessed: 2026-01-08.
- [19] International Laser Ranging Service (ILRS). “CRD - Consolidated Ranging Data Format.” Web: https://ilrs.gsfc.nasa.gov/data_and_products/formats/crd.html. Accessed: 2026-01-03.
- [20] DGFI-TUM. “EDC - Eurolas Data Center (ILRS data mirror).” Web: <https://edc.dgfi.tum.de/>. Accessed: 2026-01-03.
- [21] Pearlman, M. R., et al. “The ILRS: approaching 20 years and planning for the future.” *Journal of Geodesy* **93**, 2161-2180 (2019). DOI: 10.1007/s00190-019-01241-1.
- [22] Saastamoinen, J. “Atmospheric correction for the troposphere and stratosphere in radio ranging of satellites.” In: *The Use of Artificial Satellites for Geodesy* (Geophysical Monograph Series, Vol. 15). AGU (1972).
- [23] Niell, A. E. “Global mapping functions for the atmosphere delay at radio wavelengths.” *Journal of Geophysical Research: Solid Earth* **101**(B2), 3227-3246 (1996). DOI: 10.1029/95JB03048.
- [24] Petit, G., and Luzum, B. (eds.). *IERS Conventions (2010)*. IERS Technical Note No. 36 (2010).
- [25] Event Horizon Telescope Collaboration. “First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole.” *The Astrophysical Journal Letters* **875**, L1 (2019). DOI: 10.3847/2041-8213/ab0ec7. Accessed: 2026-01-22.
- [26] Event Horizon Telescope Collaboration. “First M87 Event Horizon Telescope Results. VI. The Shadow and Mass of the Central Black Hole.” *The Astrophysical Journal Letters* **875**, L6 (2019). DOI: 10.3847/2041-8213/ab1141.
- [27] Event Horizon Telescope Collaboration. “The persistent shadow of the supermassive black hole of M87. I. Observations, calibration, imaging, and analysis.” *Astronomy & Astrophysics* **681**, A79 (2024). DOI: 10.1051/0004-6361/202347932.
- [28] Event Horizon Telescope Collaboration. “The persistent shadow of the supermassive black hole of M87. II. Model comparisons and theoretical interpretations.” *Astronomy & Astrophysics* **693**, A265 (2025). DOI: 10.1051/0004-6361/202451296.
- [29] Event Horizon Telescope Collaboration. “First Sagittarius A* Event Horizon Telescope Results. I.” *The Astrophysical Journal Letters* **930**, L12 (2022). DOI: 10.3847/2041-8213/ac6674. Accessed: 2026-01-22.
- [30] Event Horizon Telescope Collaboration, et al. “First Sagittarius A* Event Horizon Telescope Results. V. Testing Astrophysical Models of the Galactic Center Black Hole.” *The Astrophysical Journal Letters* **930**, L16 (2022). DOI: 10.3847/2041-8213/ac6672. arXiv:2311.09478.
- [31] Vincent, F. H., et al. “Thick accretion discs and photon rings: the case for not-so-black shadows.” *Astronomy & Astrophysics* **667**, A170 (2022). DOI: 10.1051/0004-6361/202244339. arXiv:2206.12066.

- [32] Johnson, M. D., et al. “Universal interferometric signatures of a black hole’s photon ring.” *Science Advances* **6**(12), eaaz1310 (2020). DOI: 10.1126/sciadv.aaz1310. arXiv:1907.04329.
- [33] Gralla, S. E., Holz, D. E., and Wald, R. M. “Black hole photon ring shape as a test of general relativity.” *Physical Review D* **102**, 124004 (2020). DOI: 10.1103/PhysRevD.102.124004. arXiv:2008.03879.
- [34] GRAVITY Collaboration, et al. “A geometric distance measurement to the Galactic center black hole with 0.3% uncertainty.” *Astronomy & Astrophysics* **625**, L10 (2019). DOI: 10.1051/0004-6361/201935656.
- [35] GRAVITY Collaboration, et al. “Detection of the gravitational redshift in the orbit of the star S2 near the Galactic centre massive black hole.” *Astronomy & Astrophysics* **615**, L15 (2018). DOI: 10.1051/0004-6361/201833718. arXiv:1807.09409. Accessed: 2026-01-22.
- [36] GRAVITY Collaboration, et al. “Detection of the Schwarzschild precession in the orbit of the star S2 near the Galactic centre massive black hole.” *Astronomy & Astrophysics* **636**, L5 (2020). DOI: 10.1051/0004-6361/202037813. arXiv:2004.07187. Accessed: 2026-01-22.
- [37] GRAVITY Collaboration, et al. “The Flux Distribution of Sgr A*.” arXiv:2004.07185.
- [38] Wielgus, M., et al. “Millimeter light curves of Sagittarius A* observed during the 2017 Event Horizon Telescope campaign.” arXiv:2207.06829.
- [39] Johnson, M. D., et al. “The Scattering and Intrinsic Structure of Sagittarius A* at Radio Wavelengths.” *The Astrophysical Journal* **865**(2), 104 (2018). arXiv:1808.08966.
- [40] Psaltis, D., et al. “A Model for Anisotropic Interstellar Scattering and its Application to Sgr A*.” arXiv:1805.01242 (2018).
- [41] Zhu, Z., Johnson, M. D., and Narayan, R. “Testing General Relativity with the Black Hole Shadow Size and Asymmetry of Sagittarius A*: Limitations from Interstellar Scattering.” arXiv:1811.02079 (2018).
- [42] Peters, P. C., and Mathews, J. “Gravitational Radiation from Point Masses in a Keplerian Orbit.” *Physical Review* **131**, 435 (1963). DOI: 10.1103/PhysRev.131.435.
- [43] Peters, P. C. “Gravitational Radiation and the Motion of Two Point Masses.” *Physical Review* **136**, B1224 (1964). DOI: 10.1103/PhysRev.136.B1224.
- [44] Weisberg, J. M., and Huang, Y. “Relativistic Measurements from Timing the Binary Pulsar PSR B1913+16.” *The Astrophysical Journal* **829**, 55 (2016). DOI: 10.3847/0004-637X/829/1/55.
- [45] Kramer, M., et al. “Strong-field Gravity Tests with the Double Pulsar.” *Physical Review X* **11**, 041050 (2021). DOI: 10.1103/PhysRevX.11.041050. arXiv:2112.06795.
- [46] Freire, P. C. C., et al. “The relativistic pulsar-white dwarf binary PSR J1738+0333 II. The most stringent test of scalar-tensor gravity.” *Monthly Notices of the Royal Astronomical Society* (2012). DOI: 10.1111/j.1365-2966.2012.21253.x. arXiv:1205.1450. Accessed: 2026-01-08.
- [47] Antoniadis, J., et al. “A Massive Pulsar in a Compact Relativistic Binary.” *Science* **340**, 448-450 (2013). DOI: 10.1126/science.1233232. arXiv:1304.6875. Accessed: 2026-01-08.
- [48] Abbott, B. P., et al. “Observation of Gravitational Waves from a Binary Black Hole Merger.” *Physical Review Letters* **116**, 061102 (2016). DOI: 10.1103/PhysRevLett.116.061102. Accessed: 2026-01-06.

- [49] Gravitational Wave Open Science Center (GWOSC). Event API / strain data. Web: <https://gwosc.org/eventapi/>. Accessed: 2026-01-06.
- [50] Martinelli, M., et al. “Euclid: Forecast constraints on the cosmic distance duality relation with complementary external probes.” *Astronomy & Astrophysics* (2021). DOI: 10.1051/0004-6361/202039637. arXiv:2007.16153. Accessed: 2026-01-08.
- [51] Blondin, S., et al. “Time Dilation in Type Ia Supernova Spectra at High Redshift.” Accepted for publication in *The Astrophysical Journal* (2008). arXiv:0804.3595. Accessed: 2026-01-09.
- [52] Avgoustidis, A., et al. “Constraints on the CMB temperature-redshift dependence from SZ and distance measurements.” arXiv:1112.1862 (2011). Accessed: 2026-01-09.
- [53] Alam, S., et al. “The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample.” *Monthly Notices of the Royal Astronomical Society* **470**, 2617-2652 (2017). DOI: 10.1093/mnras/stx721. arXiv:1607.03155. Accessed: 2026-01-09.
- [54] Ross, A. J., et al. “COMBINEDDR12 correlation function multipoles (post-reconstruction) data package.” SDSS SAS (DR12 / BOSS / papers / clustering). Web: https://data.sdss.org/sas/dr12/booss/papers/clustering/Ross_etal_2016_COMBINEDDR12_corrfunc.zip. Accessed: 2026-01-11.
- [55] Satpathy, S., et al. “COMBINEDDR12 full-shape correlation function multipoles (pre-reconstruction) data package.” SDSS SAS (DR12 / BOSS / papers / clustering). Web: https://data.sdss.org/sas/dr12/booss/papers/clustering/Satpathy_etal_2016_COMBINEDDR12_full_shape_corrfunc_multipoles.zip. Accessed: 2026-01-11.